ggplot2

Ajustando os seus gráficos

James Sampaio, PhD

ggplot2

Visualizando conjuntos de dados

- Título
- Sistema de coordenadas
- Escalas
- Temas
- Eixos
- Legendas

Dados sobre a população do Texas

- Usaremos um gráfico baseado neste conjunto de dados como exemplo. Ele exibe as populações dos condados do Texas.

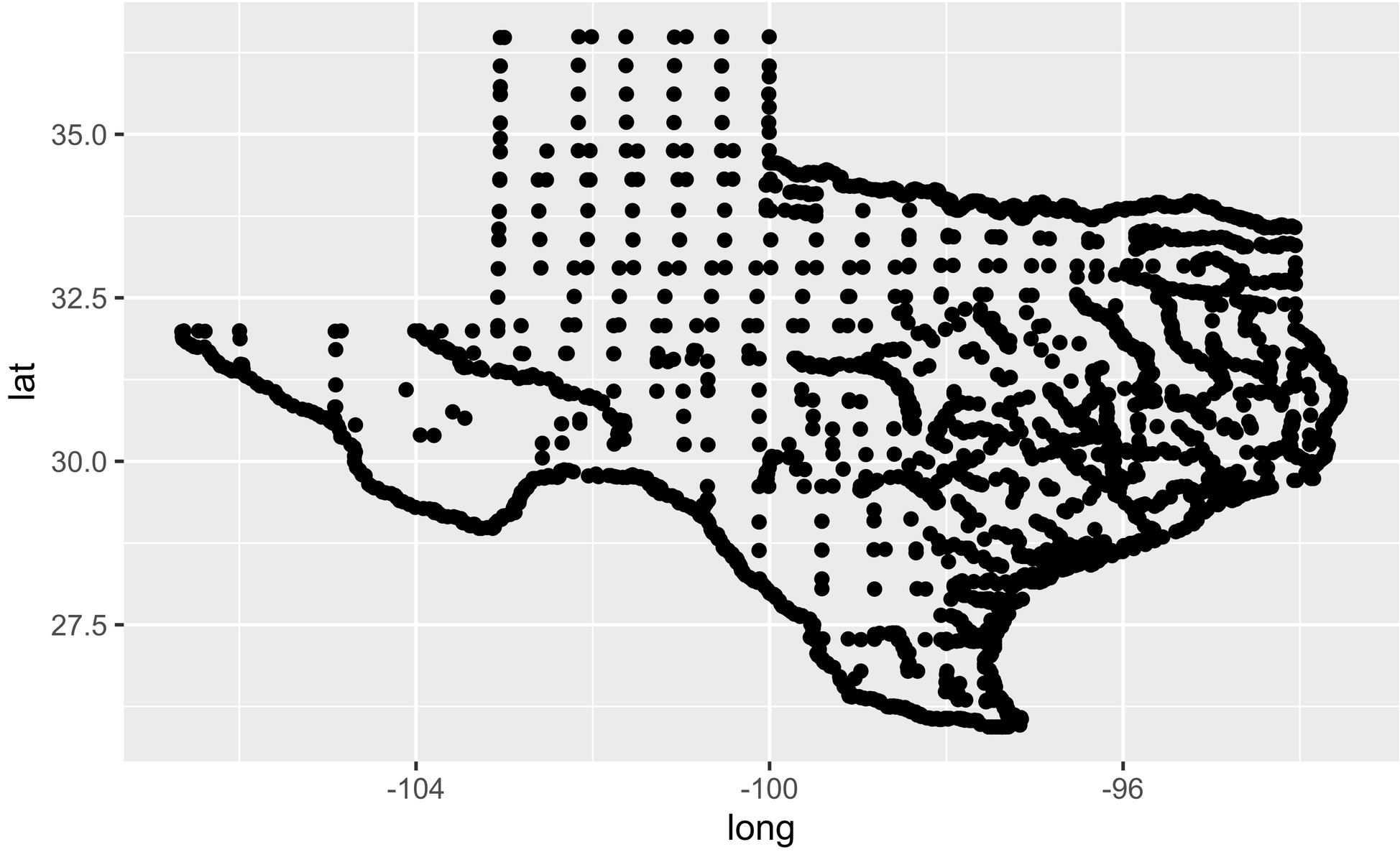
```
texas <- read.csv("texas.csv")
```

Dados sobre a população do Texas

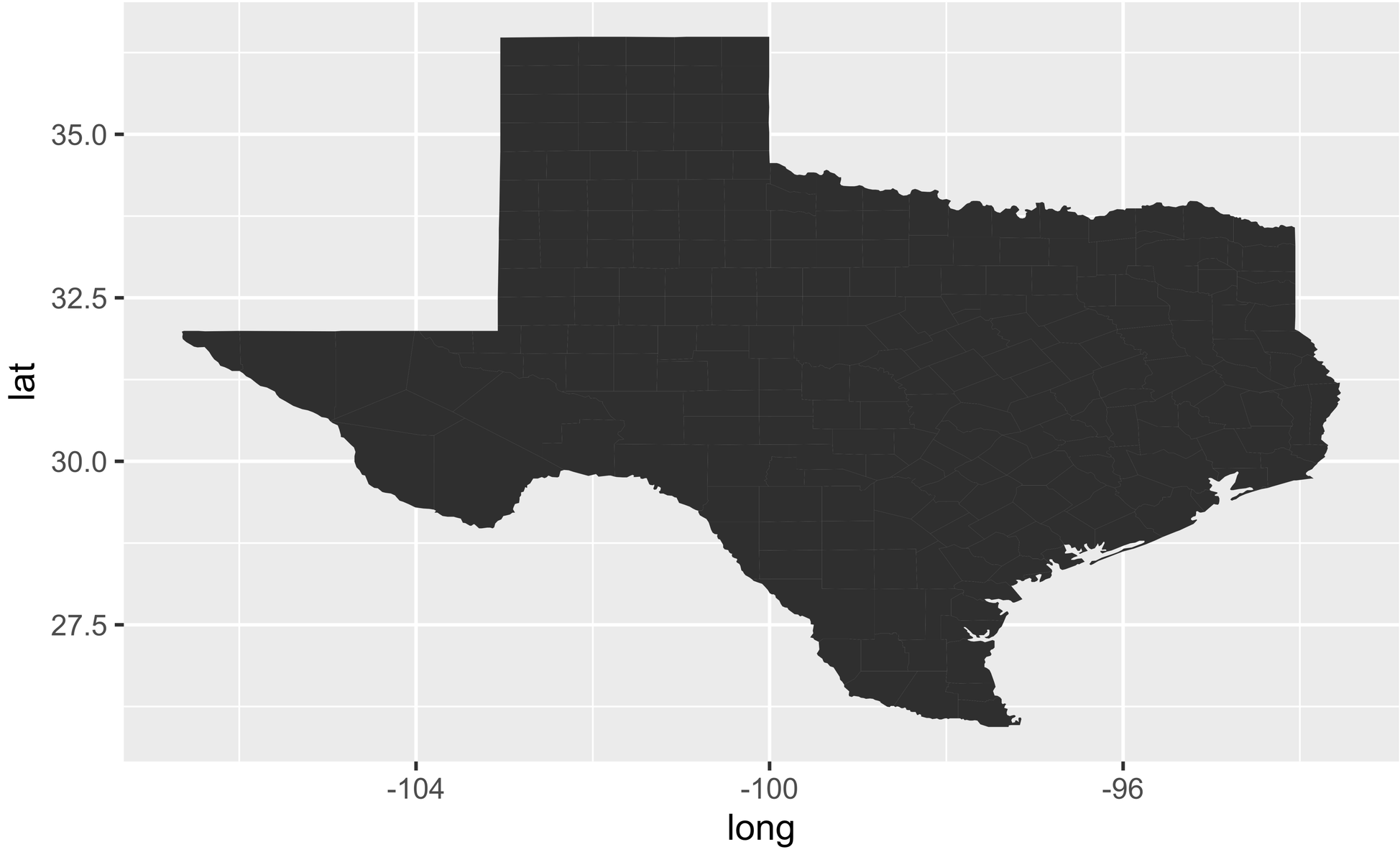
- Usaremos um gráfico baseado neste conjunto de dados como exemplo. Ele exibe as populações dos condados do Texas.

```
texas <- read.csv("texas.csv")
```

```
> head(texas)
       long      lat group order state   county   pop   bin
1 -95.75271 31.53560     1     1 texas anderson 56474 < 1e5
2 -95.76989 31.55852     1     2 texas anderson 56474 < 1e5
3 -95.76416 31.58143     1     3 texas anderson 56474 < 1e5
4 -95.72979 31.58143     1     4 texas anderson 56474 < 1e5
5 -95.74698 31.61008     1     5 texas anderson 56474 < 1e5
6 -95.72405 31.63873     1     6 texas anderson 56474 < 1e5
```




```
ggplot( texas, aes(long, lat) ) + geom_point()
```

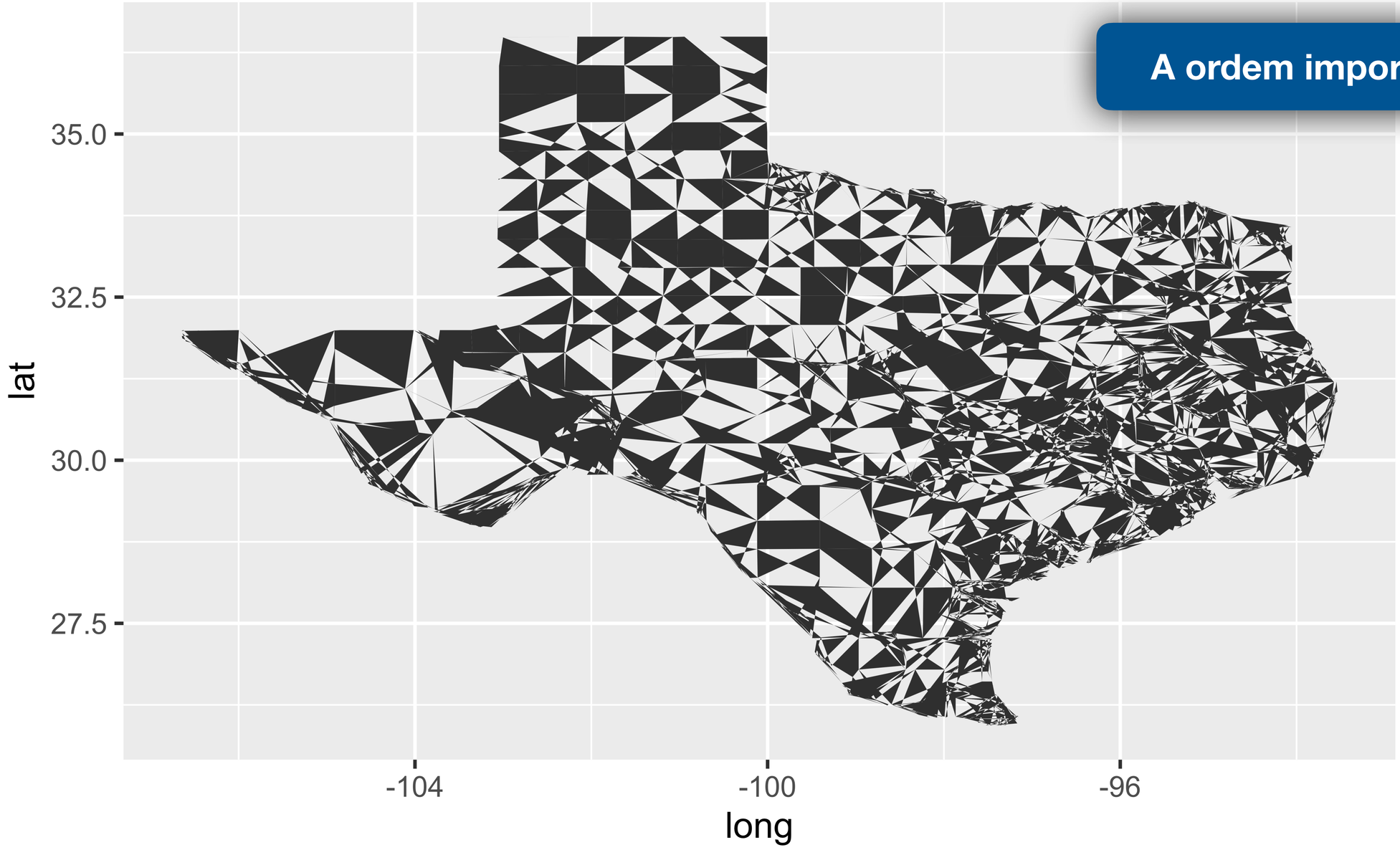



```
ggplot( texas, aes(long, lat) ) + geom_polygon( aes(group = group) )
```

Agrupa os pontos em diferentes polígonos

Determina a ordem de conexão dos pontos

```
> head(texas)
       long      lat group order state   county   pop    bin
1 -95.75271 31.53560     1     1 texas anderson 56474 < 1e5
2 -95.76989 31.55852     1     2 texas anderson 56474 < 1e5
3 -95.76416 31.58143     1     3 texas anderson 56474 < 1e5
4 -95.72979 31.58143     1     4 texas anderson 56474 < 1e5
5 -95.74698 31.61008     1     5 texas anderson 56474 < 1e5
6 -95.72405 31.63873     1     6 texas anderson 56474 < 1e5
```



A ordem importa

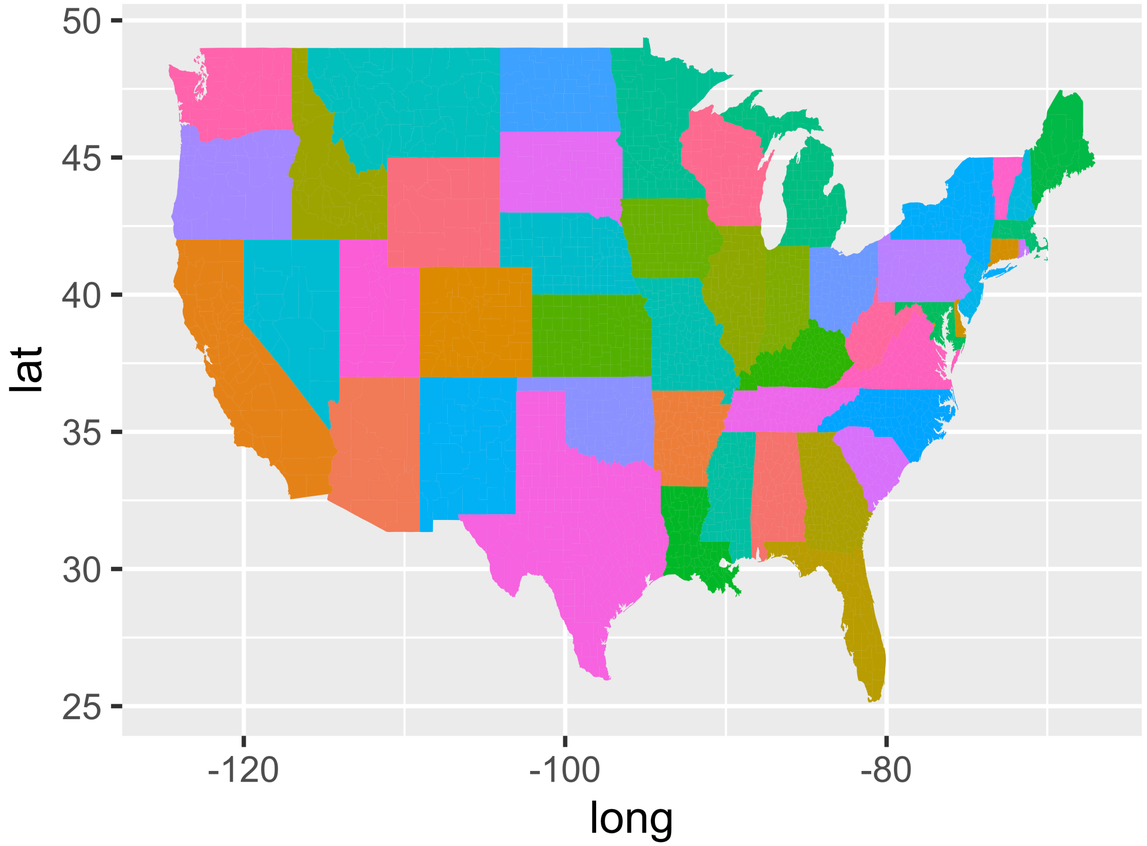
```
texas_samp <- texas[sample( nrow(texas) ), ]
ggplot( texas_samp, aes(long, lat) ) + geom_polygon( aes(group = group) )
```

Pacote maps

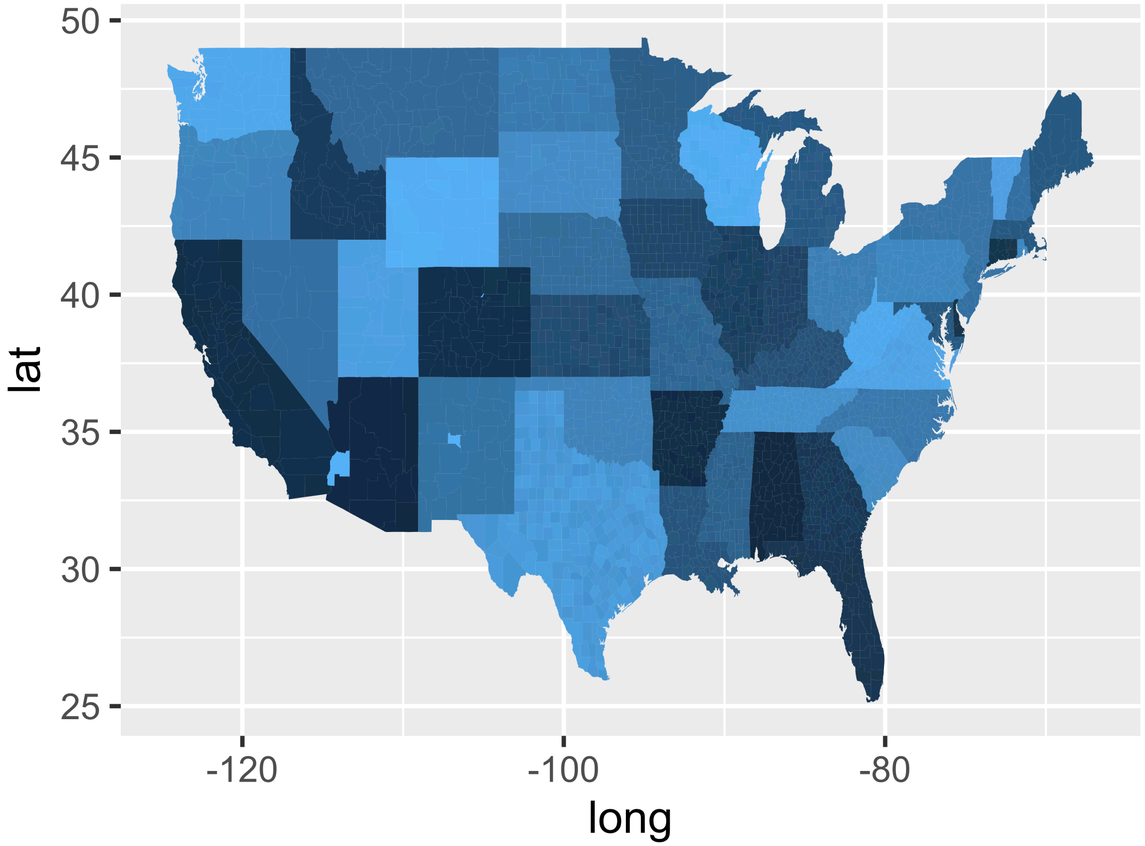
```
install.packages("maps")
library(maps)

counties <- map_data("county")

c_map <- ggplot( counties, aes(long, lat) )
```



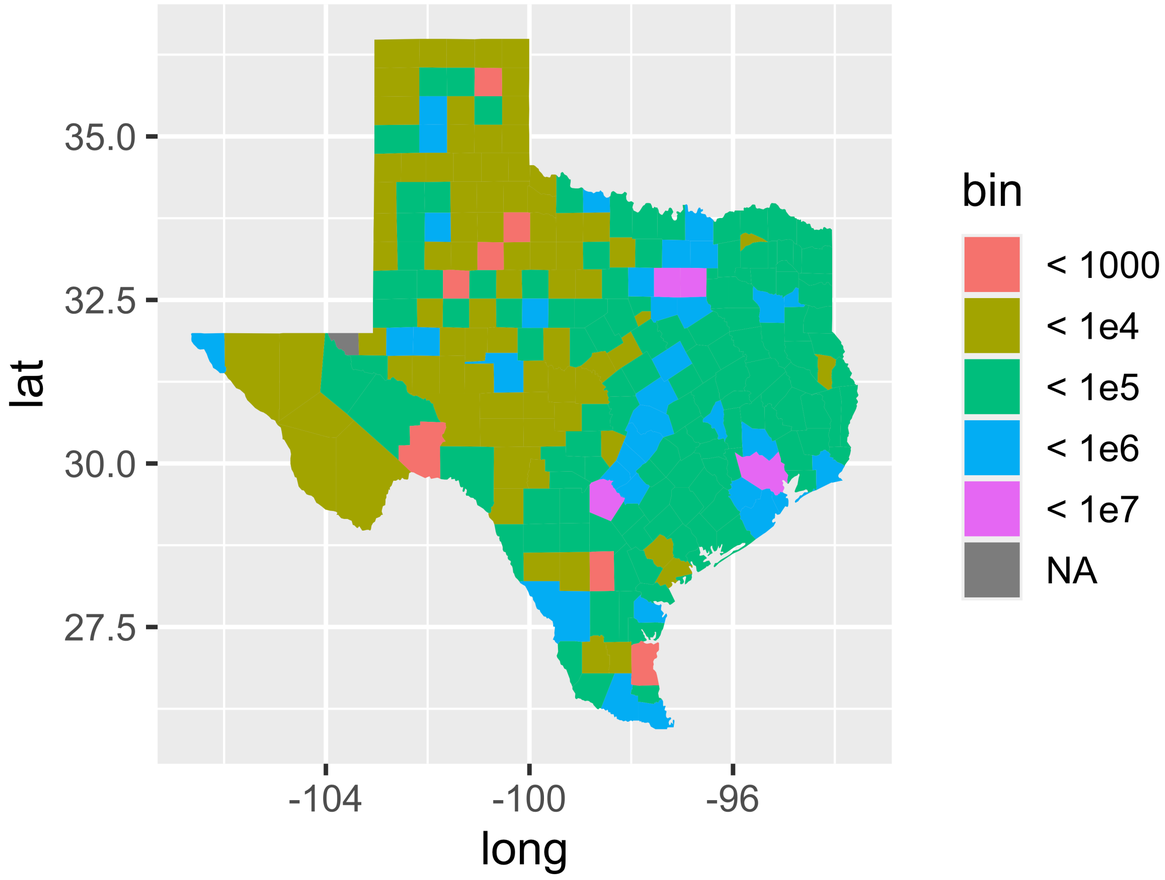

```
c_map +
geom_polygon( aes(group = group,
                  fill = region) )
```




```
c_map +
geom_polygon( aes(group = group,
                  fill = group) )
```

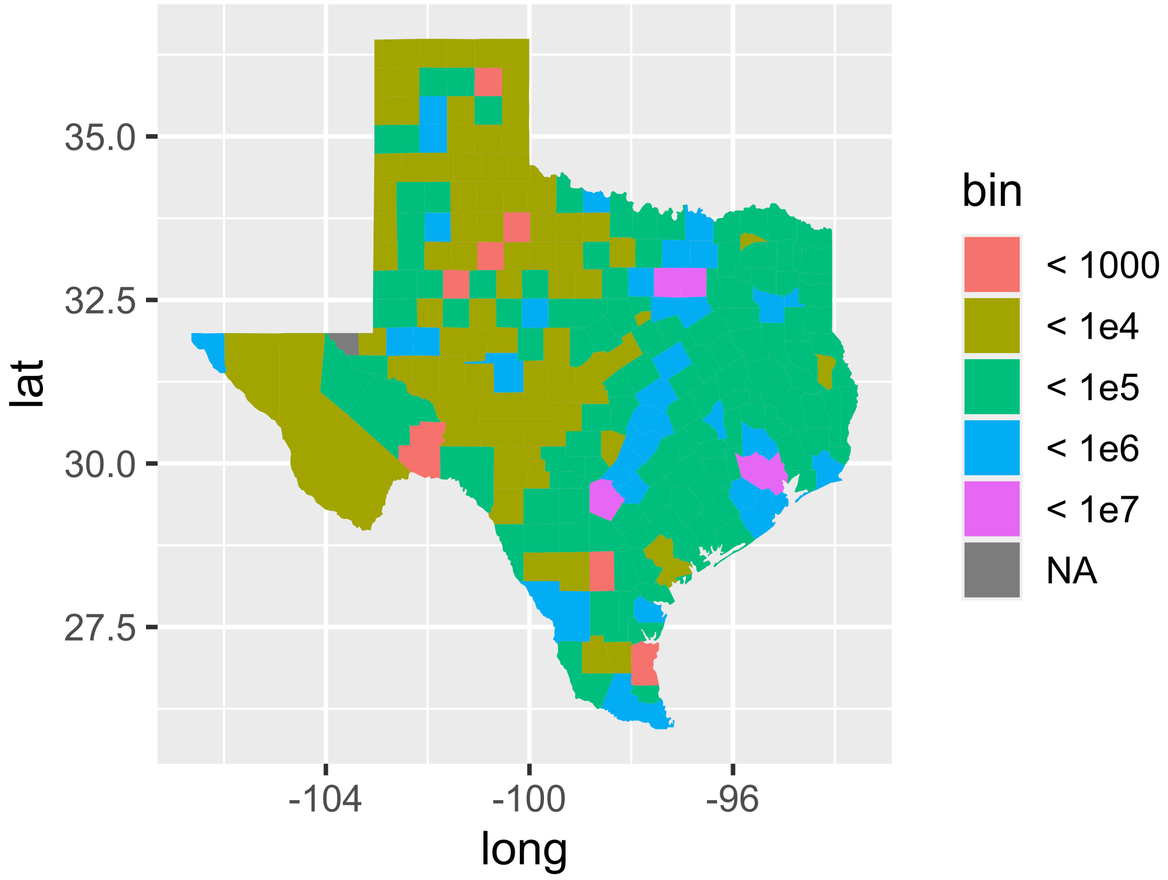
SUA VEZ

Use `texas` para recriar o gráfico abaixo



SUA VEZ

Use `texas` para recriar o gráfico abaixo

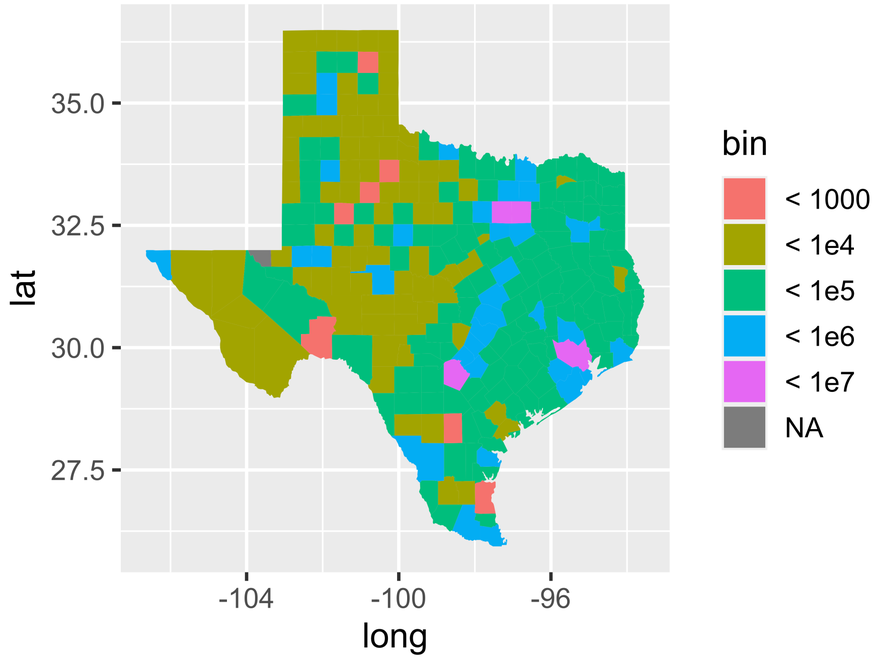


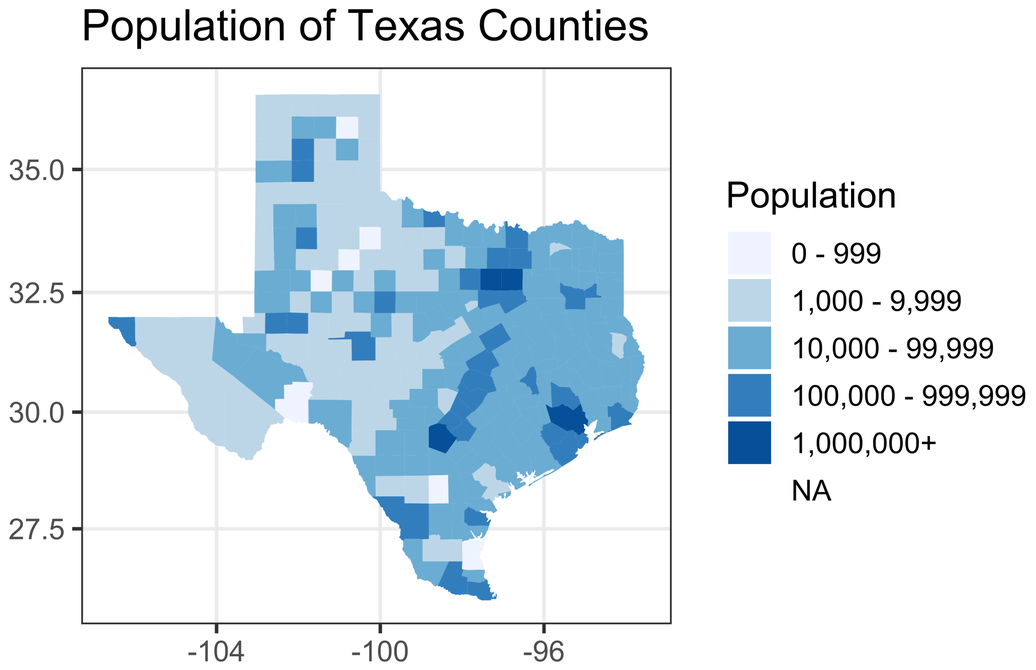

```
ggplot( texas, aes(long, lat) ) +
  geom_polygon( aes(group = group,
                    fill = bin) )
```

Iremos passear pela personalização do gráfico base abaixo

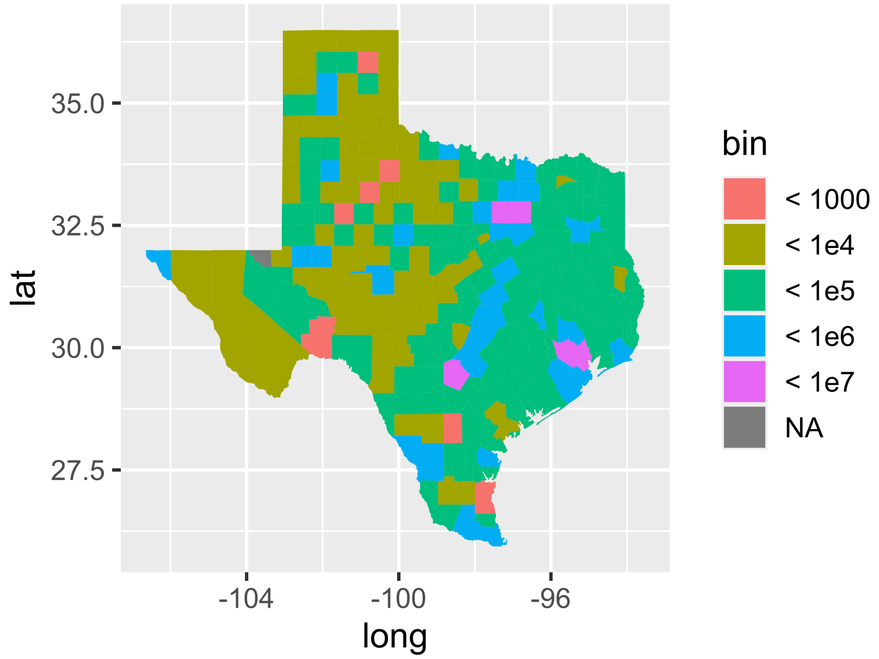
```
tx <- ggplot( texas, aes(long, lat) ) +
        geom_polygon( aes(group = group, fill = bin) )
```

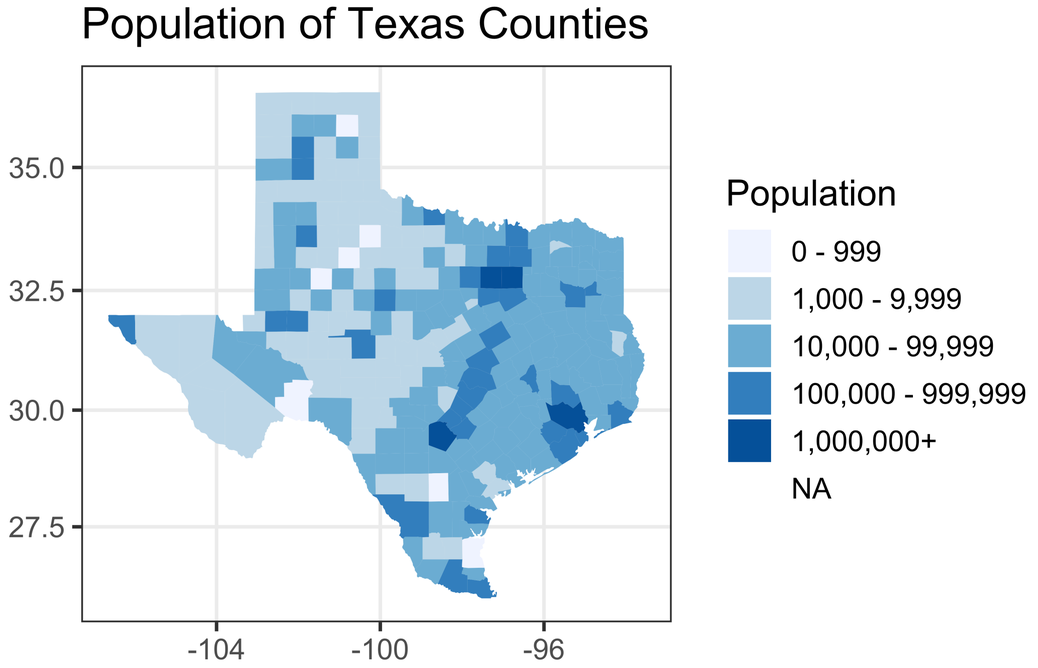
Qual você prefere?





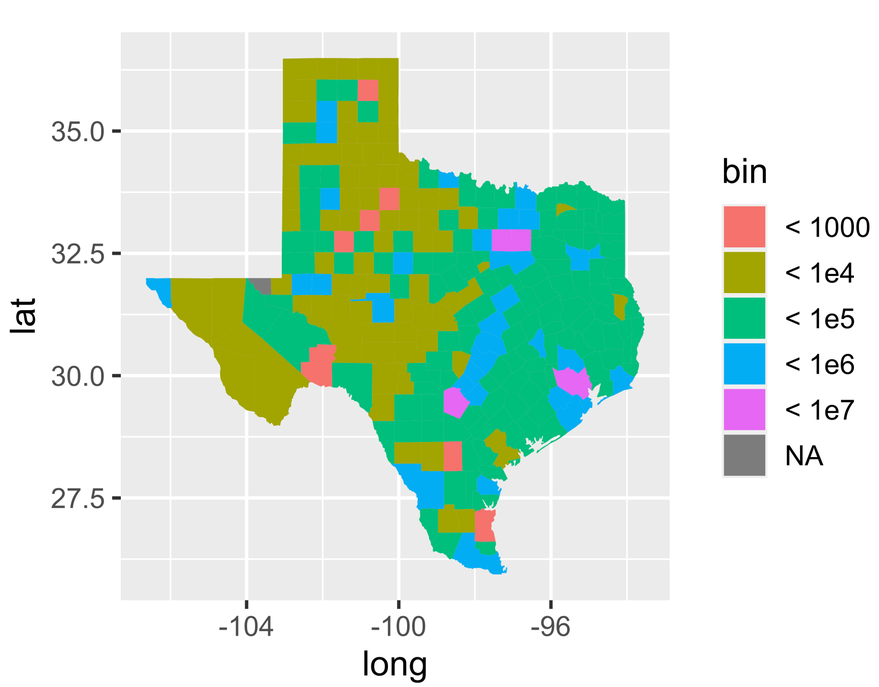
Qual você prefere?

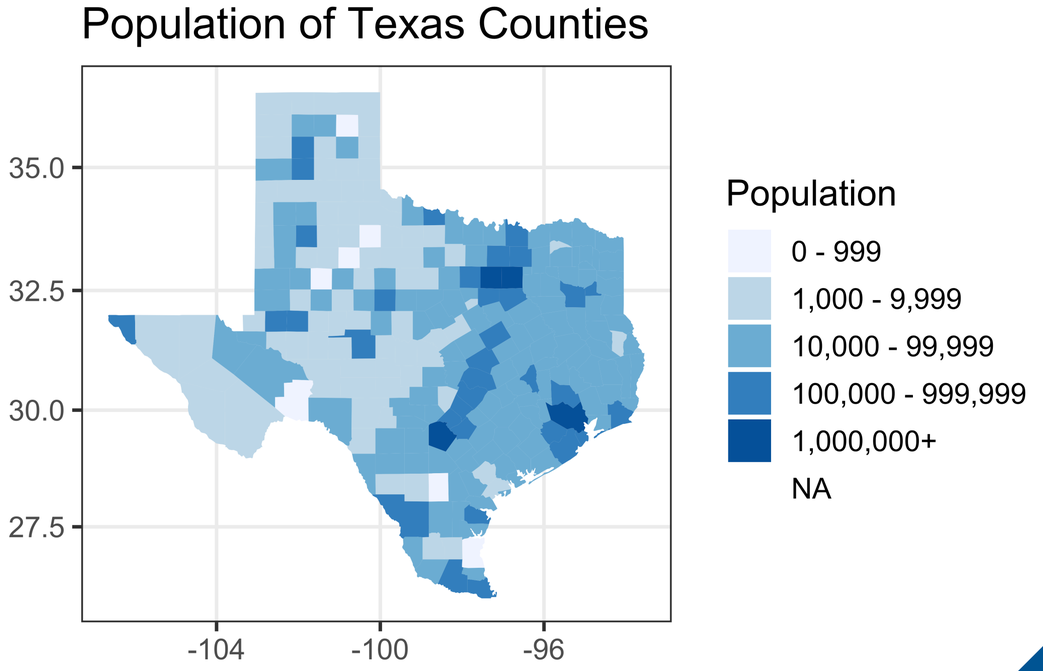




- Título
- Sistema de coordenadas correto
- Esquema de cores
- Fundo branco
- Rótulos dos eixos
- Rótulos das legendas

Qual você prefere?





- Título
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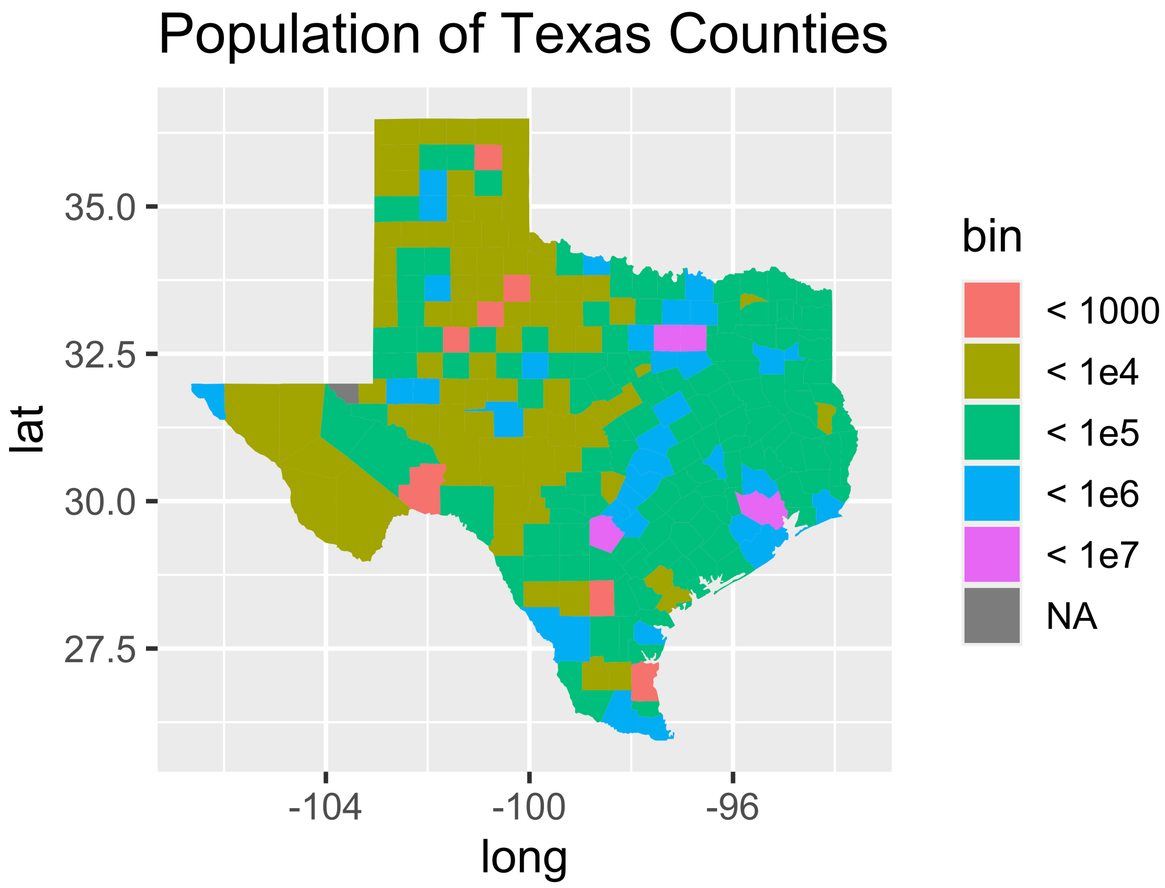
Nosso objetivo

Título

ggtitle

Adiciona um título ao seu gráfico

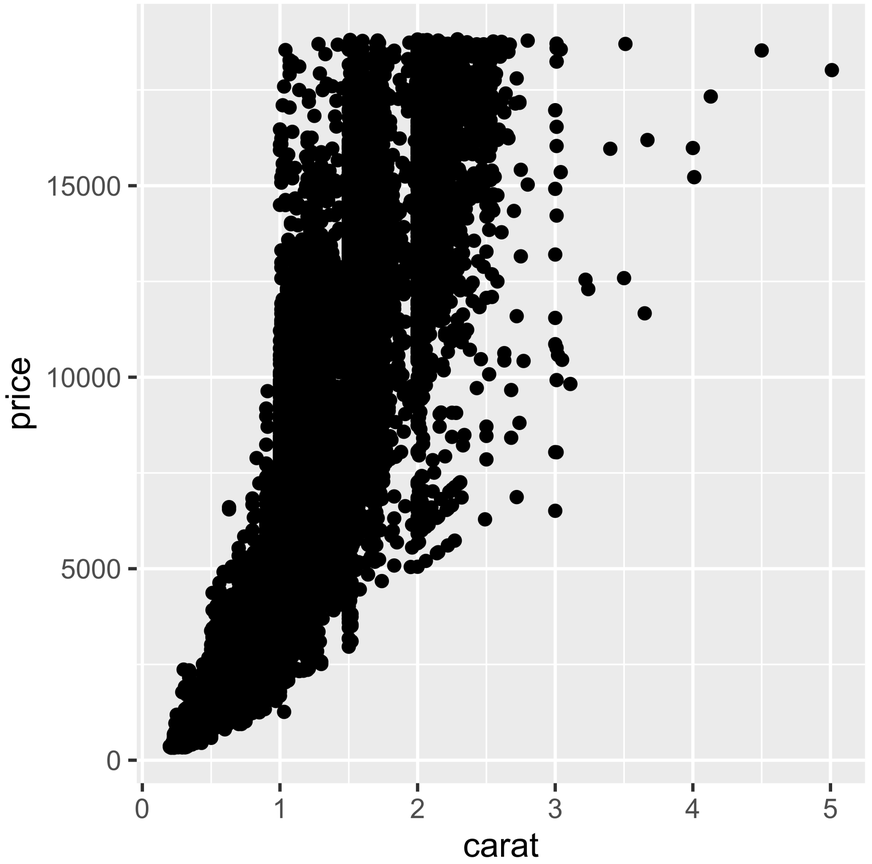



```
tx +
  ggtitle("Population of Texas Counties")
```

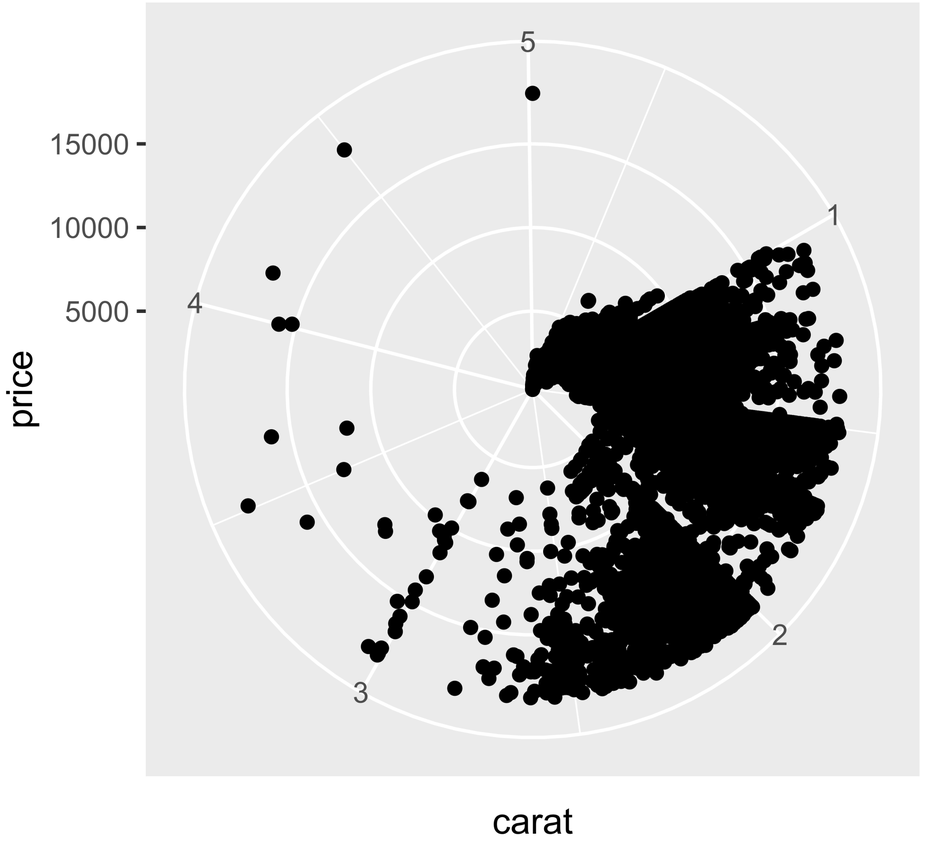
Sistema de coordenadas

Sistema de coordenadas

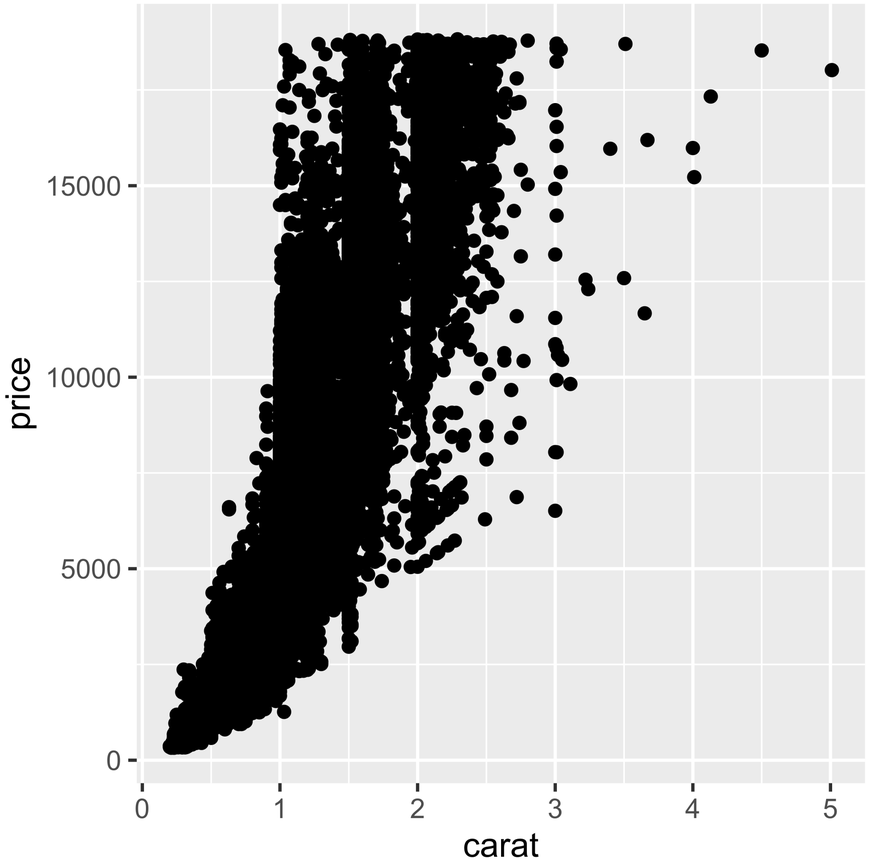
```
cp <- ggplot( diamonds, aes(carat, price) ) +
        geom_point()

class(cp)
[1] "gg"     "ggplot"
```




```
cp + coord_cartesian()
```

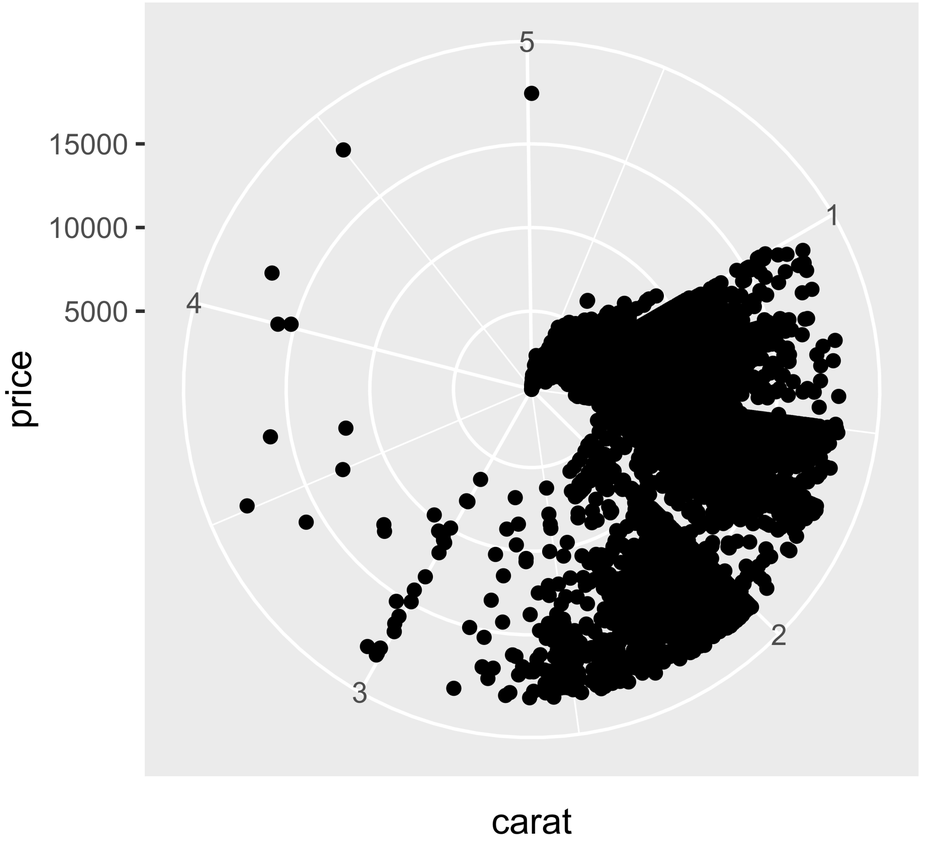



```
cp + coord_polar()
```



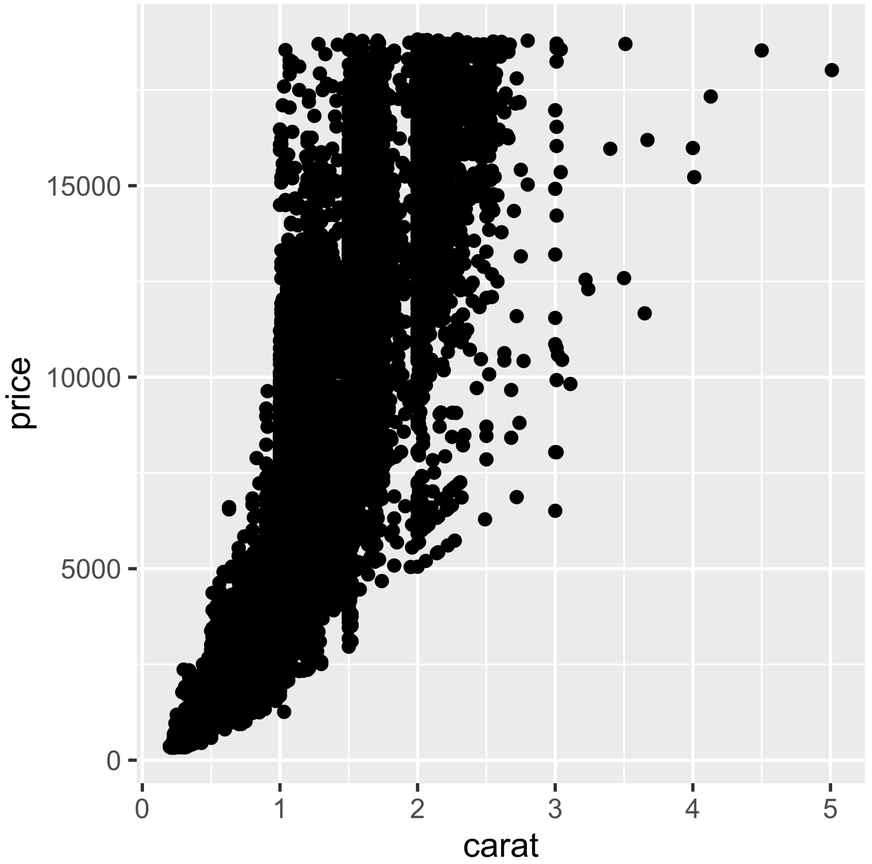

```
cp + coord_cartesian()
```

Função coordenada



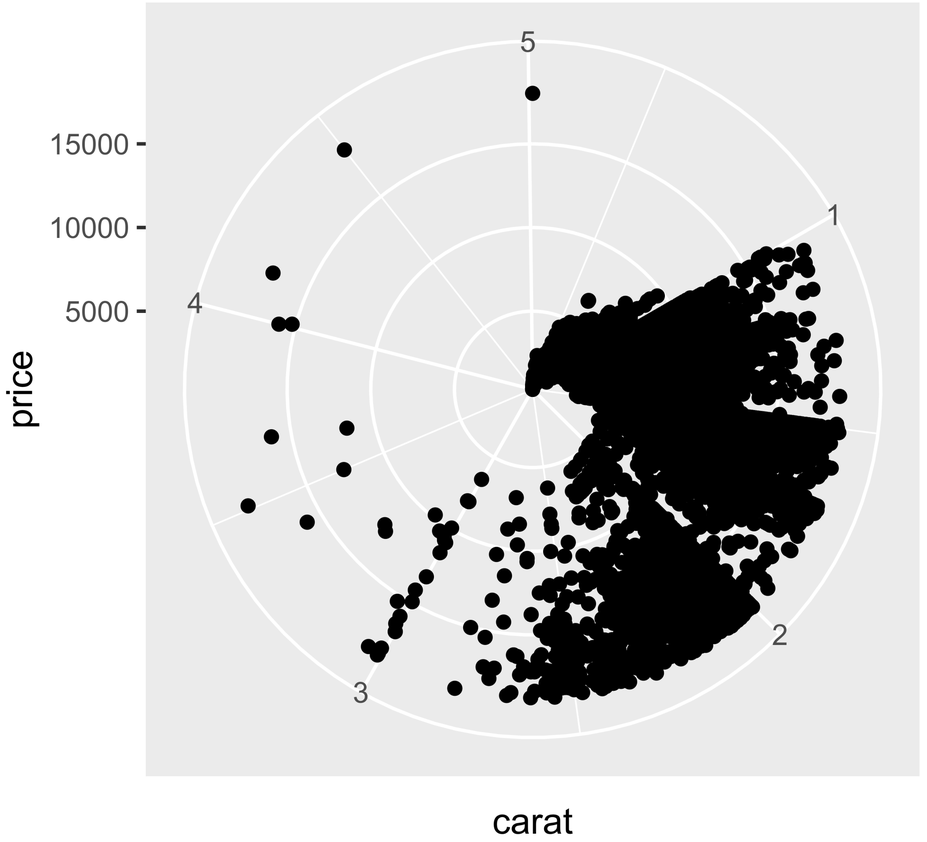

```
cp + coord_polar()
```

Função coordenada



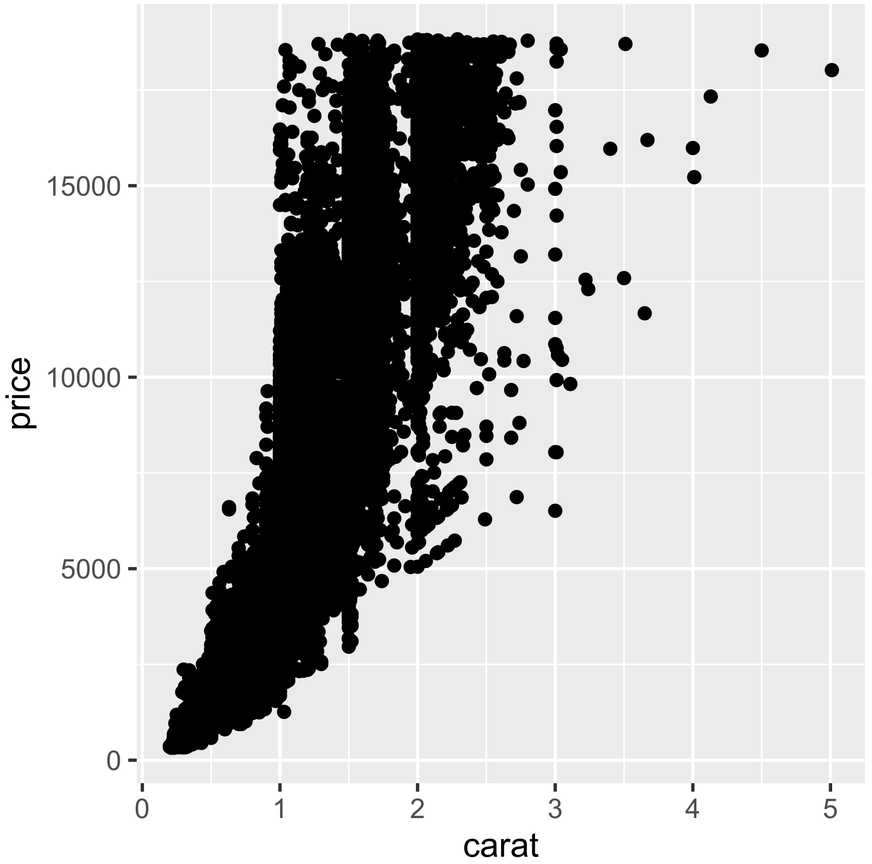

```
cp + coord_cartesian()
```

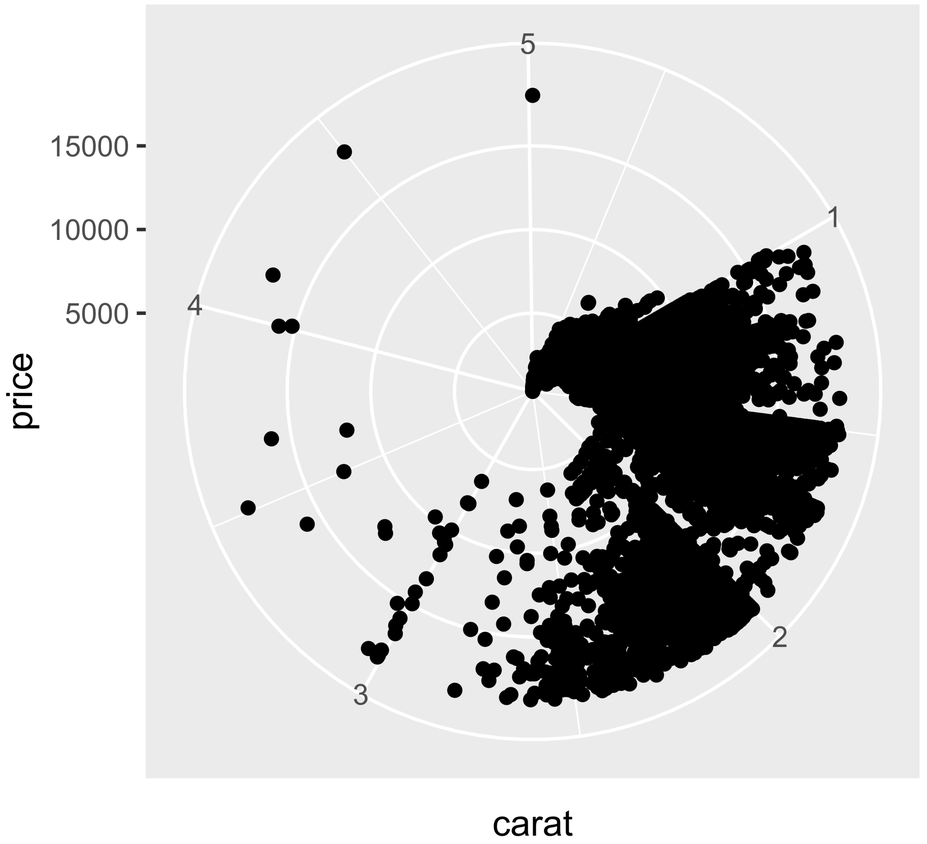
Tipo




```
cp + coord_polar()
```

Tipo

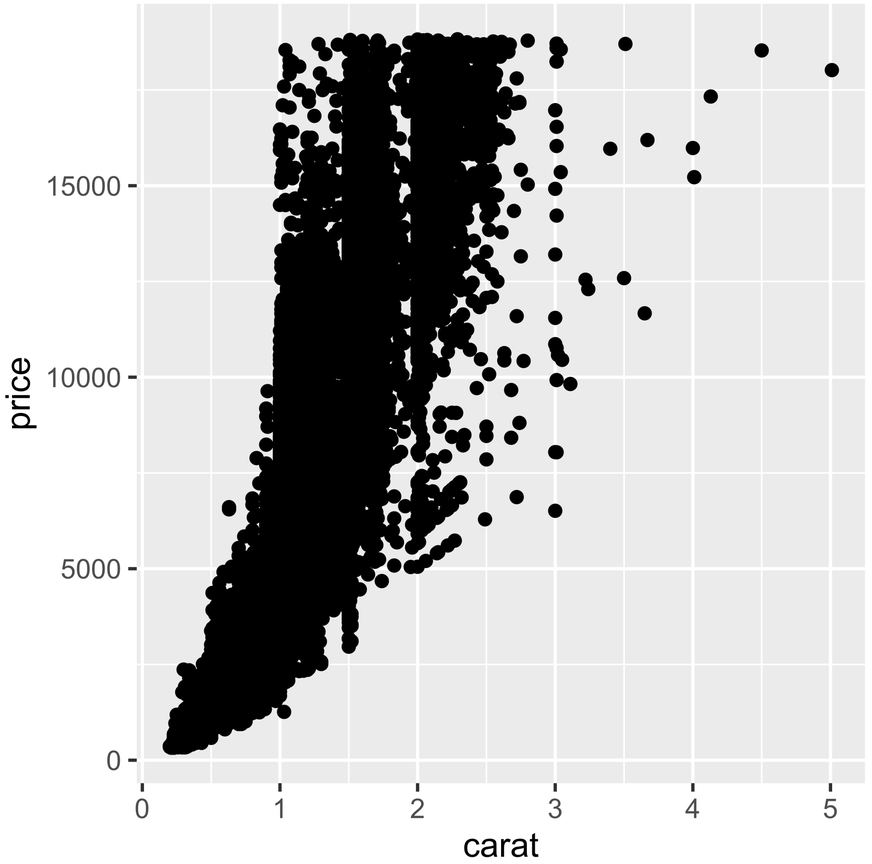




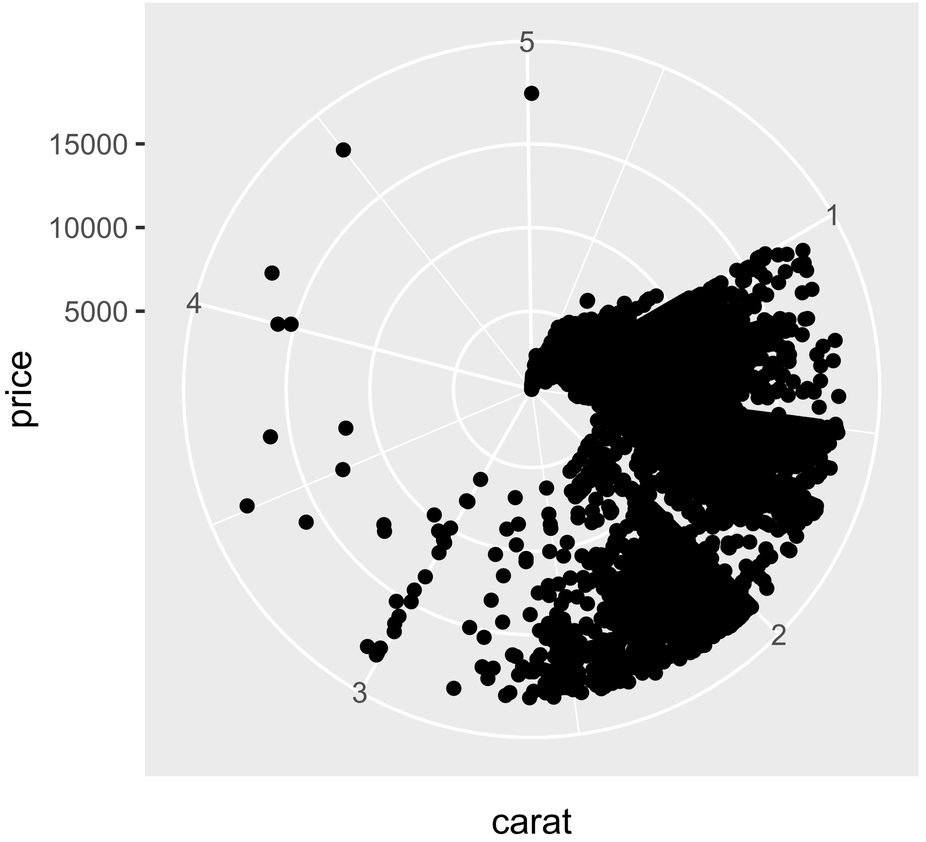

```
cp + coord_cartesian()
```

```
cp + coord_polar()
```

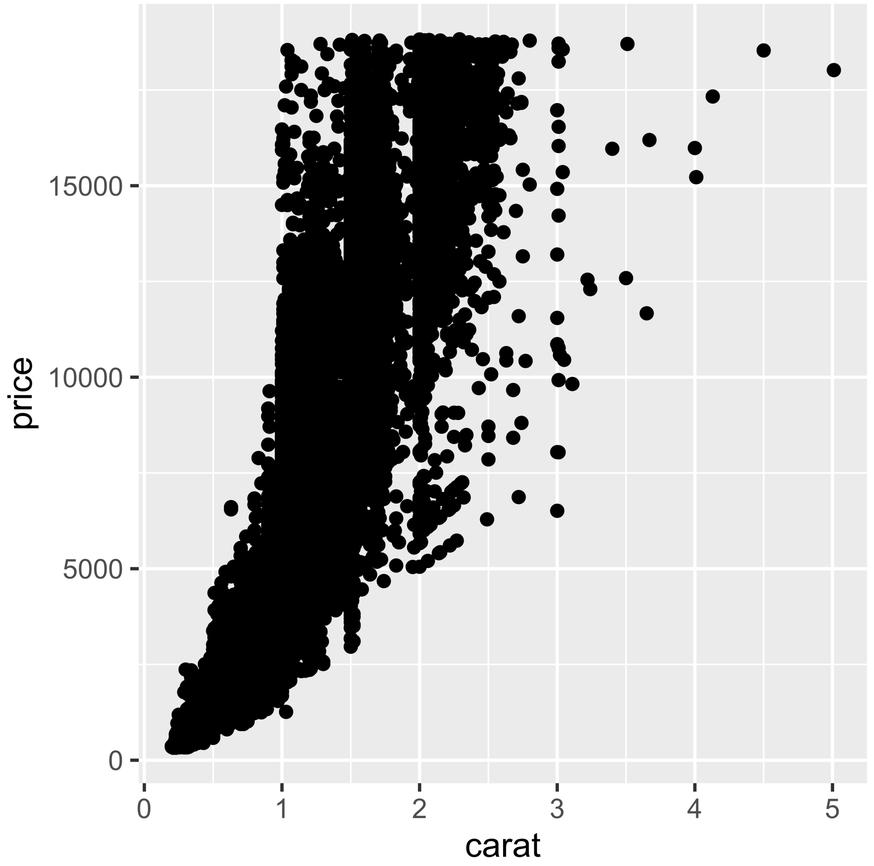
Padrão

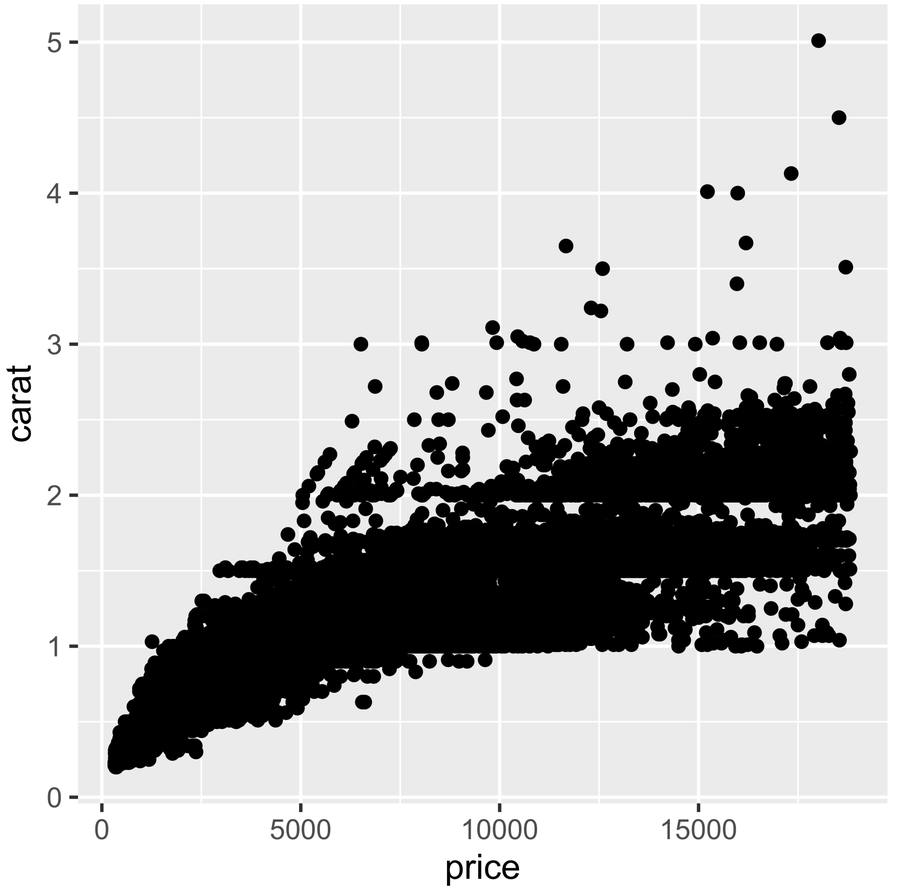


cp



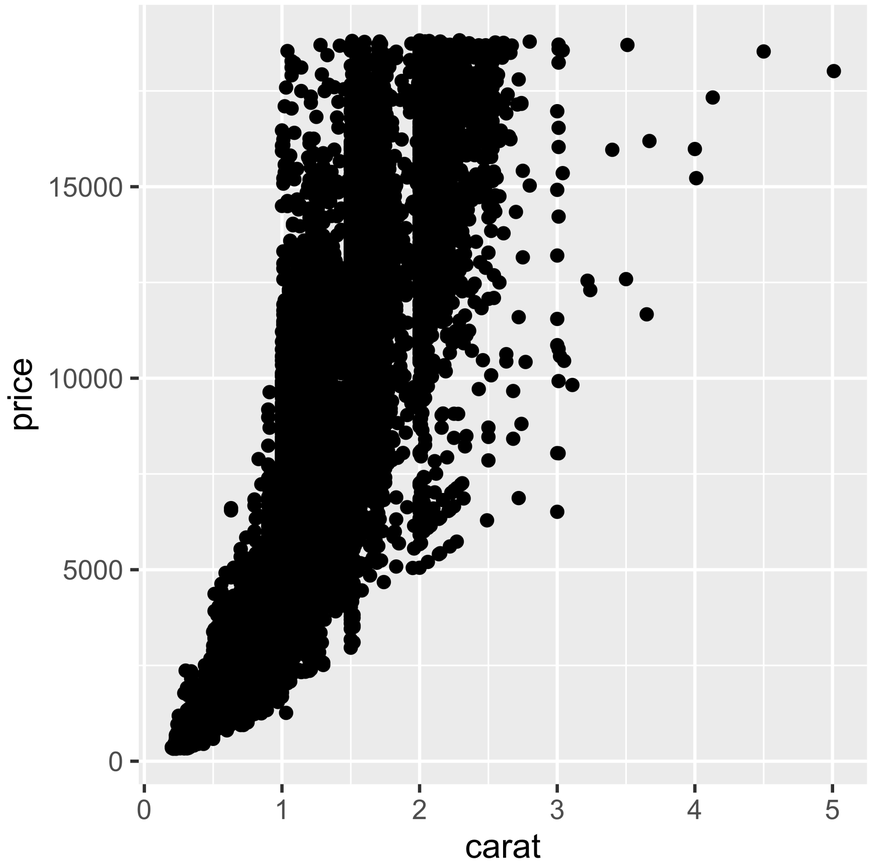
cp + coord_polar()

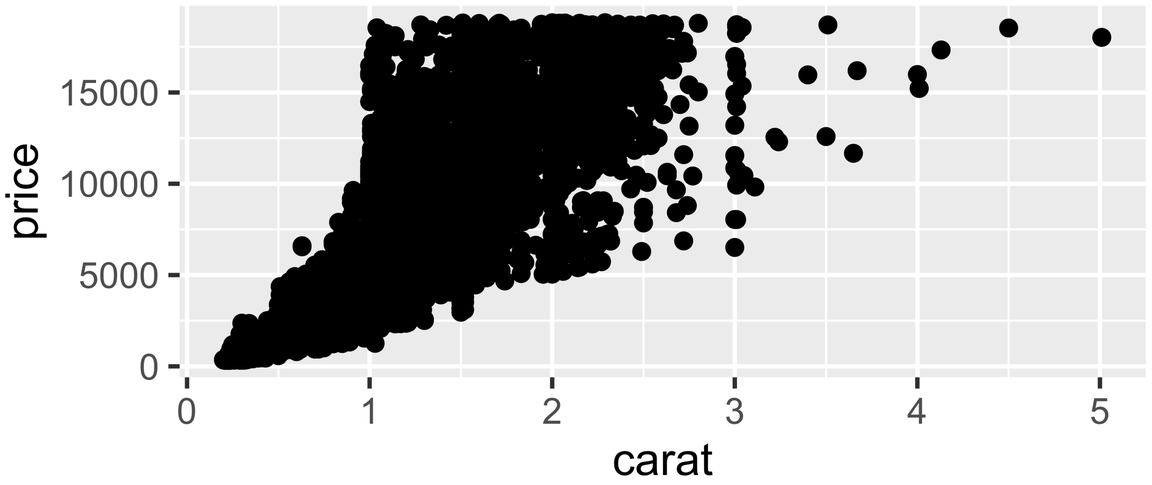





```
cp
```

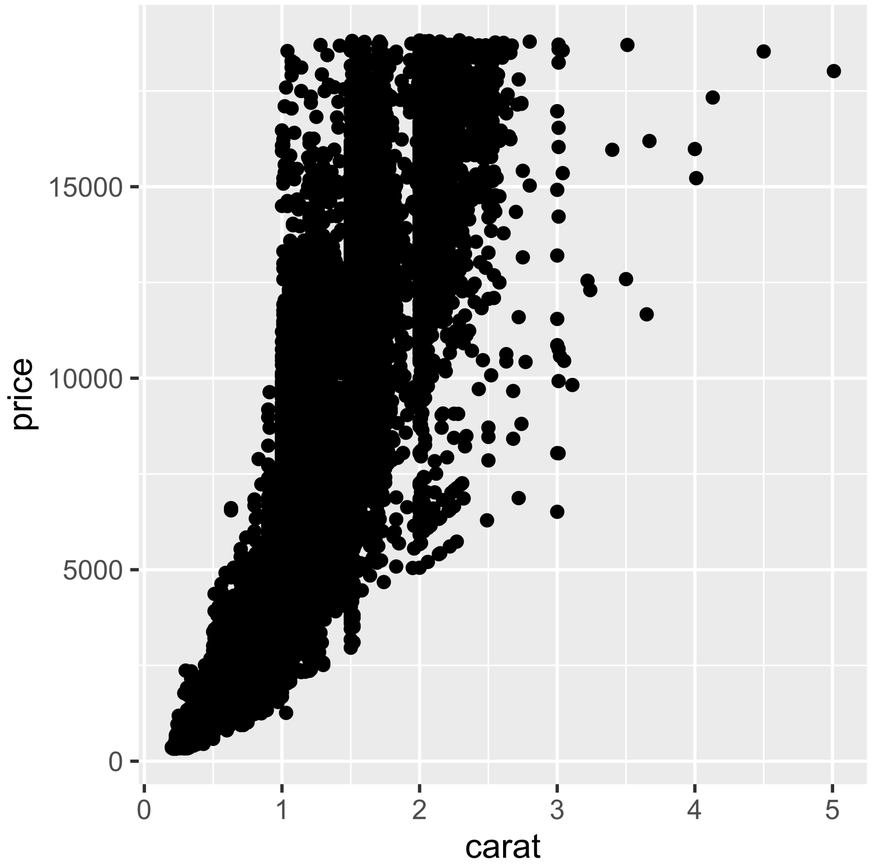
```
cp + coord_flip()
```

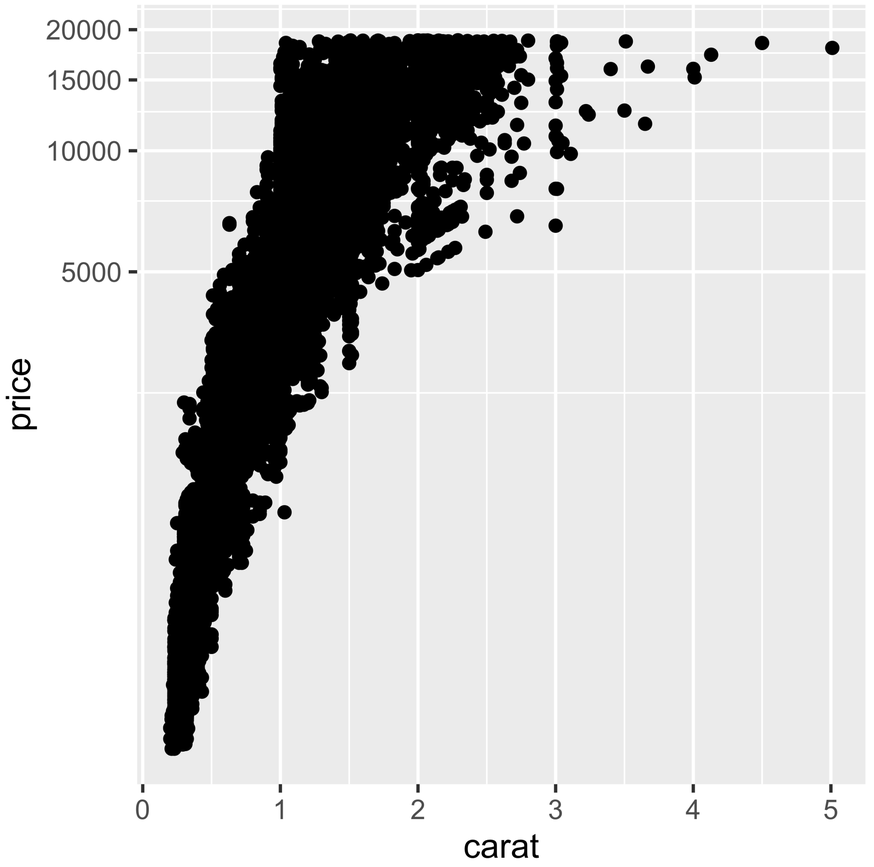





```
cp
```

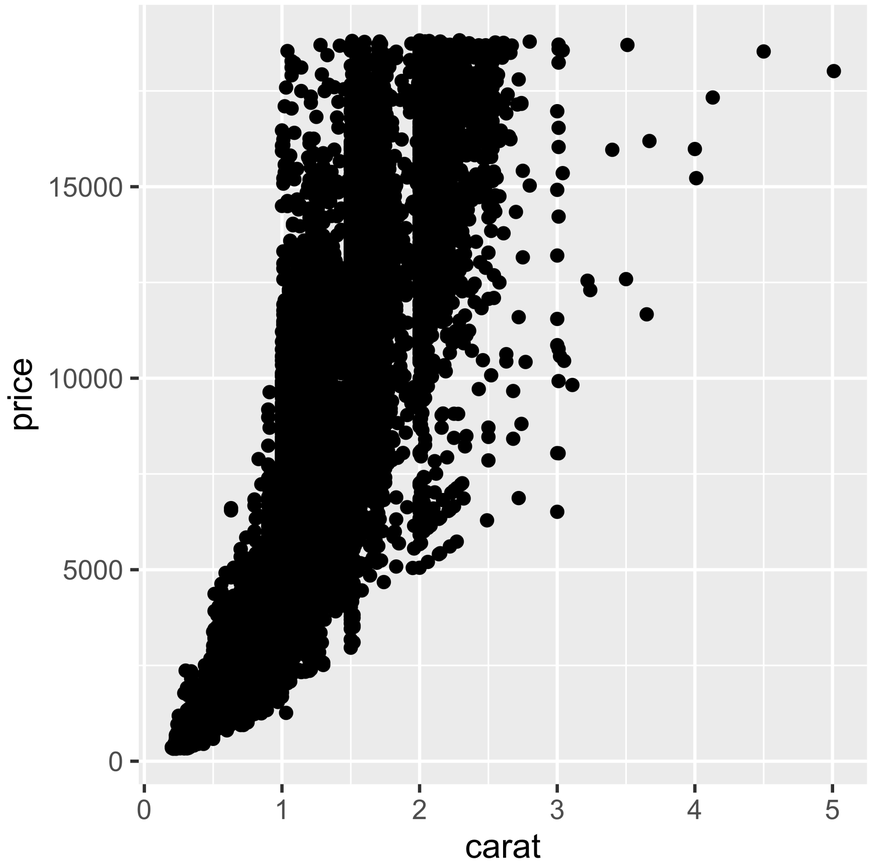
```
cp + coord_fixed( ratio = 1/10000 )
```

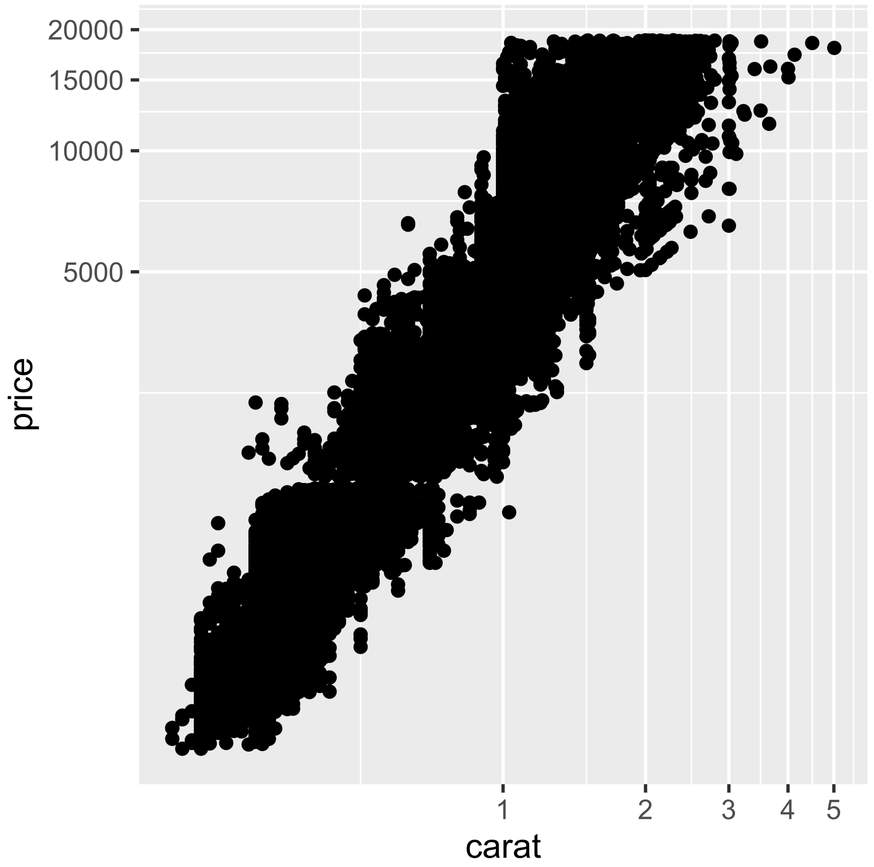





```
cp
```

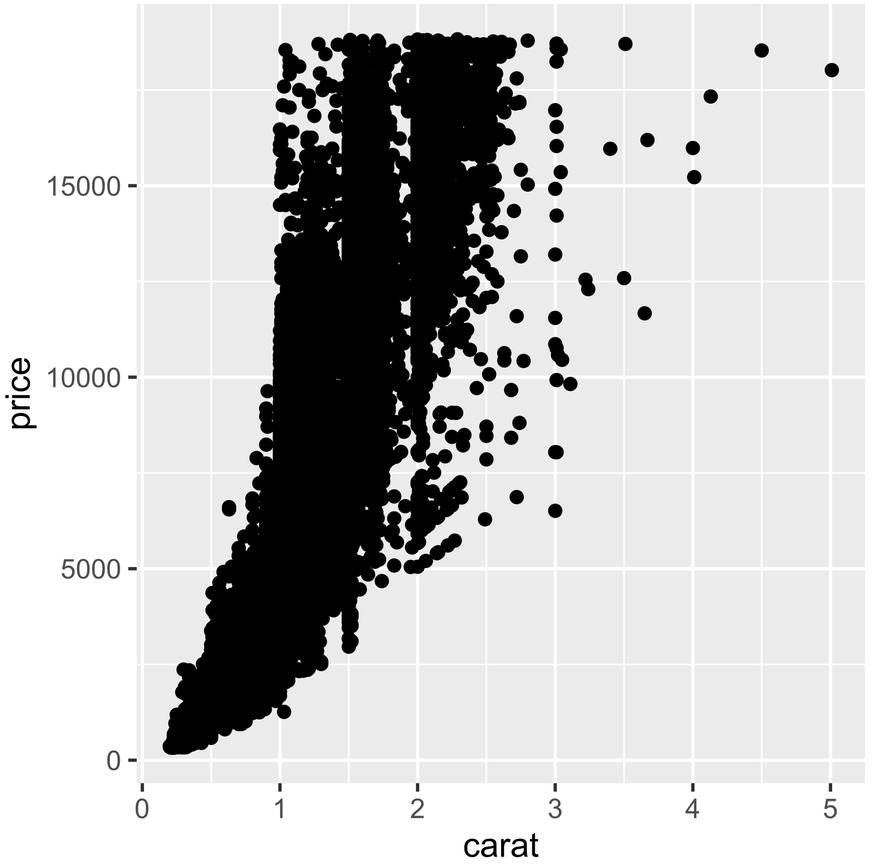
```
cp + coord_trans( y = "log10" )
```

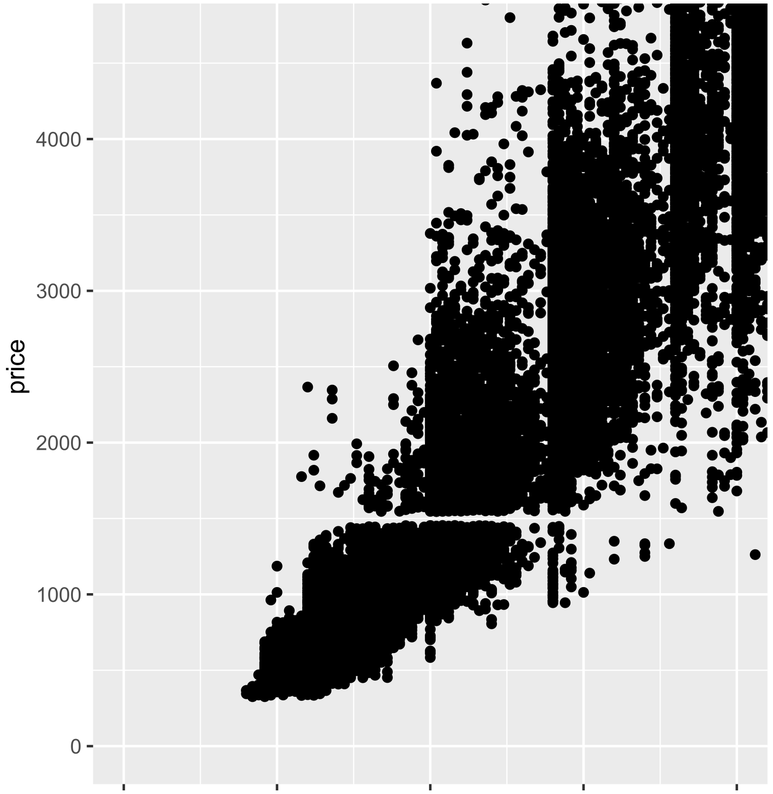





```
cp
```

```
cp + coord_trans( y = "log10",
                  x = "log10")
```






```
cp
```

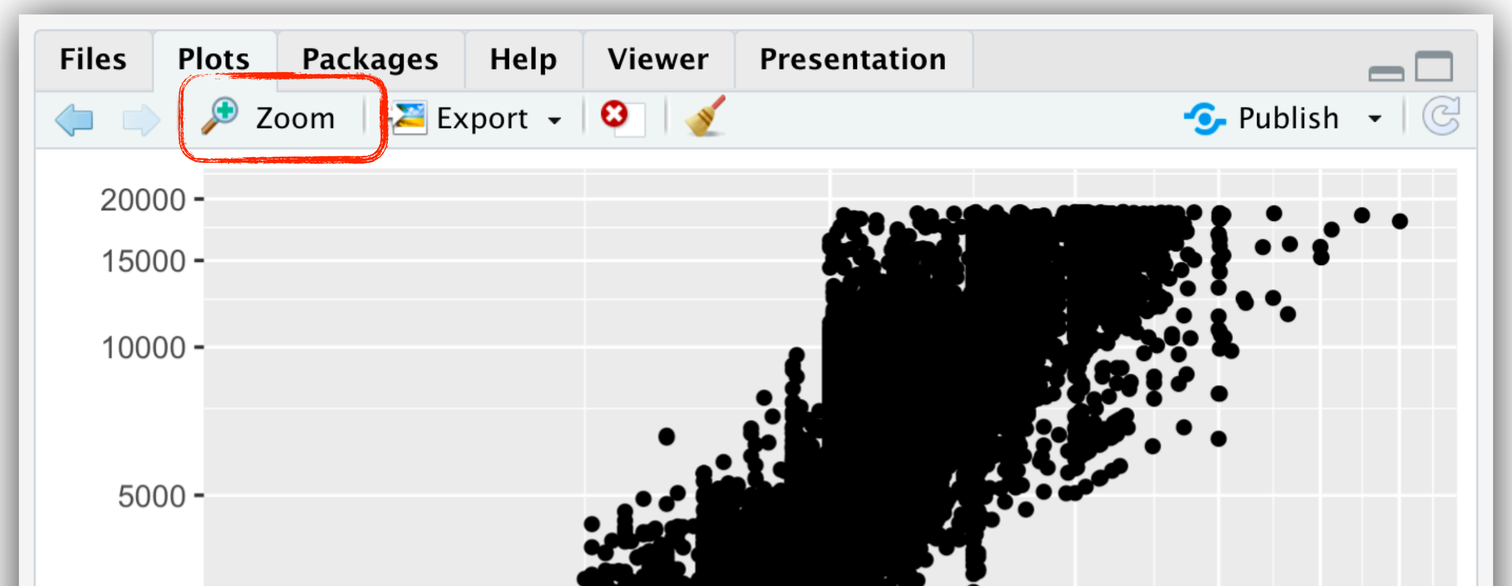
```
cp + coord_cartesian( ylim = c(0,5000),
                      xlim = c(0, 1) )
```

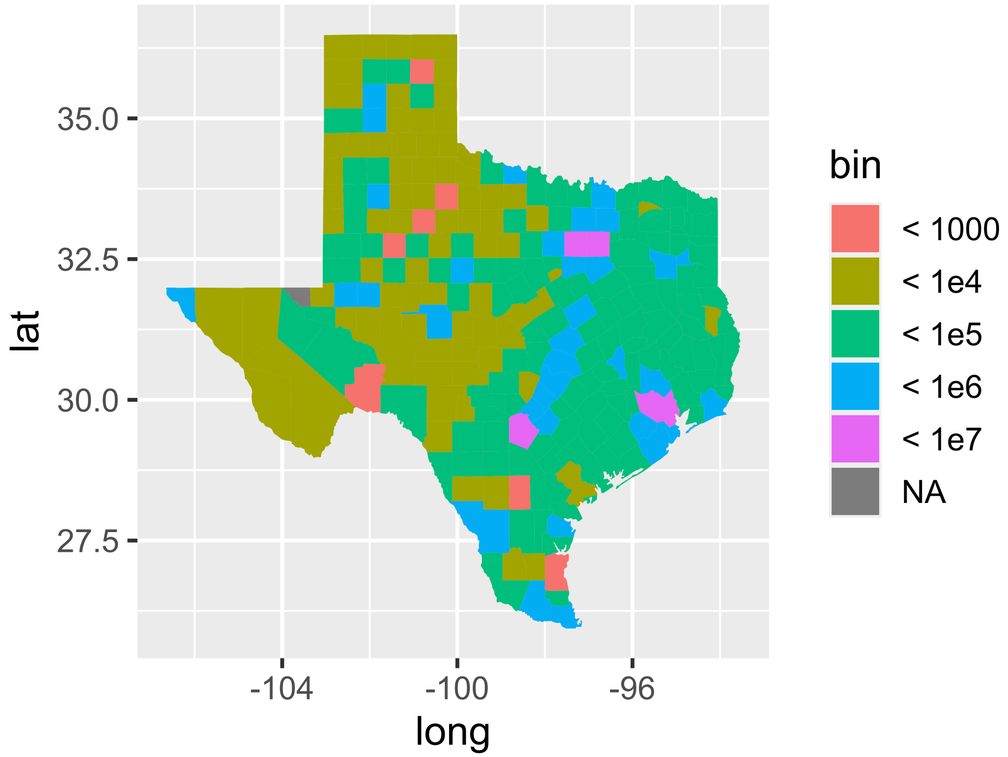
Zoom

SUA VEZ

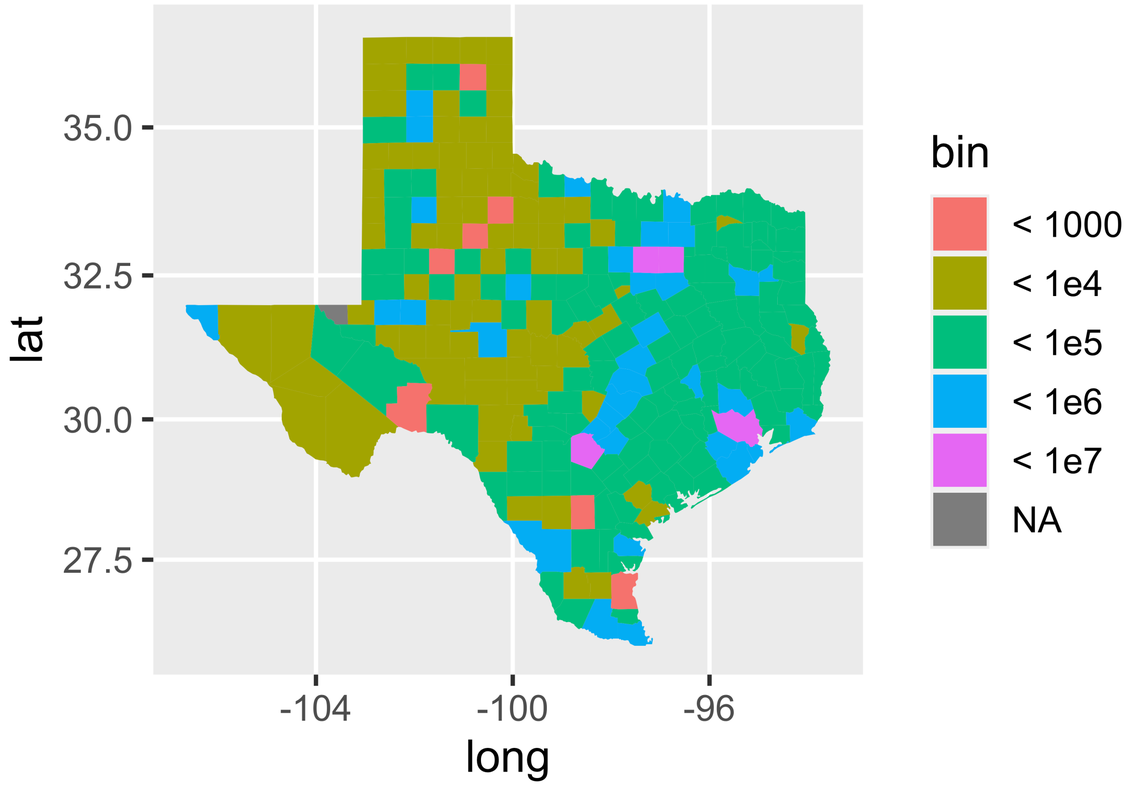
Existe uma coordenada `coord_map`**.** Adicione a `tx` e abra o gráfico em uma nova janela por meio do botão zoom.

Brinque com a escala do gráfico. Você consegue dizer o que faz o comando `coord_map`?

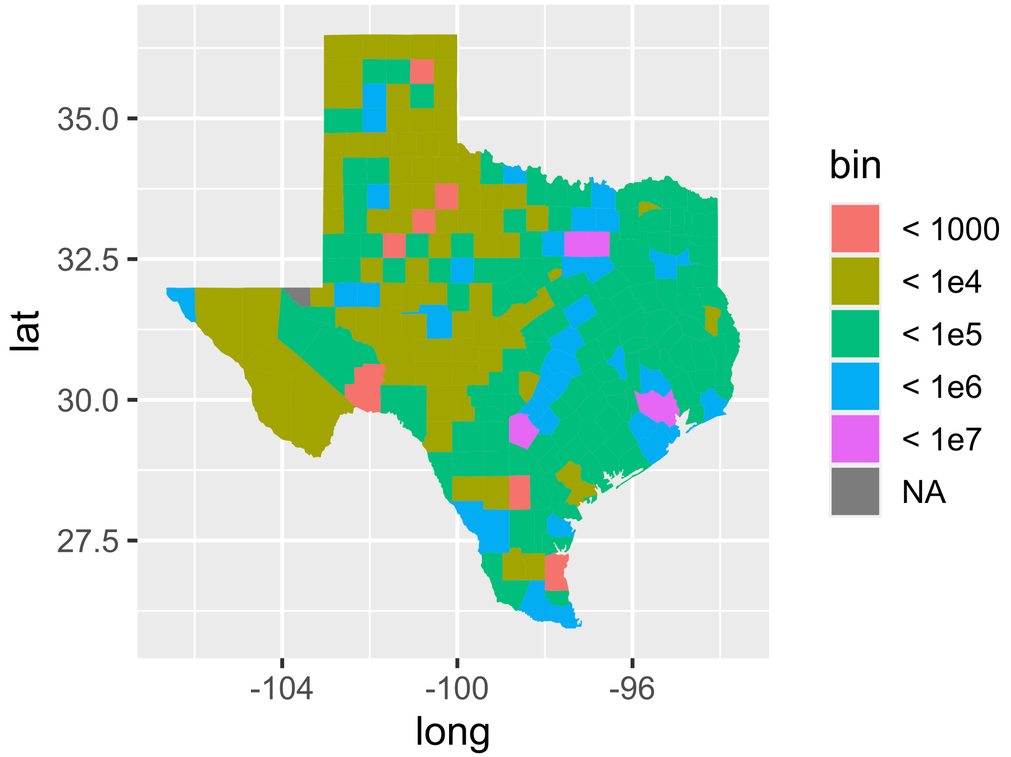




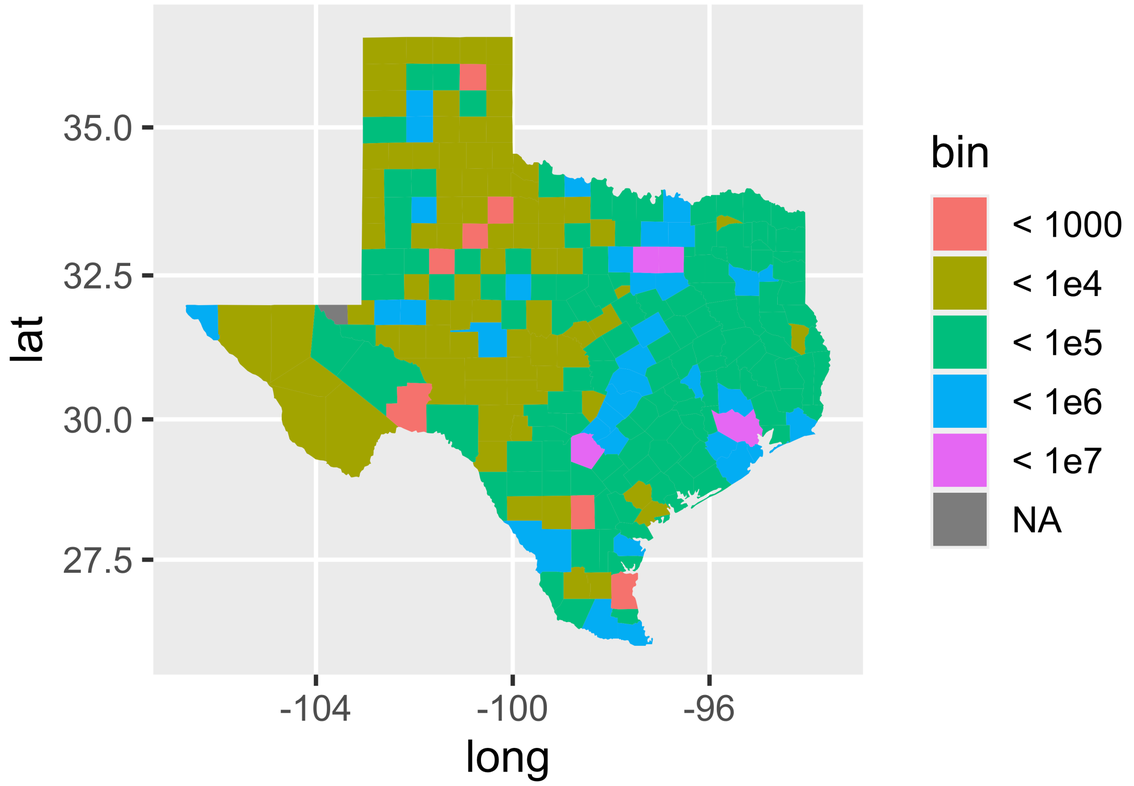
`tx`



`tx + coord_map()`




```
tx
```




```
tx + coord_map()
```

Mercador

Gráfico pizza

Um gráfico pizza é um gráfico de barras segmentado em coordenadas polares

Gráfico pizza

Um gráfico pizza é um gráfico de barras segmentado em coordenadas polares

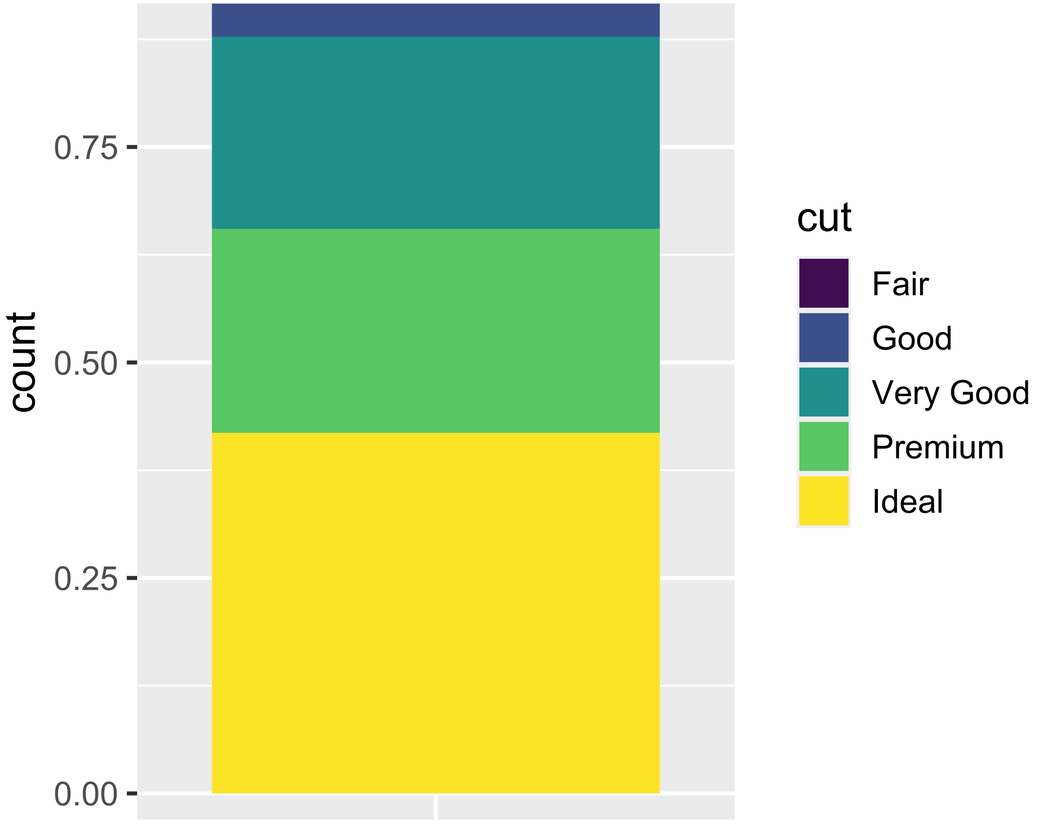
```
D_color <- filter(diamonds, color == "D")
```

Gráfico pizza

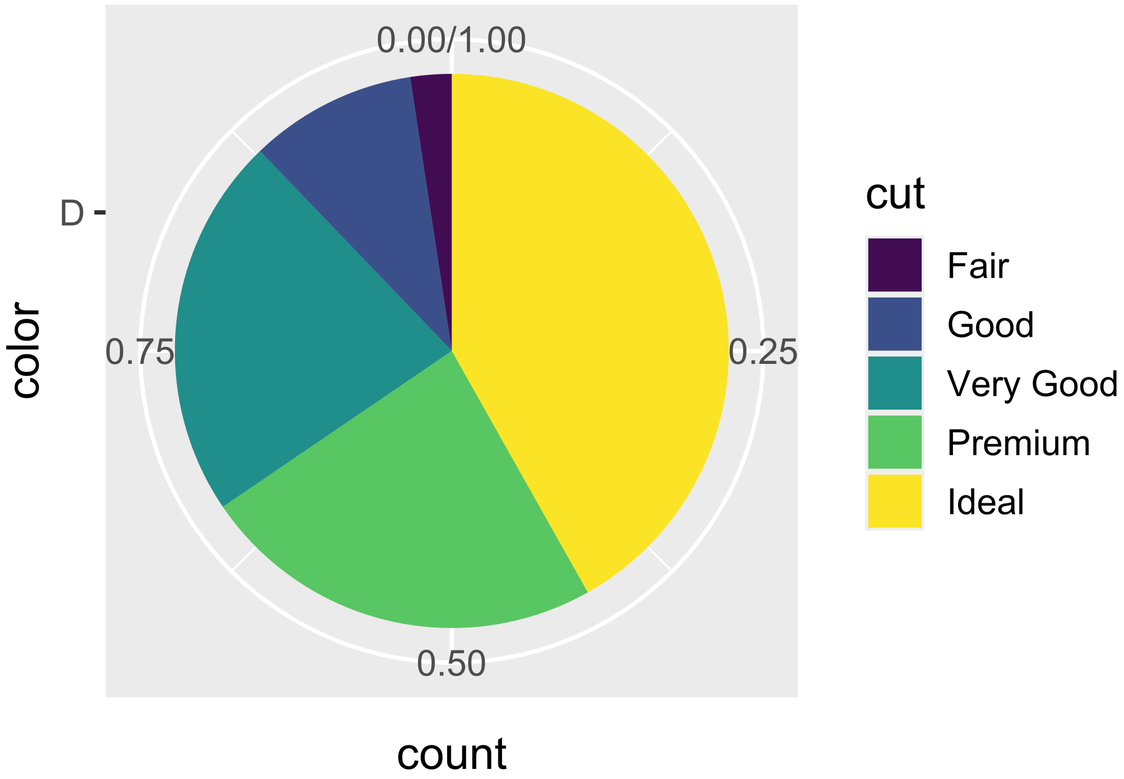
Um gráfico pizza é um gráfico de barras segmentado em coordenadas polares

```
D_color <- filter(diamonds, color == "D")
```

```
color_cut <- ggplot( D_color, aes(color) ) +
                geom_bar( aes(fill = cut), position = "fill" )
```




```
color_cut
```




```
color_cut +
  coord_polar( theta = "y" )
```

Escalas

Escalas

scale

- **Mapeamento estético**

  Qual variável mapear esteticamente

- **Escala**

  Como mapear a estética da variável

Escalas

scale

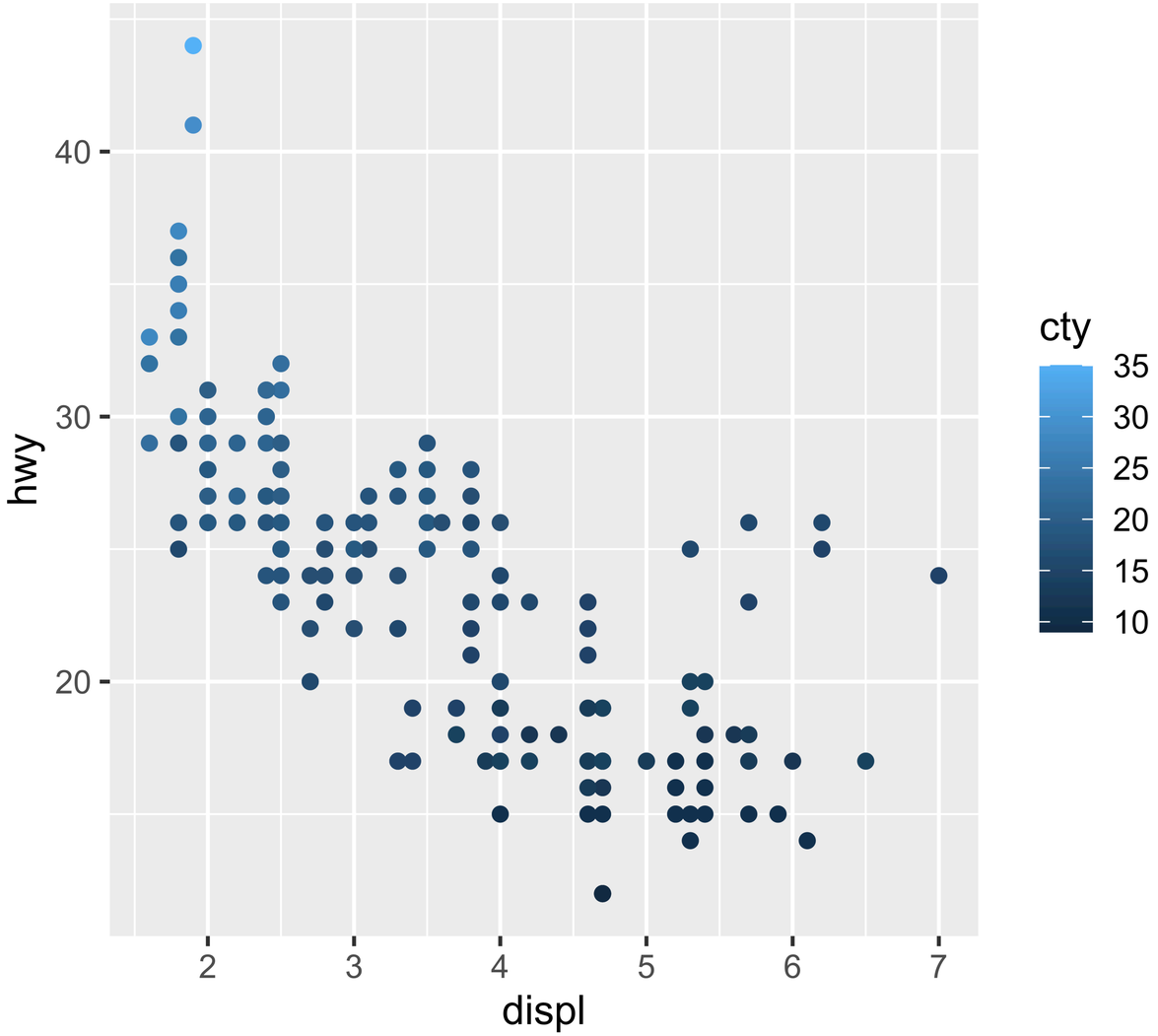
Sempre começa com scale

Estética a ser mapeada

Nome do objeto escala a ser ajustada

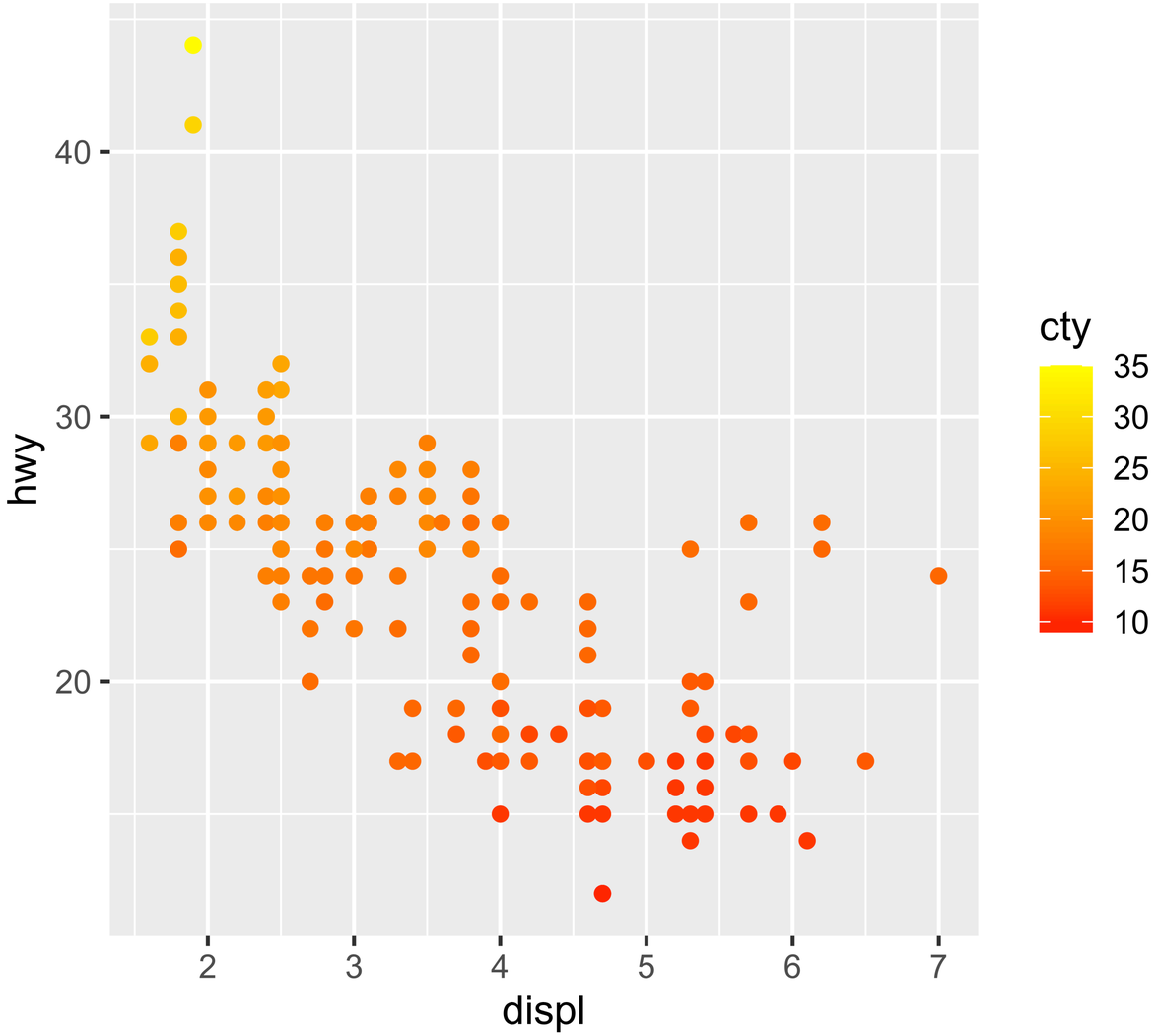
```
scale_aesthetic_name()
```

scale_color_gradient



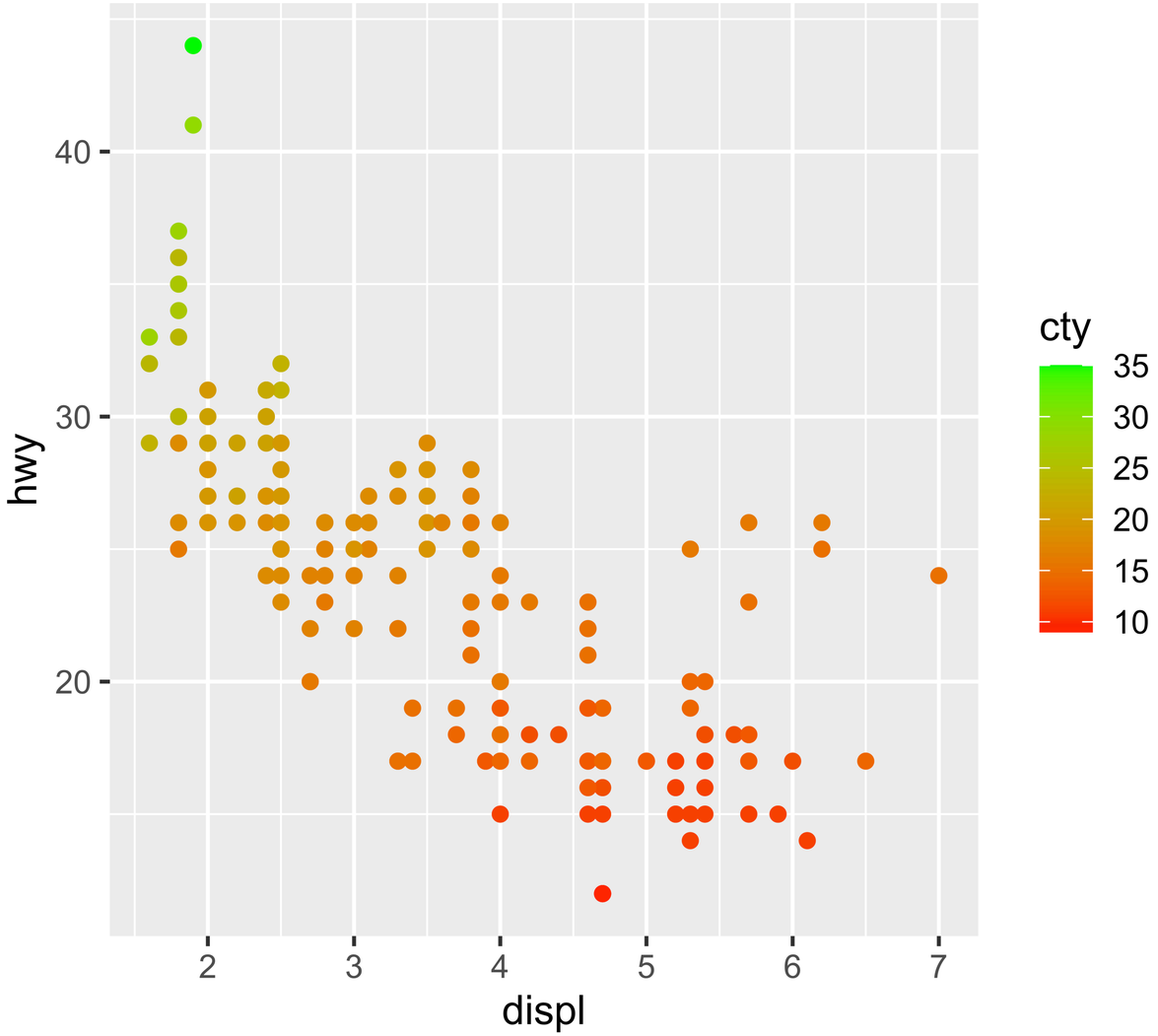

```
ggplot( mpg, aes(displ, hwy) ) +
  geom_point( aes(color = cty) )
```

scale_color_gradient



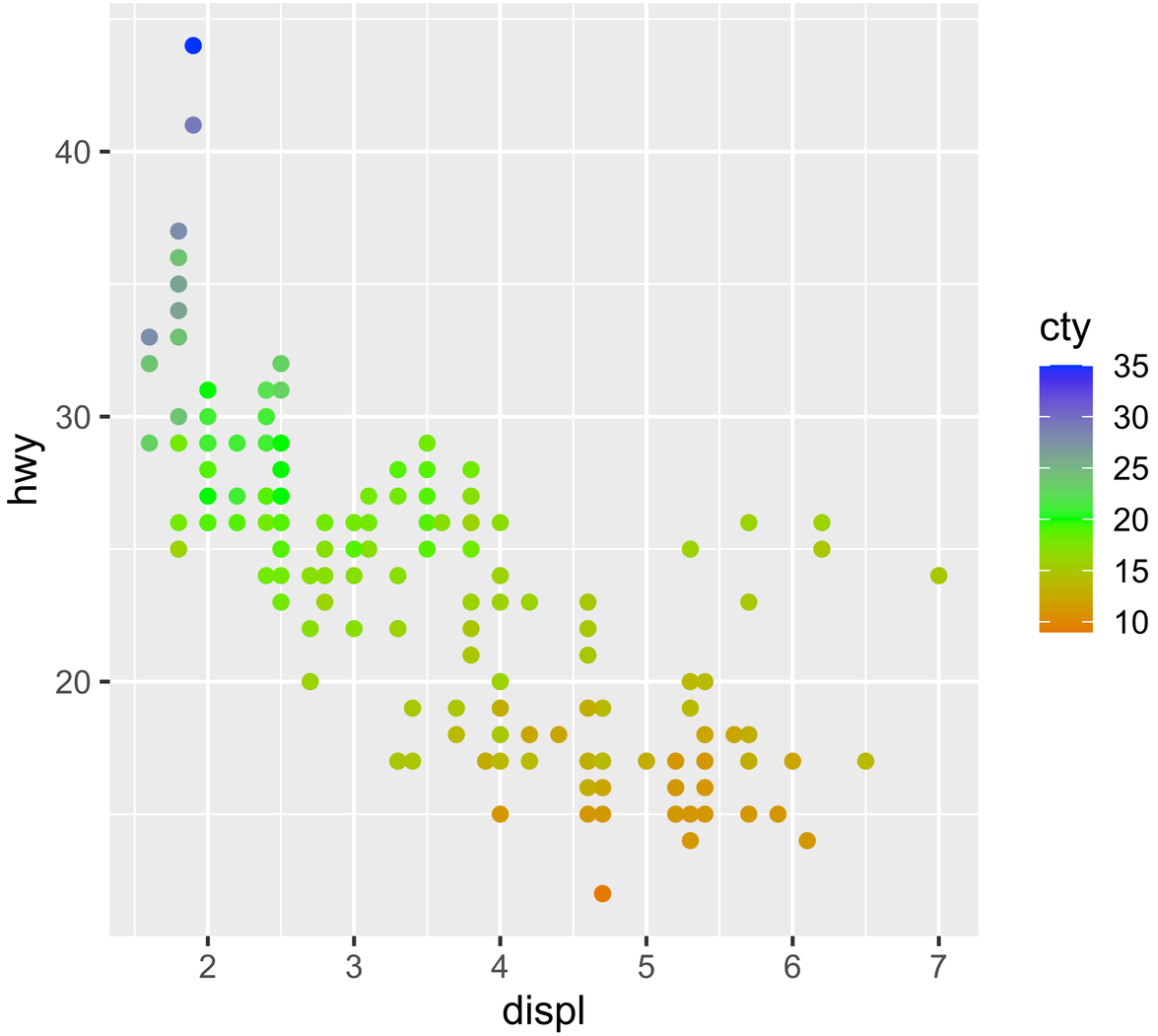

```
ggplot( mpg, aes(displ, hwy) ) +
  geom_point( aes(color = cty) ) +
   scale_color_gradient(
     low = "red",
     high = "yellow"
   )
```

scale_color_gradient



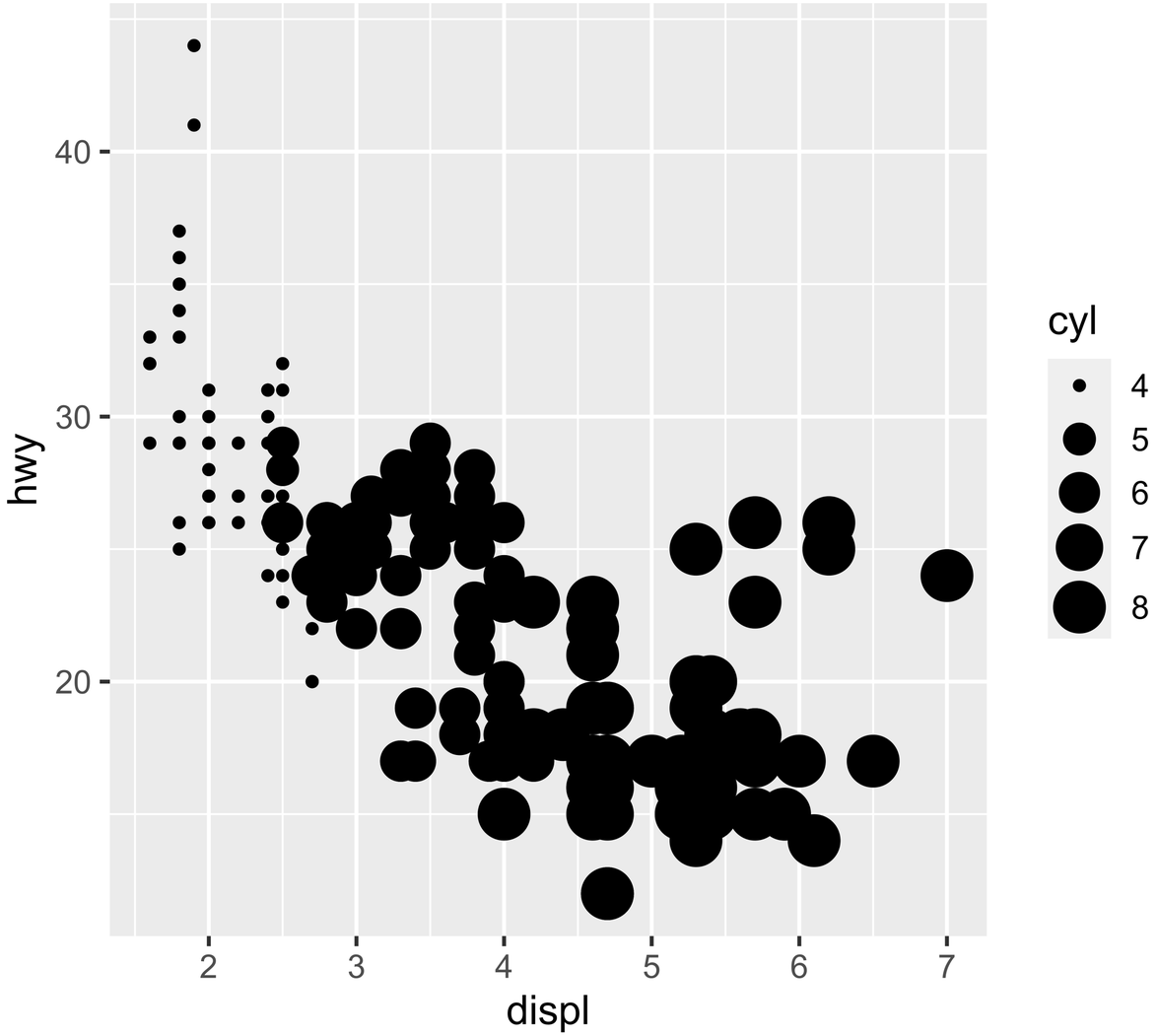

```
ggplot( mpg, aes(displ, hwy) ) +
  geom_point( aes(color = cty) ) +
   scale_color_gradient(
     low = "red",
     high = "green"
   )
```

scale_color_gradient2



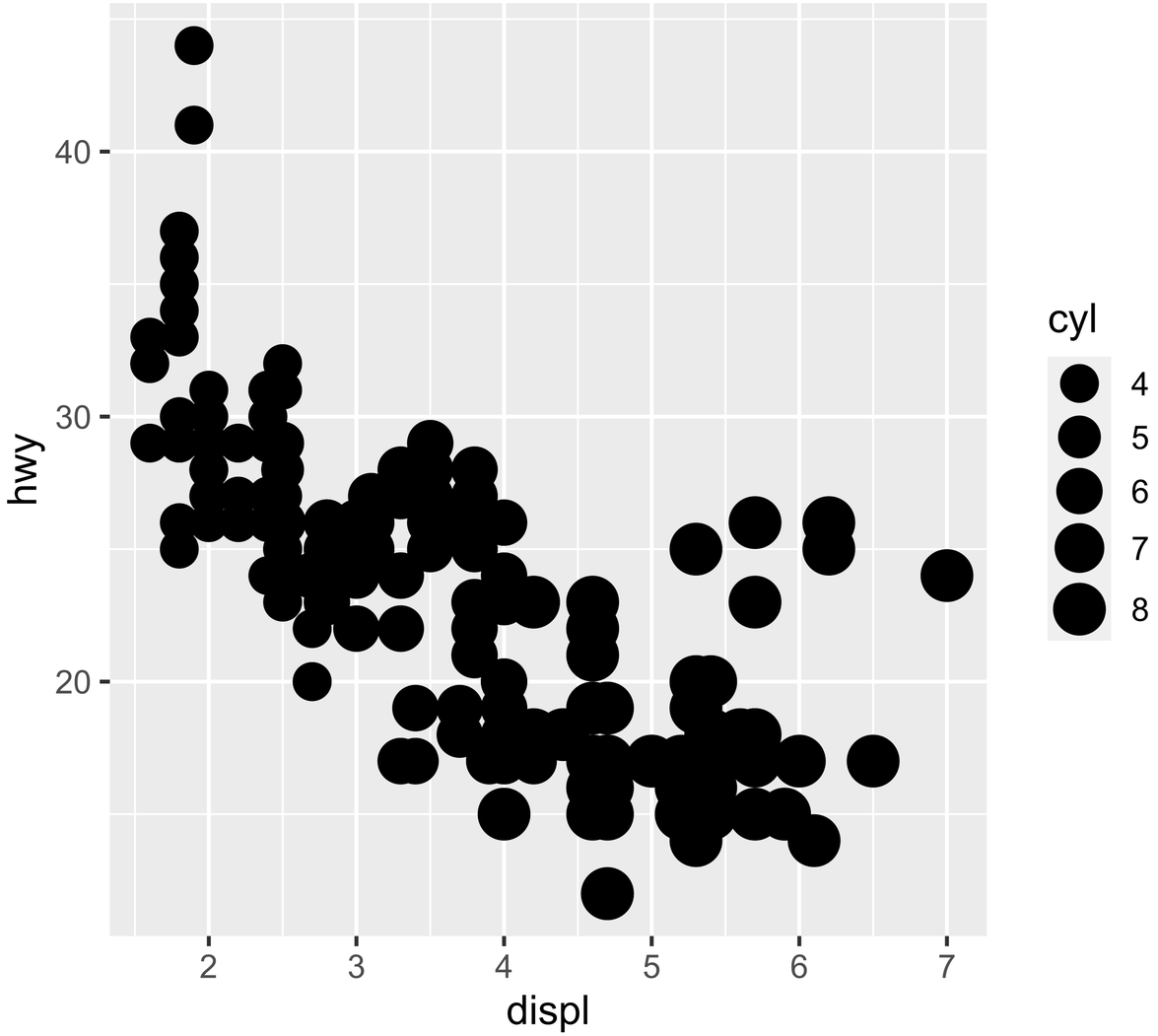

```
ggplot( mpg, aes(displ, hwy) ) +
  geom_point( aes(color = cty) ) +
   scale_color_gradient2(
     low = "red",
     mid = "green",
     high = "blue",
     midpoint = 20
   )
```

scale_size_area



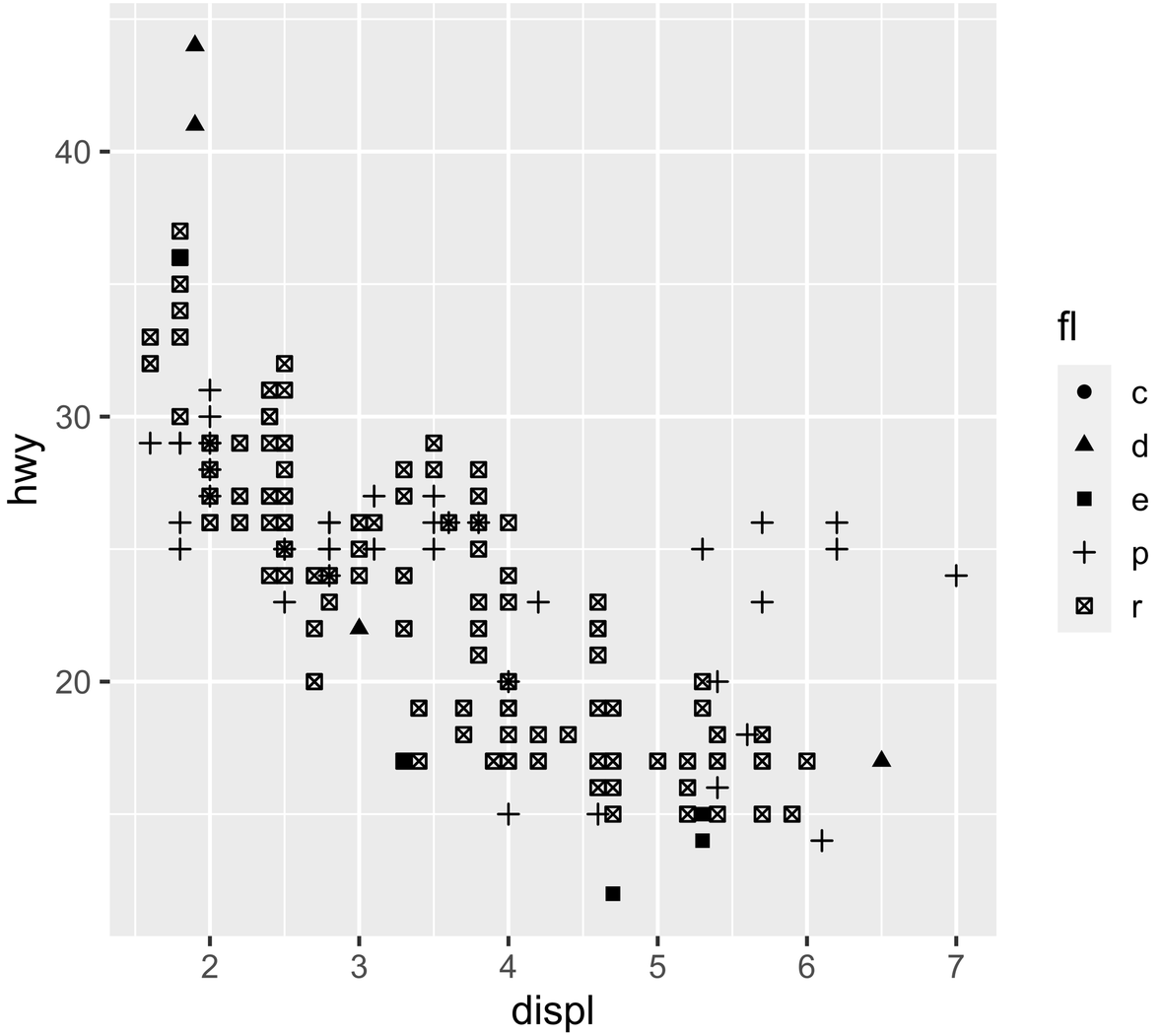

```
ggplot( mpg, aes(displ, hwy) ) +
  geom_point( aes(size = cyl) )
```

scale_size_area



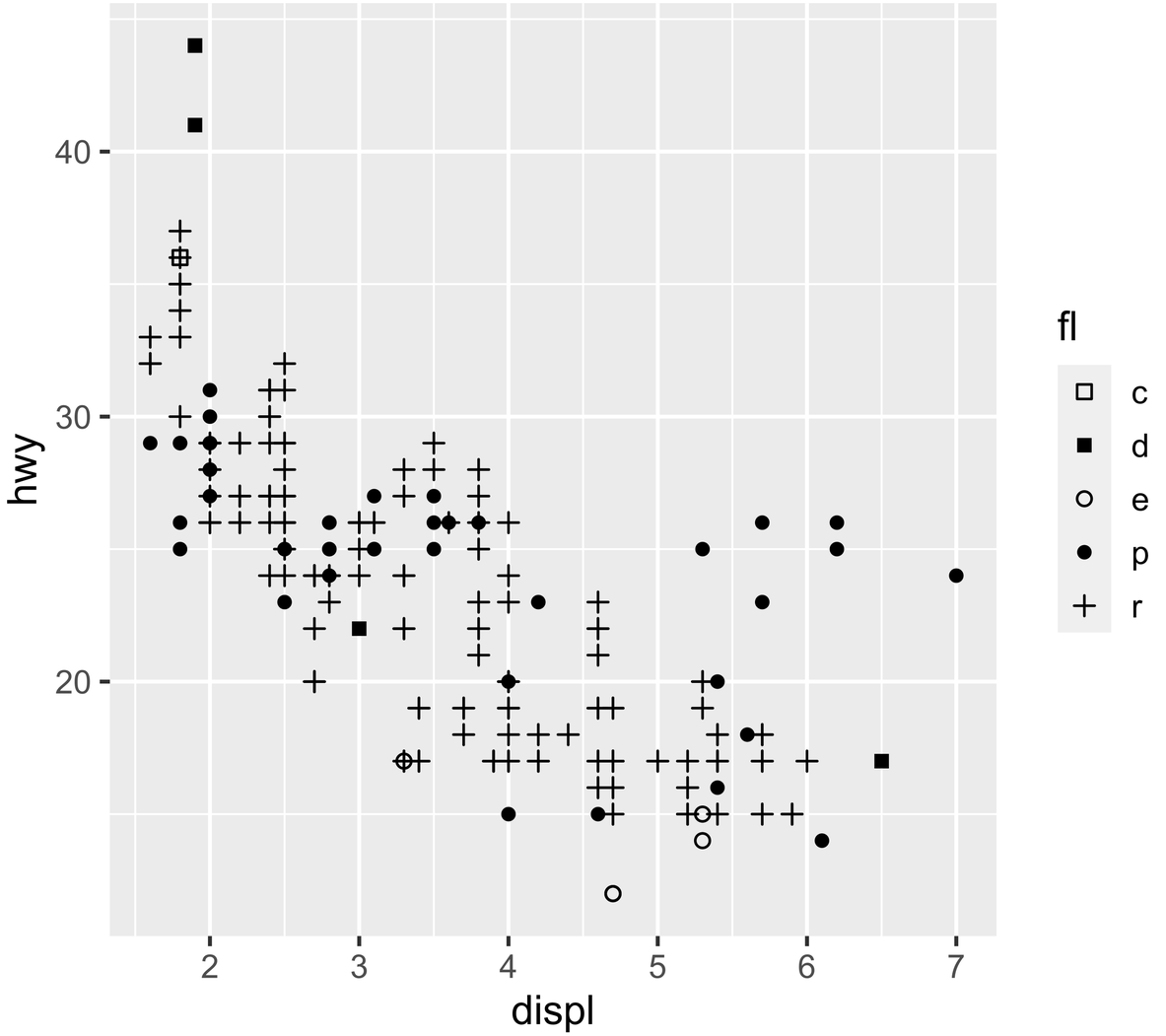

```
ggplot( mpg, aes(displ, hwy) ) +
  geom_point( aes(size = cyl) )
  scale_size_area()
```

scale_shape_manual



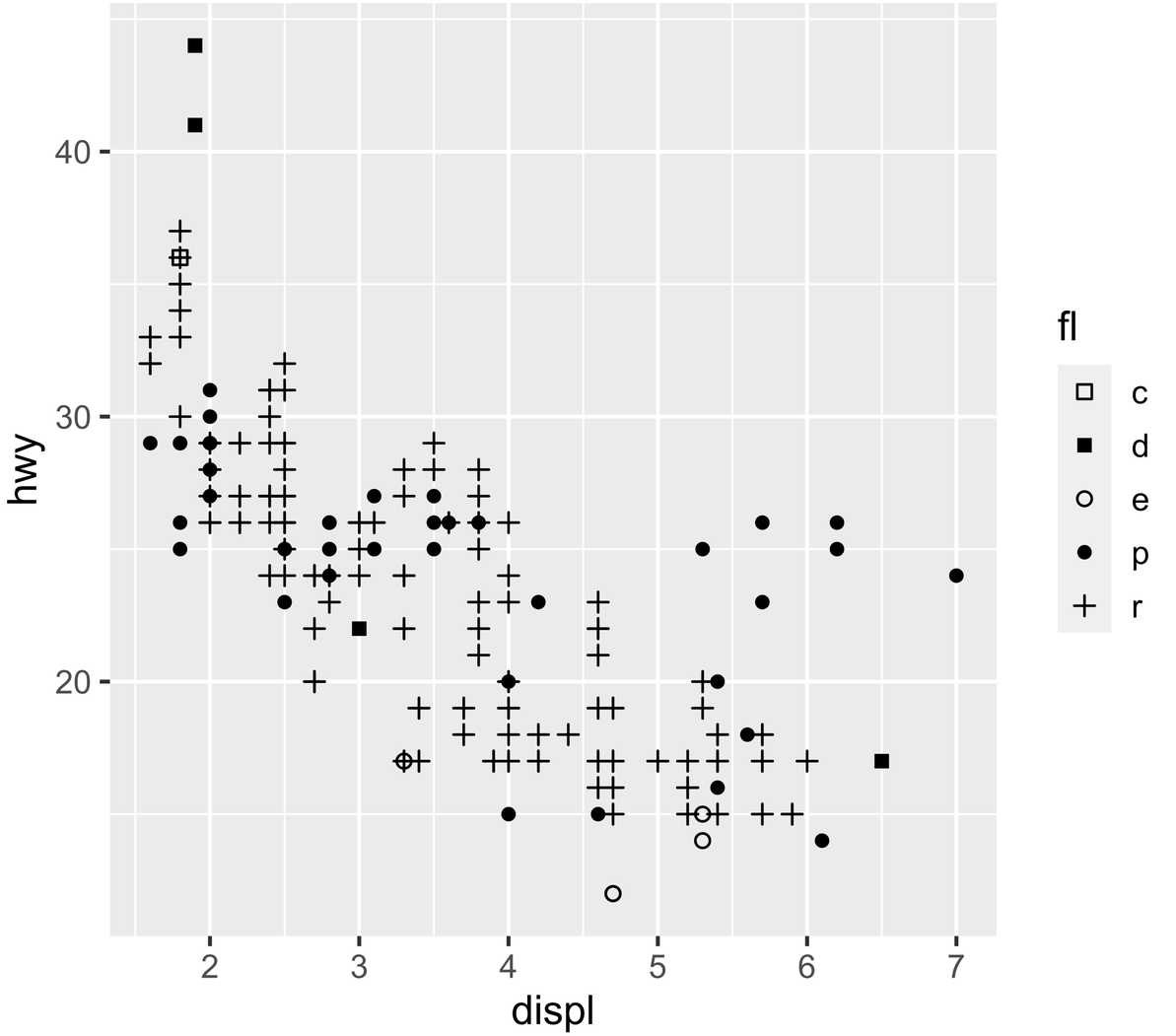

```
ggplot( mpg, aes(displ, hwy) ) +
  geom_point( aes(shape = fl) )
```

scale_shape_manual

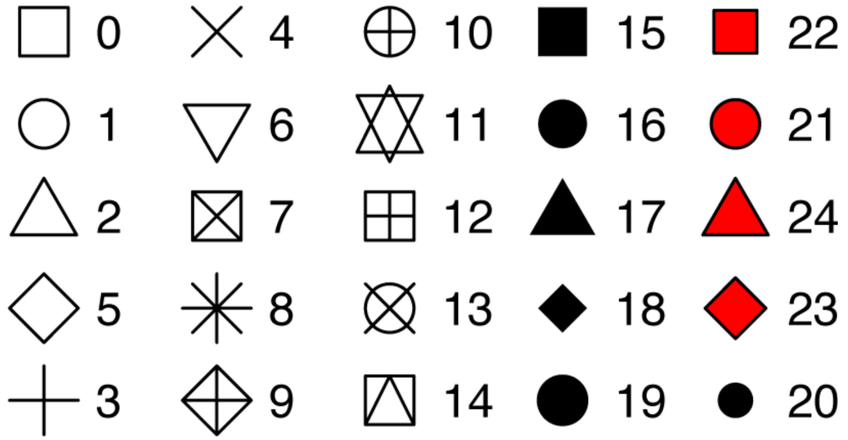



```
ggplot( mpg, aes(displ, hwy) ) +
  geom_point( aes(shape = fl) ) +
   scale_shape_manual(
     values = c(0, 15, 1, 16, 3)
   )
```

scale_shape_manual




```
ggplot( mpg, aes(displ, hwy) ) +
  geom_point( aes(shape = fl) ) +
   scale_shape_manual(
     values = c(0, 15, 1, 16, 3)
   )
```



Cores

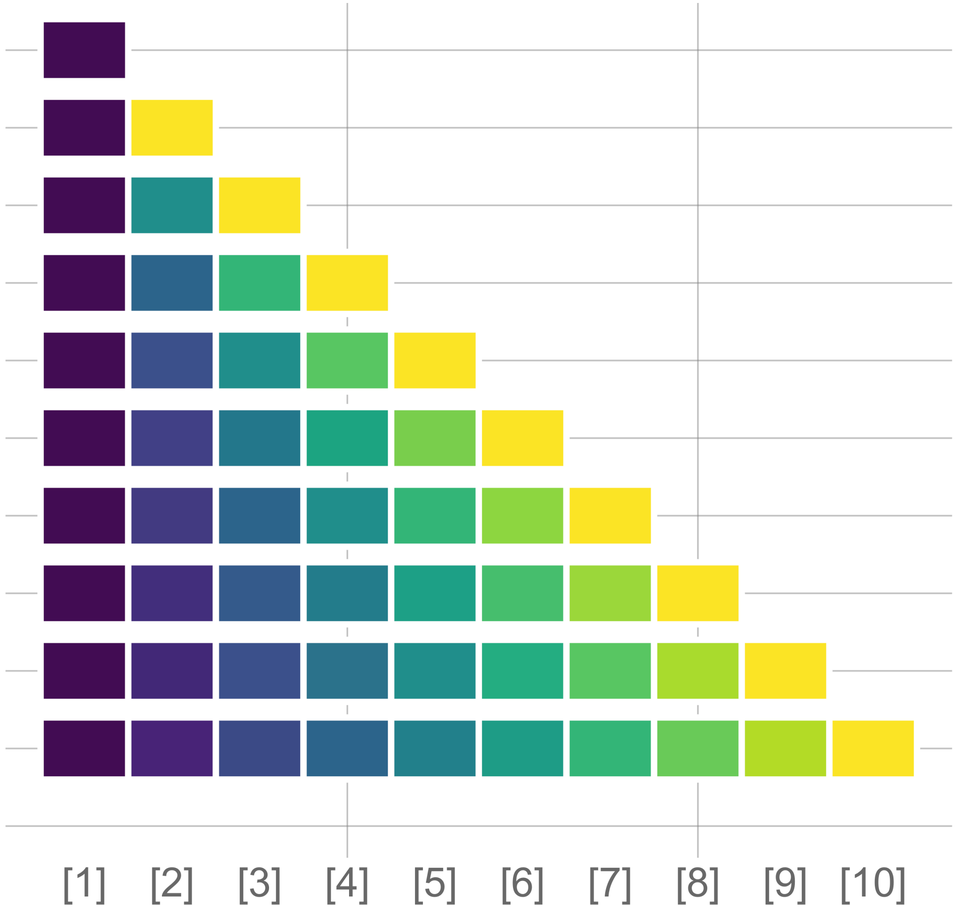
Cores

- **Cores** representa a estética mais utilizada no ggplot depois da estética posição.

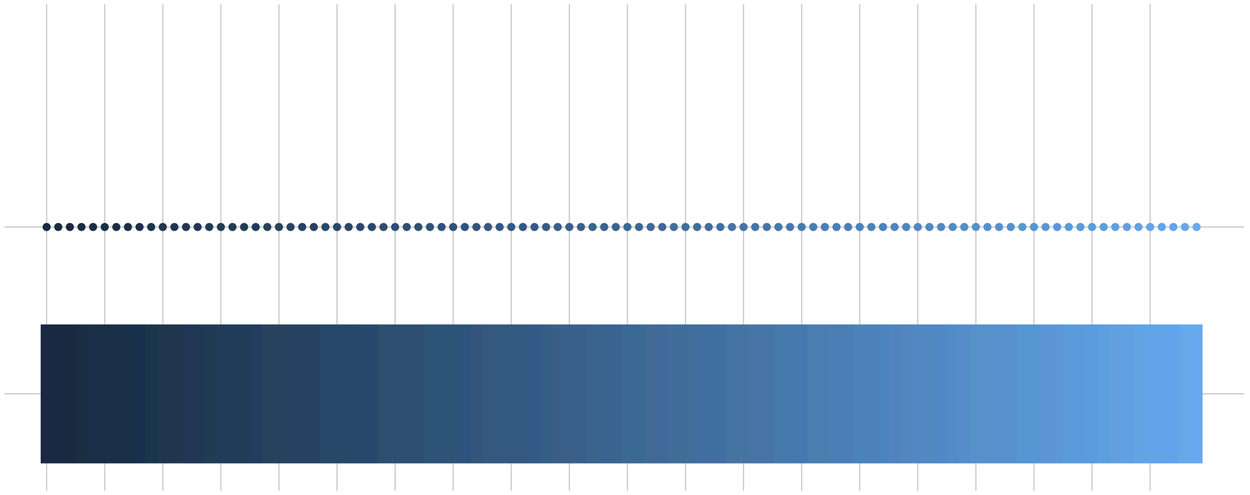
- É, também, a mais fácil de ser mal utilizada

Escala de cores padrão

Paleta de Cores Viridis (Discreta)



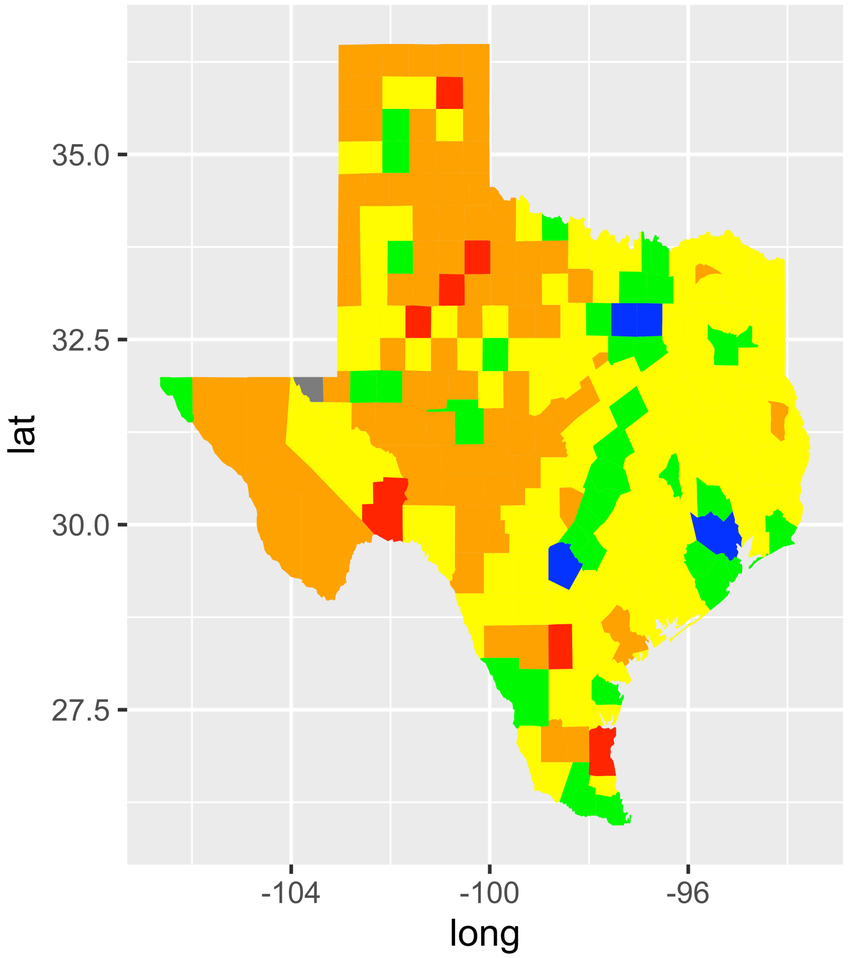
Paleta de Cores (Contínua)



Escalas de cores

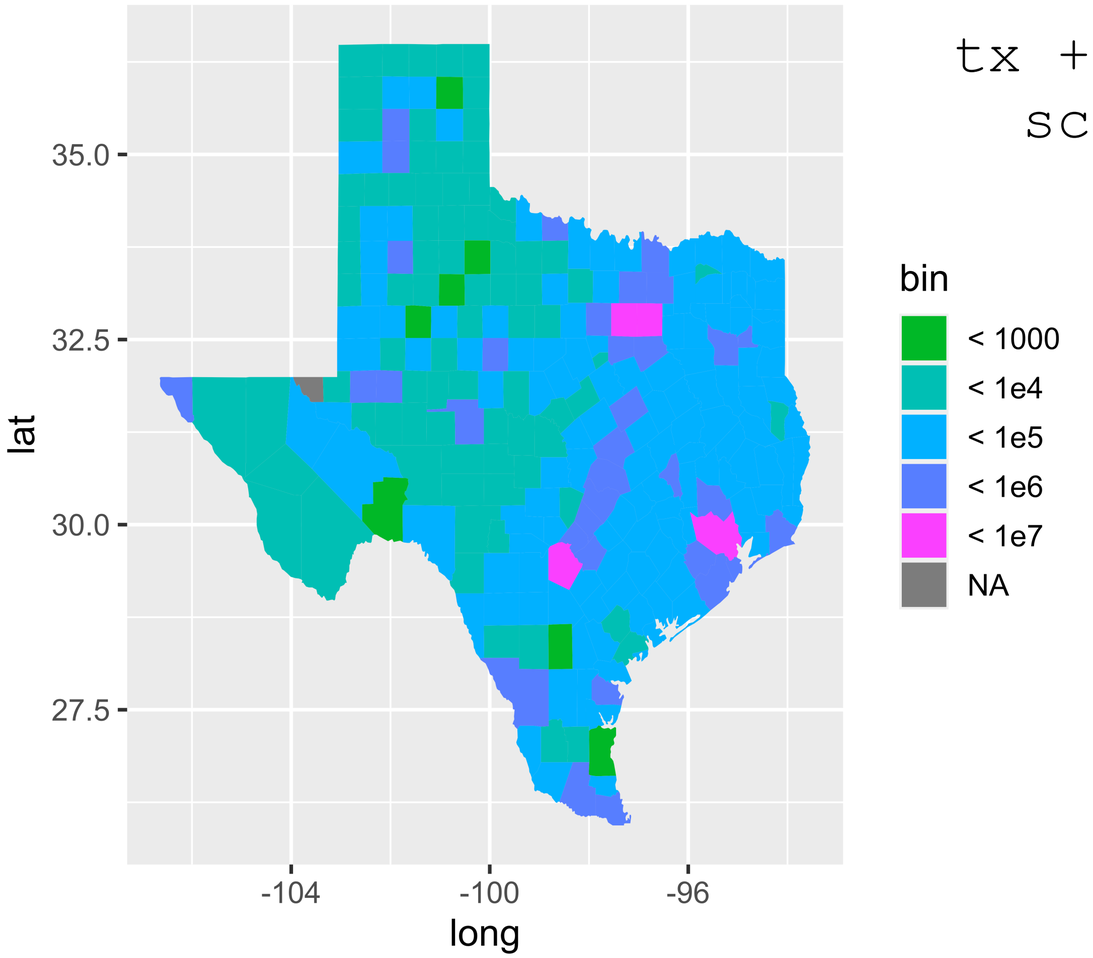
Escala a ser ajustada	Tipo de dados	O que cria?	Customização adicional
`scale_*_brewer`	Discreto	Belo esquema de cores	palette
`scale_*_manual`	Discreto	Exatamente, as cores que você listar	values
`scale_*_hue`	Discreto	Matizes uniformemente espaçados em todo o espaço de cores HCL	h, c, l
`scale_*_grey`	Discreto / Contínuo	Tons de cinza	
`scale_*_gradient`	Contínuo	Gradiente entre uma cor baixa e uma cor alta	low, high
`scale_*_gradient2`	Contínuo	Gradiente com médios neutros e extremos coloridos opostos	low, mid, high, midpoint
`scale_*_gradientn`	Contínuo	Gradiente que se move através de uma gama de cores	colours

personalizado != bonito




```
tx +
  scale_fill_manual(
    values =  c("red", "orange", "yellow",
                "green", "blue") )
```

personalizado != bonito



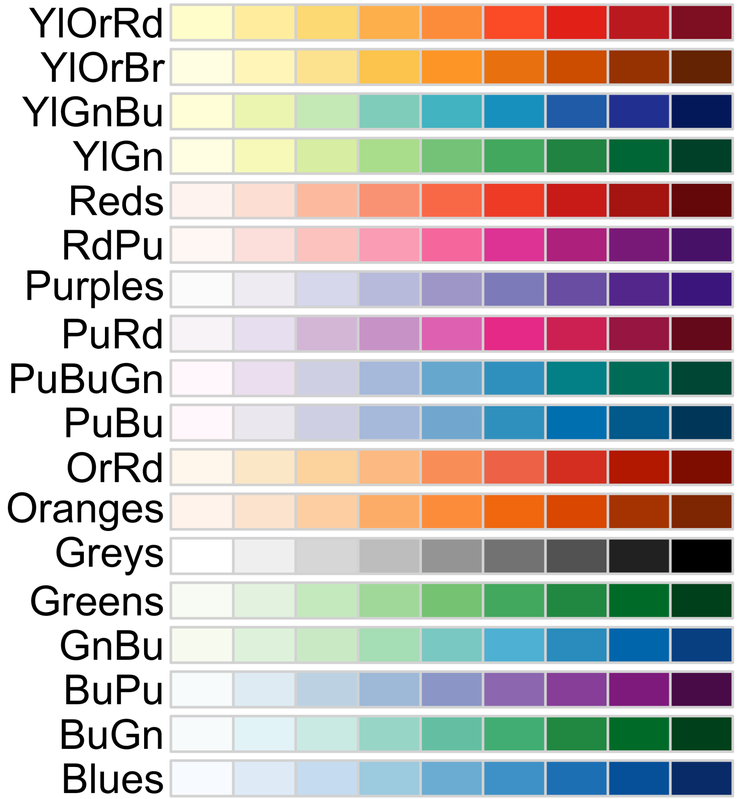

```
tx +
  scale_fill_hue( h = c(150, 300),
                  c = 150, l = 60 )
```

scale_*_brewer

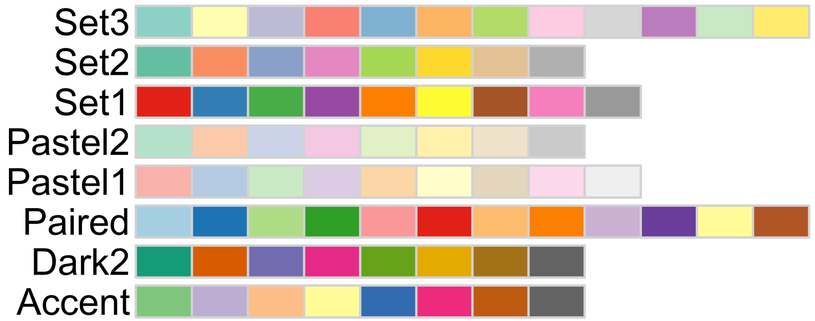
scale_*_brewer

```
library(RColorBrewer)
display.brewer.all()
```

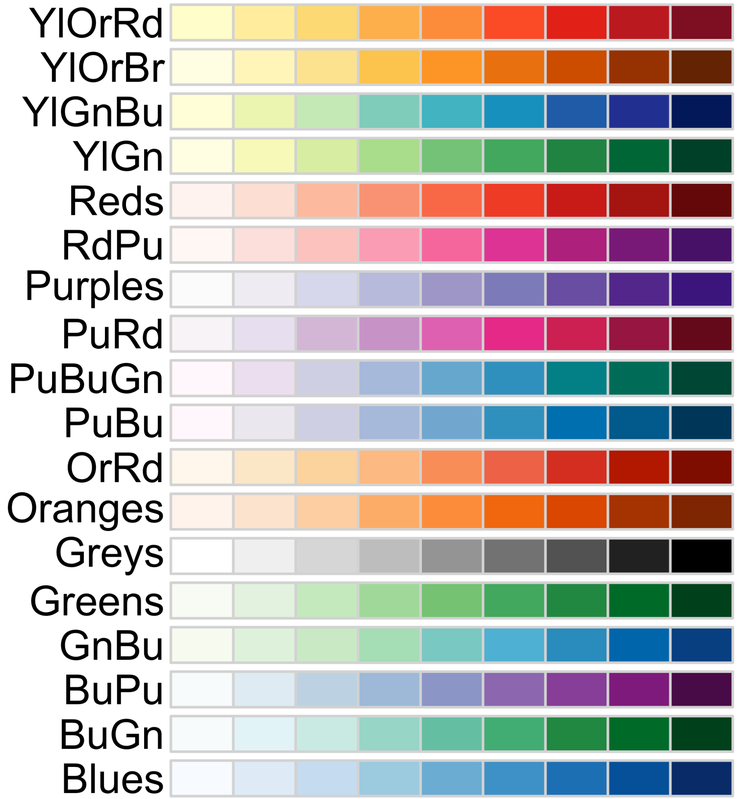
scale_*_brewer



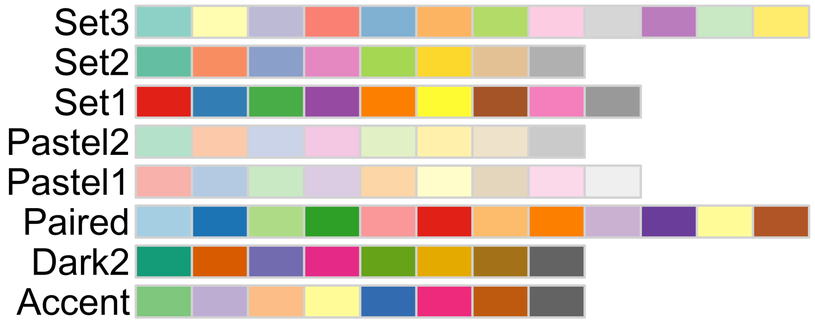

```
library(RColorBrewer)
display.brewer.all()
```



scale_*_brewer

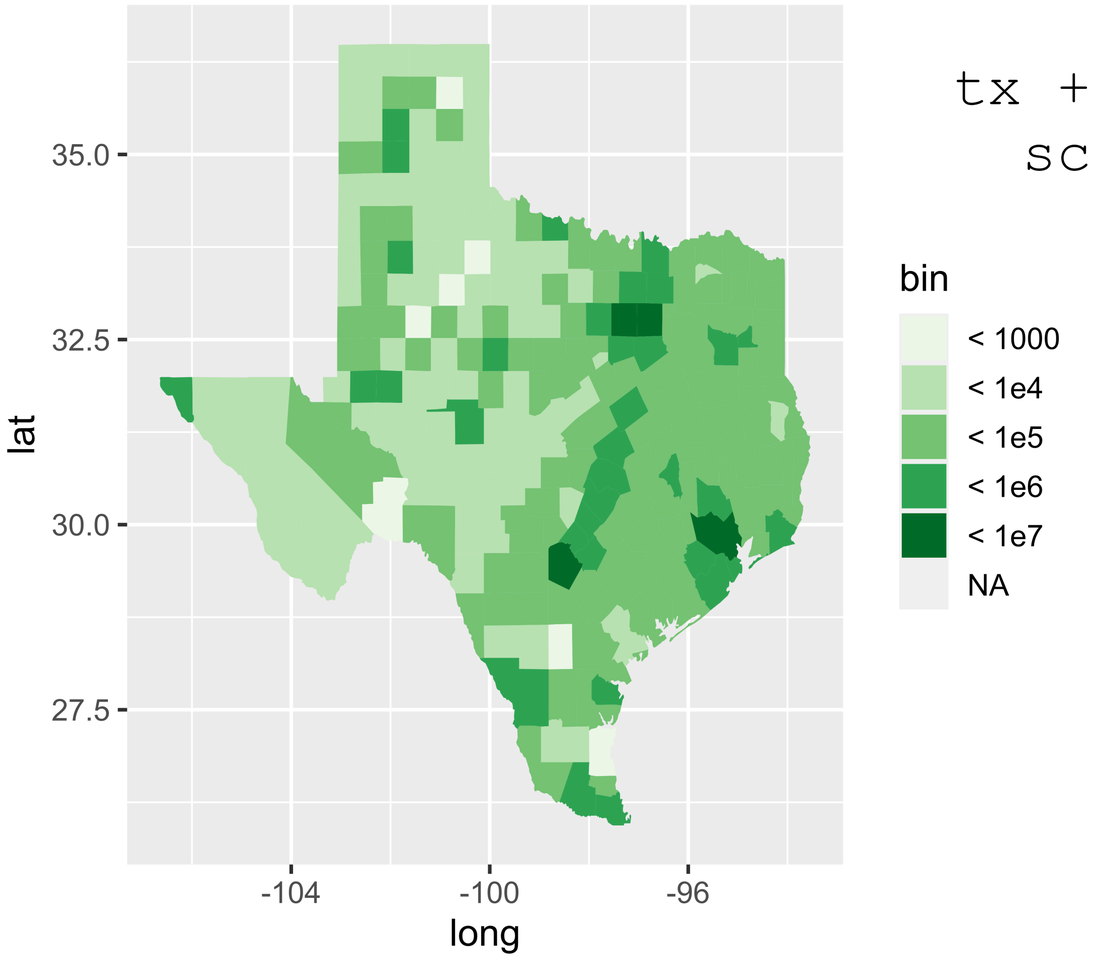



```
library(RColorBrewer)
display.brewer.all()
```



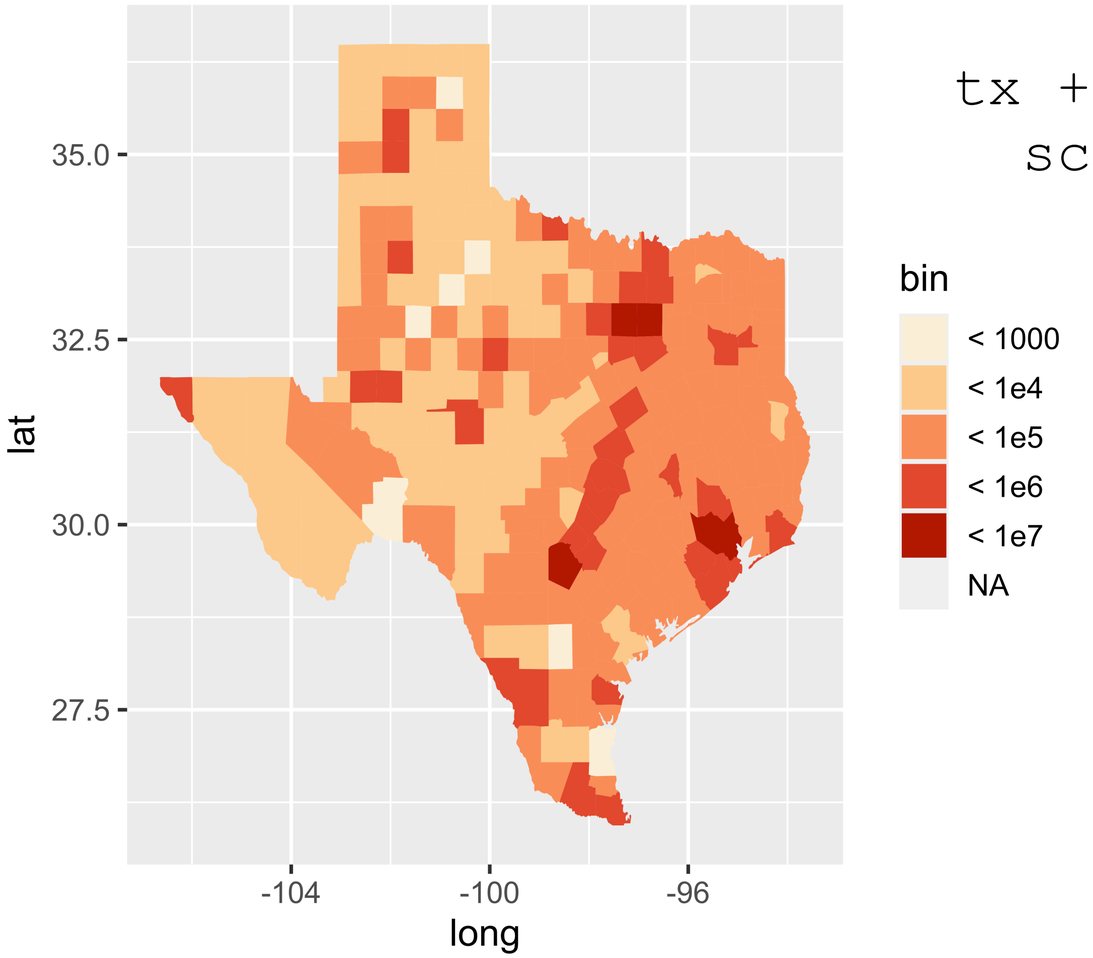

```
scale_color_brewer
scale_fill_brewer
```

scale_fill_brewer



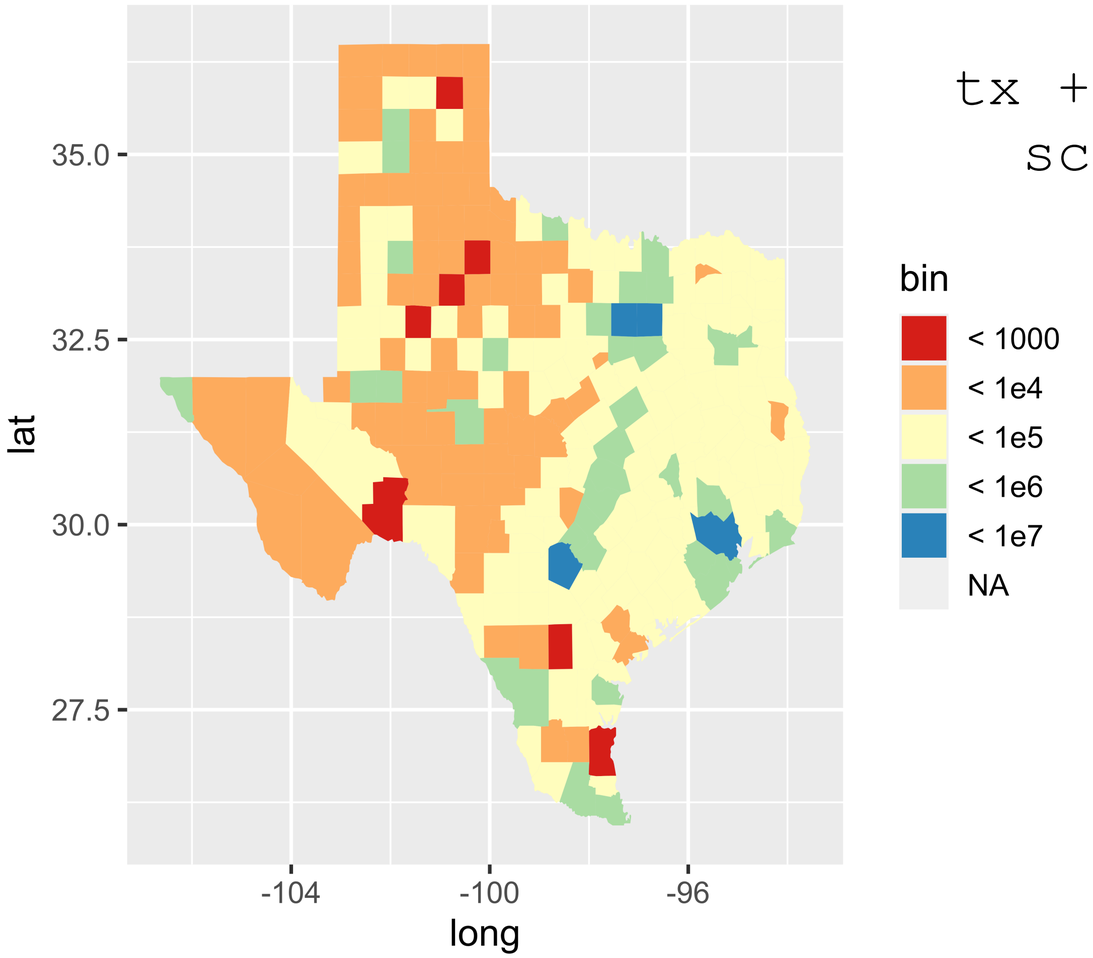

```
tx +
  scale_fill_brewer( palette = "Greens")
```

scale_fill_brewer




```
tx +
  scale_fill_brewer( palette = "OrRd")
```

scale_fill_brewer




```
tx +
  scale_fill_brewer( palette = "Spectral")
```

Existem escalas para todos as estéticas do ggplot2, não apenas para cores.
Para ver a lista completa com exemplos, visite

https://ggplot2.tidyverse.org/reference/index.html

Scales

Scales control the details of how data values are translated to visual properties. Override the default scales to tweak details like the axis labels or legend keys, or to use a completely different translation from data to aesthetic. `labs()` and `lims()` are convenient helpers for the most common adjustments to the labels and limits.

`labs()` `xlab()` `ylab()` `ggtitle()`
Modify axis, legend, and plot labels

`lims()` `xlim()` `ylim()`
Set scale limits

`expand_limits()`
Expand the plot limits, using data

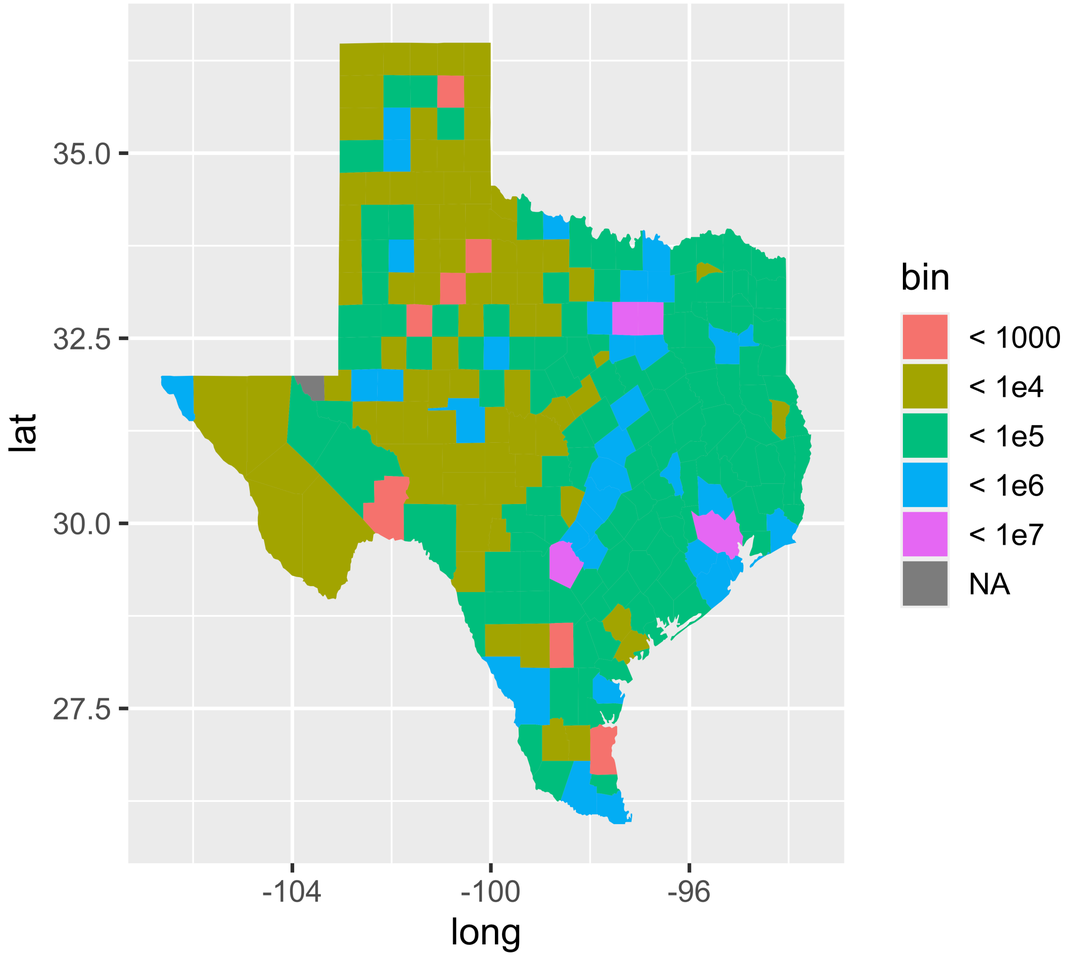
`expansion()` `expand_scale()`
Generate expansion vector for scales

Temas

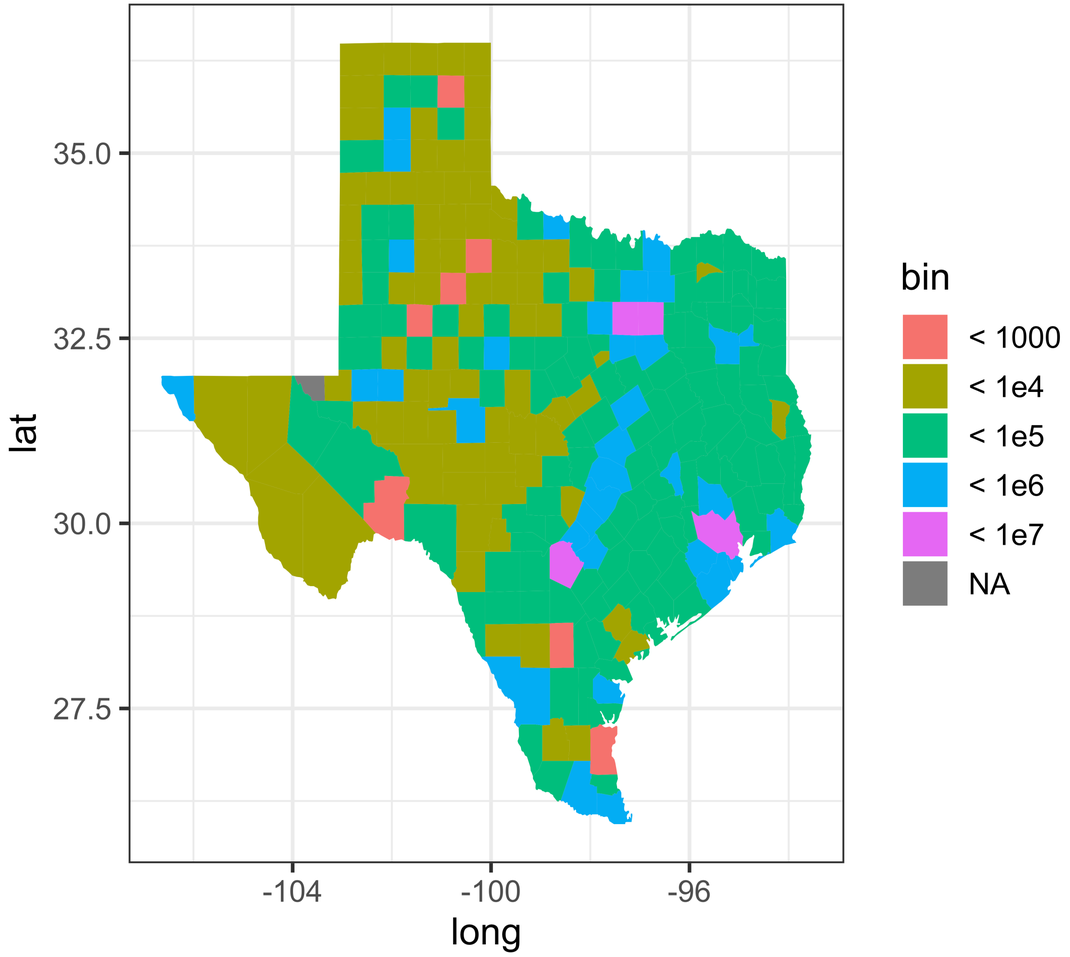
Temas

- Os temas controlam a aparência das partes não associadas às geometrias *geoms*

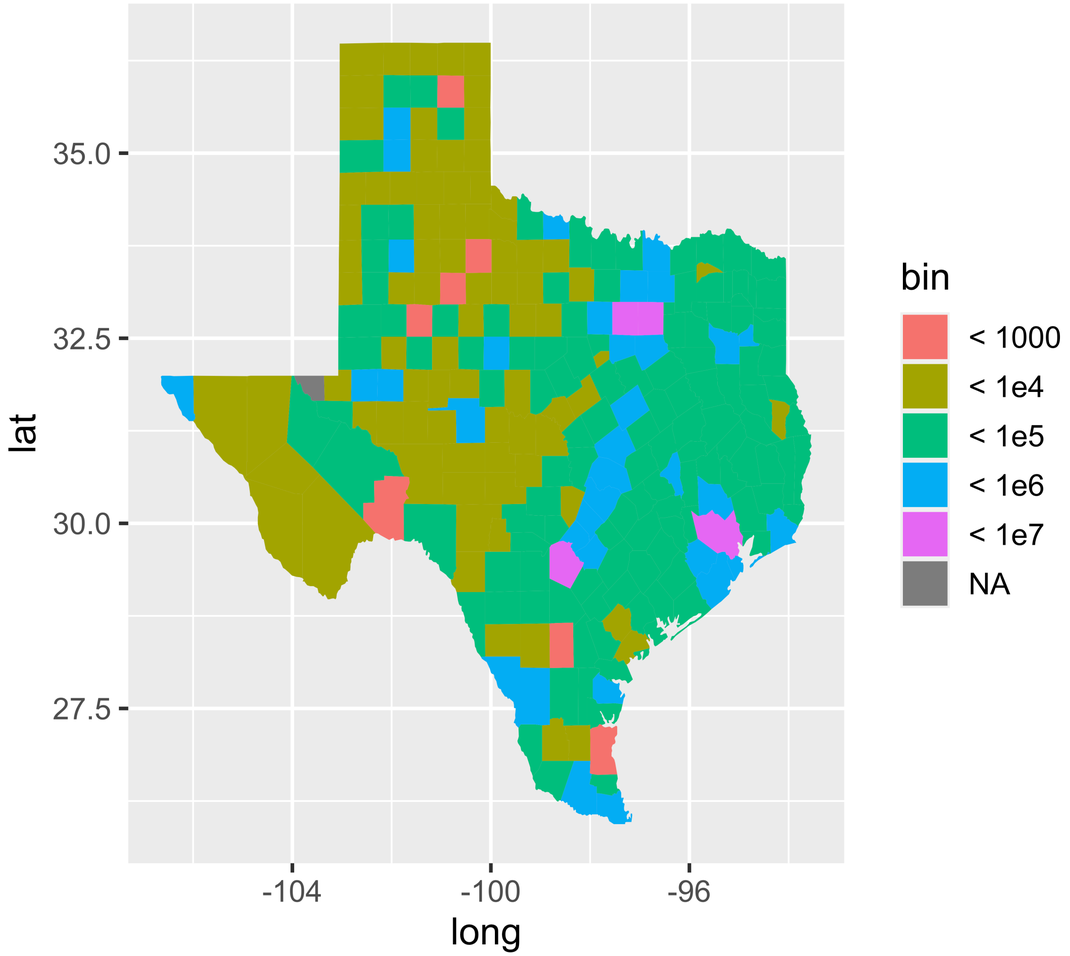
- Em outras palavras, controla a aparência visual do gráfico. Aquelas que não envolvem mapeamento dos dados.




```
tx + theme_grey()
```

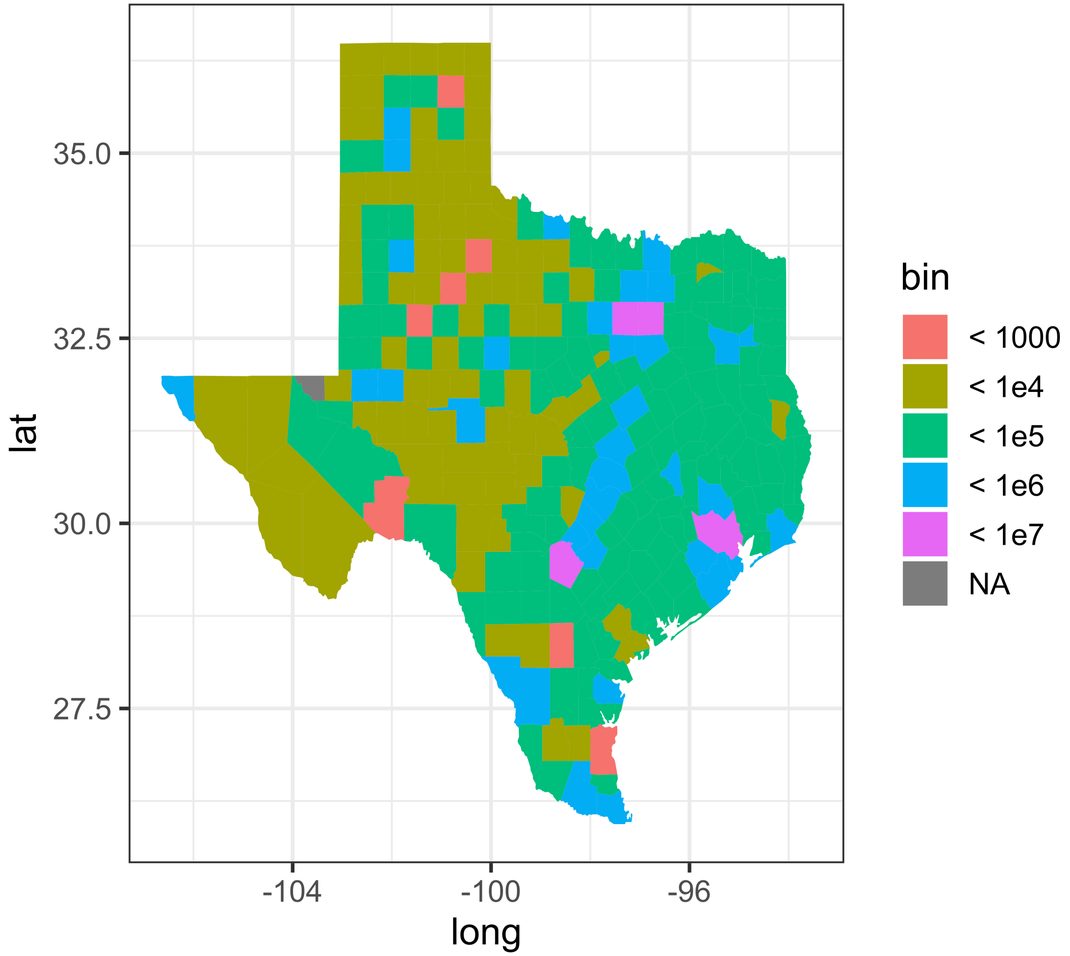



```
tx + theme_bw()
```

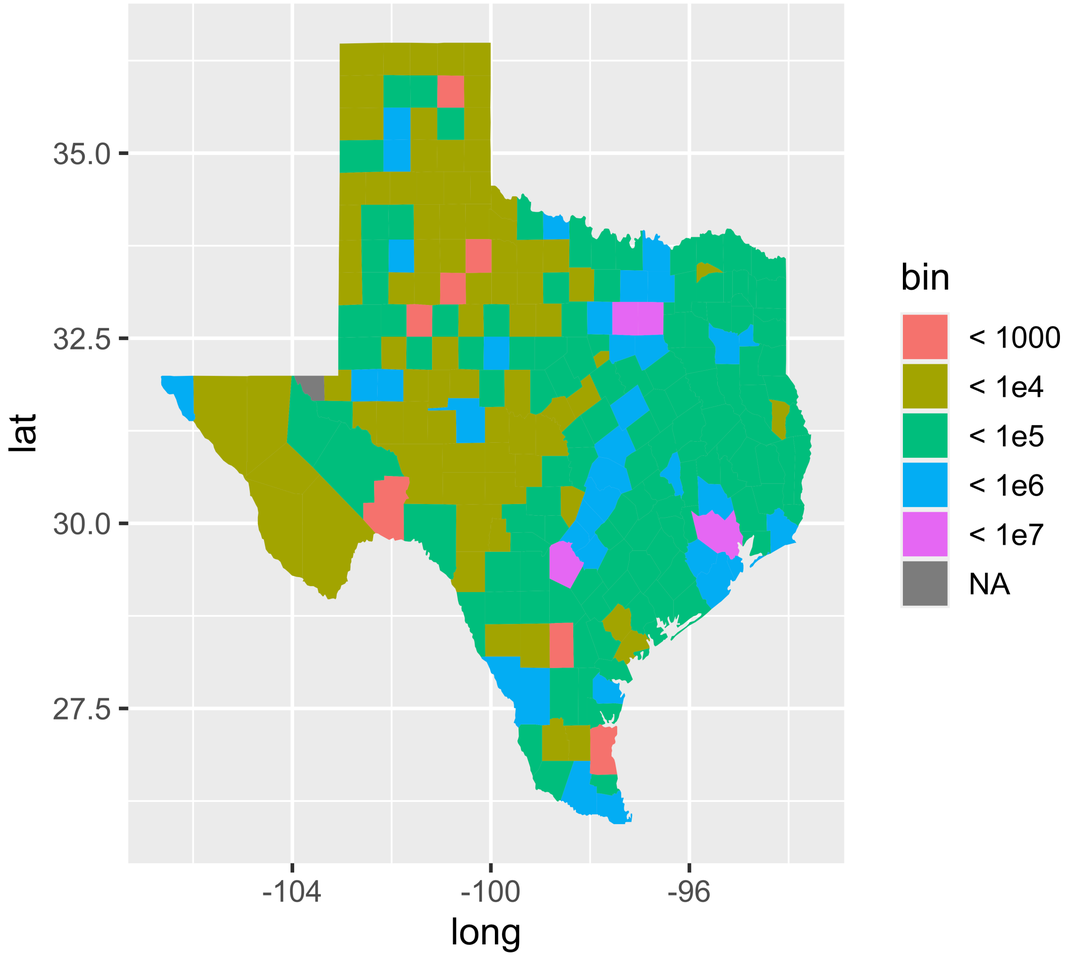



```
tx + theme_grey()
```

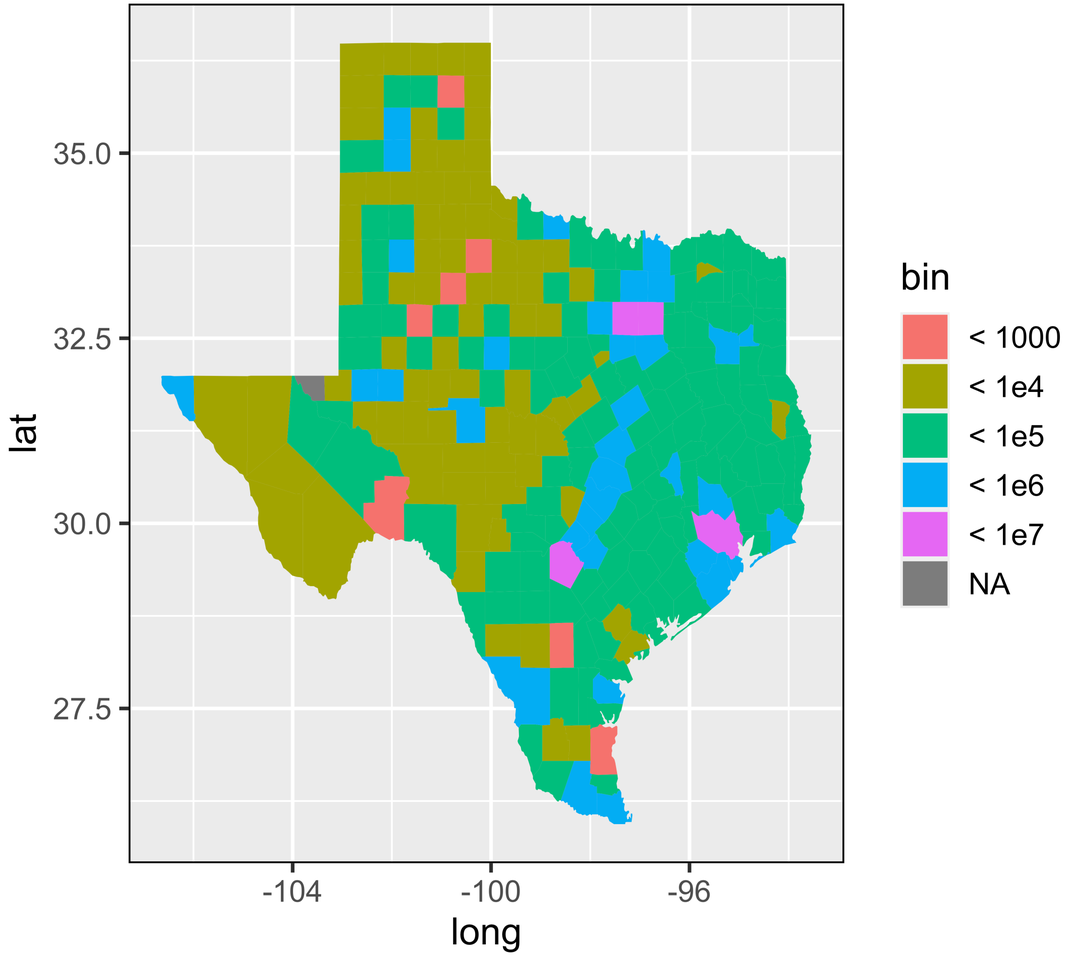
Padrão




```
tx + theme_bw()
```




```
tx + theme_grey()
```




```
tx + theme( panel.border =
              element_rect(
                colour = "black",
                fill = NA) )
```

Elementos

Axis: axis.line, axis.text.x, axis.text.y, axis.ticks, axis.title.x, axis.title.y

Legend: legend.background, legend.key, legend.text, legend.title

Panel: panel.background, panel.border, panel.grid.major, panel.grid.minor

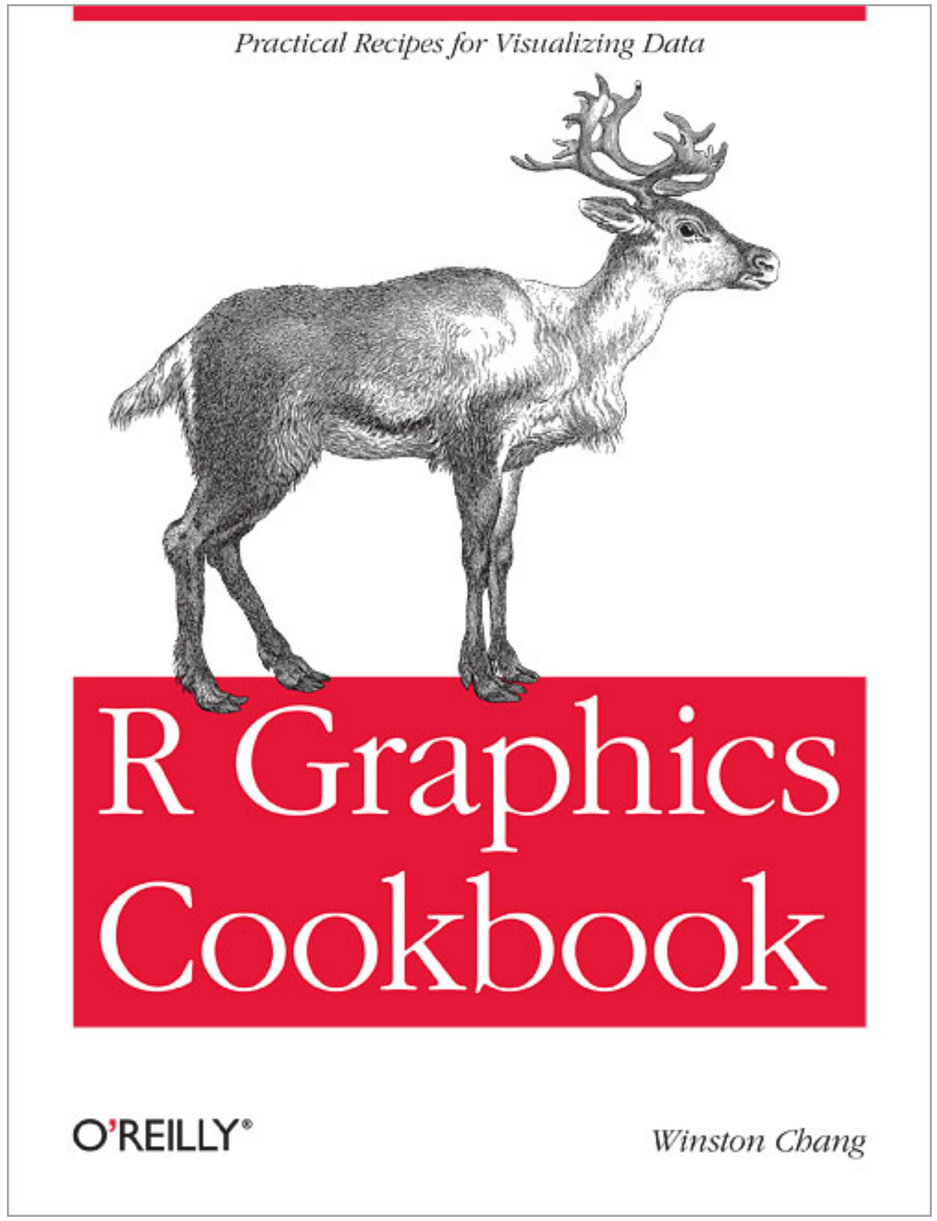
Strip: strip.background, strip.text.x, strip.text.y

?theme

Temas

Para aprender como alterar elementos individuais de um tema, recomendo o
R Graphics Cookbook
de *Winston Chang*

https://r-graphics.org/



ggthemes

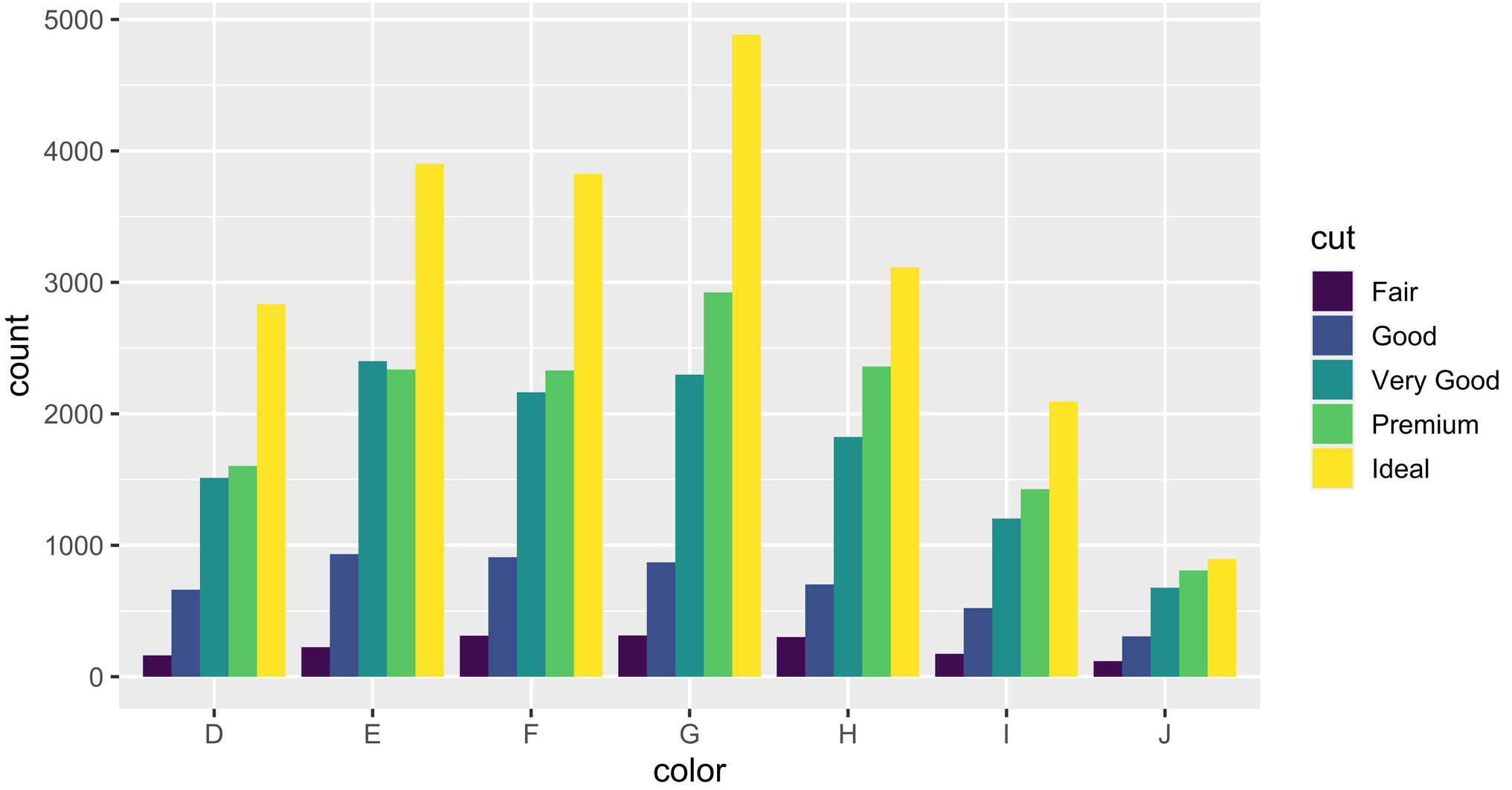
- Você pode utilizar o **ggthemes** se quiser implementar temas pré-pronto. Muitos temas famosos estão disponíveis.

```
install.packages("ggthemes")
library(ggthemes)
```

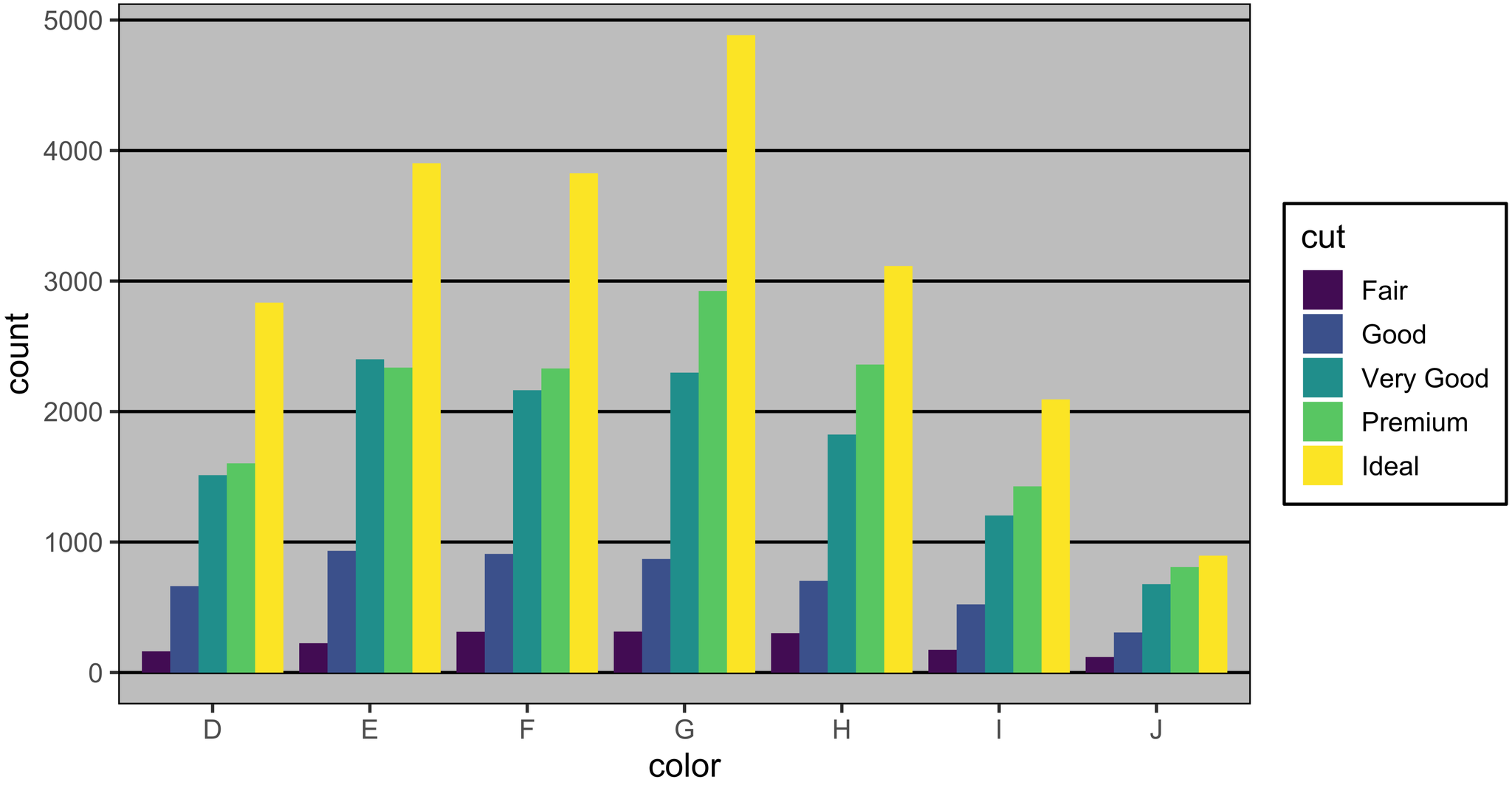
ggthemes

- Você pode utilizar o **ggthemes** se quiser implementar temas pré-pronto. Muitos temas famosos estão disponíveis.
- Para facilitar a visualização dos códigos, vamos definir um gráfico base abaixo

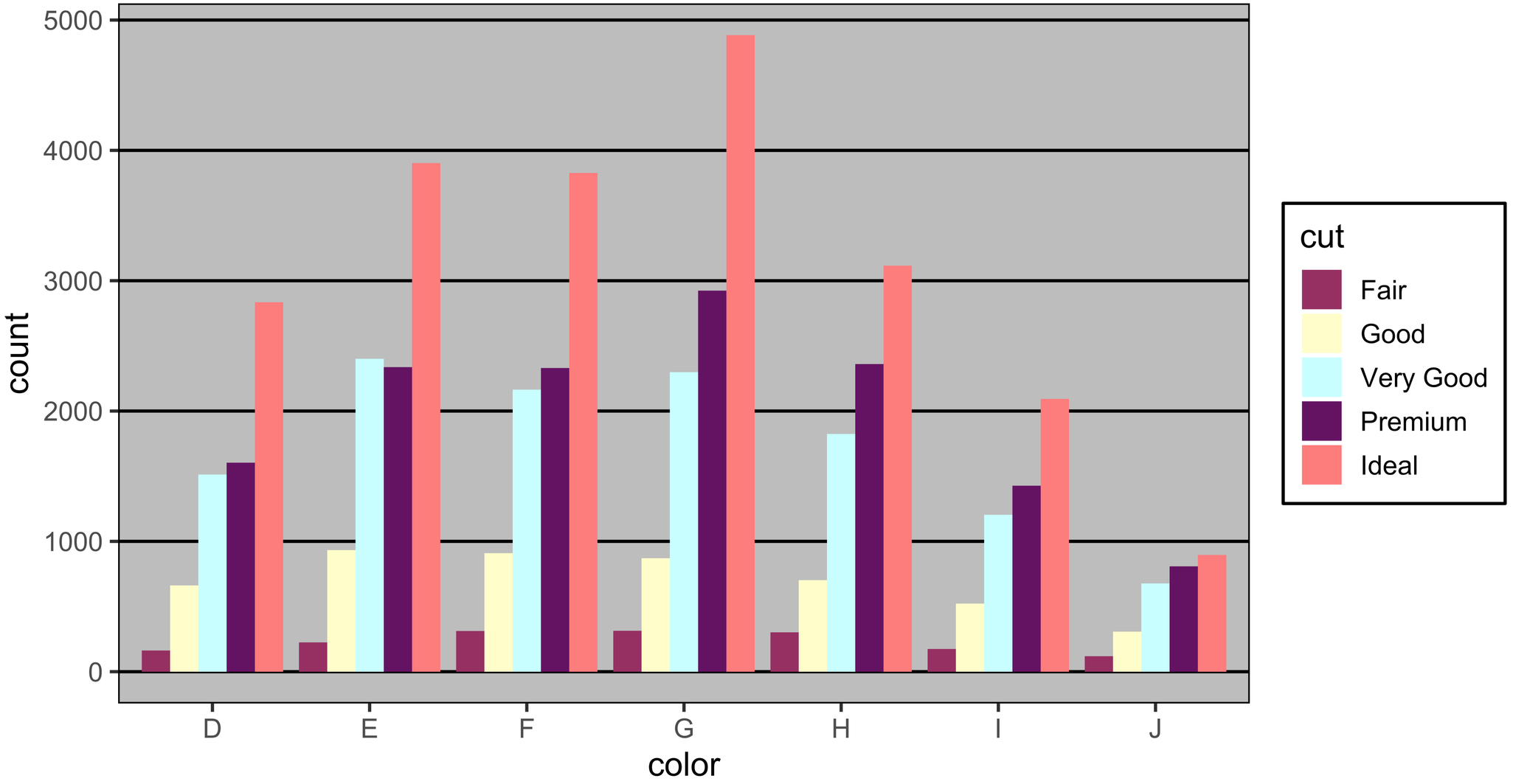
```
p <- ggplot( diamonds, aes(color) ) +
       geom_bar( aes(fill = cut), position = "dodge" )
```


5000
4000
3000
2000
1000
0
count
D
E
F
G
H
I
J
color
cut
Fair
Good
Very Good
Premium
Ideal

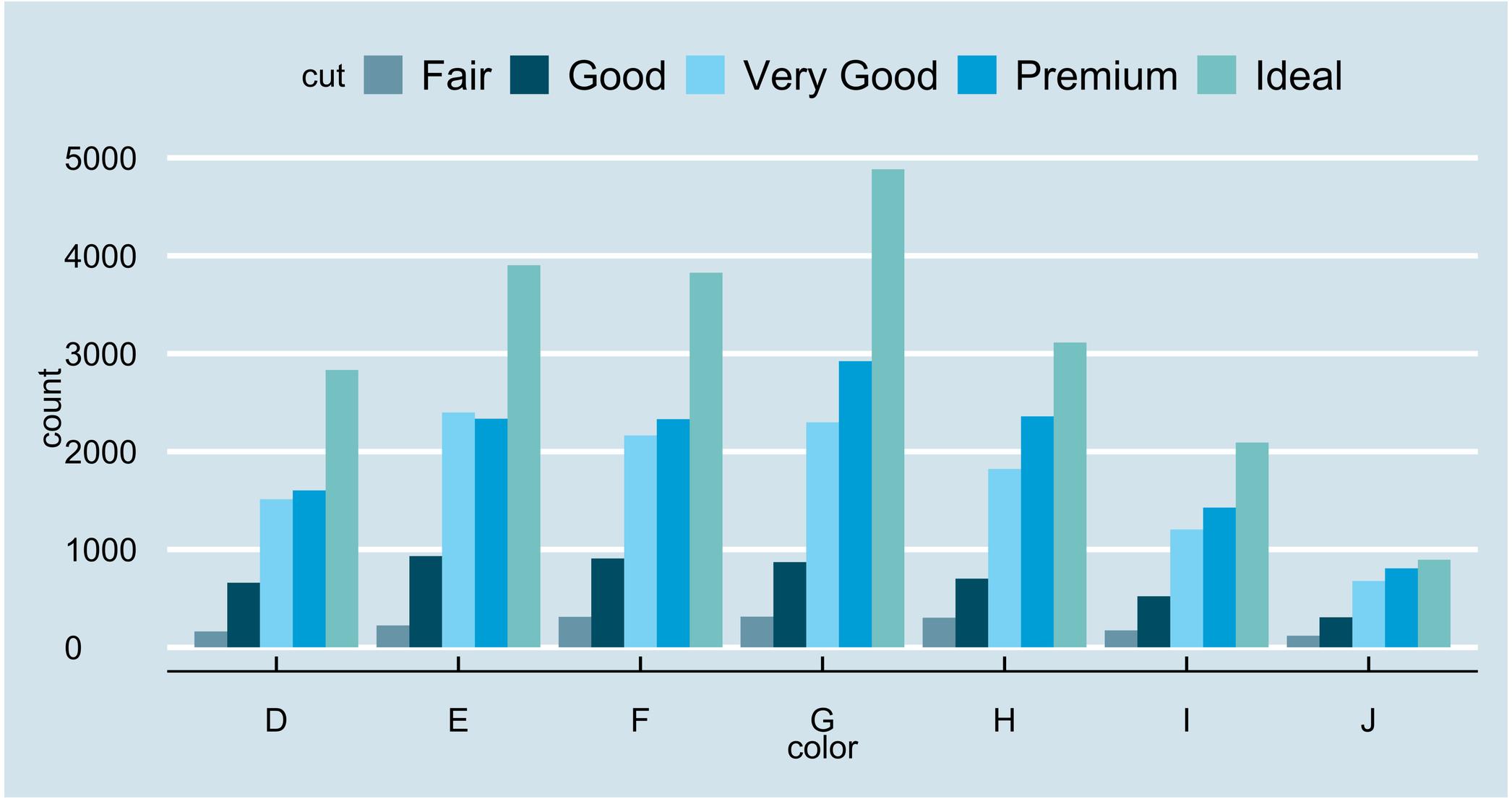
```
p
```



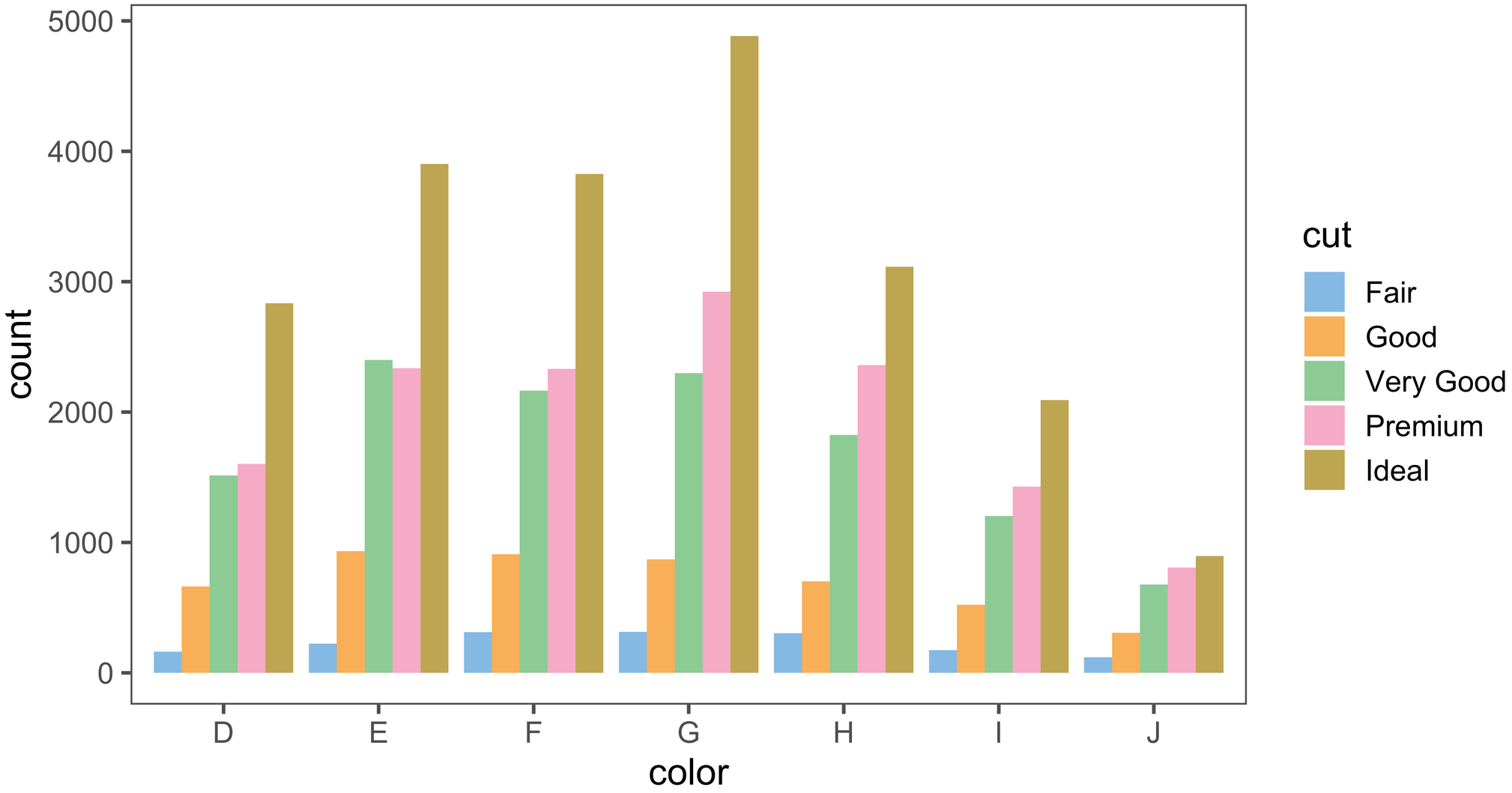

```
p + theme_excel()
```



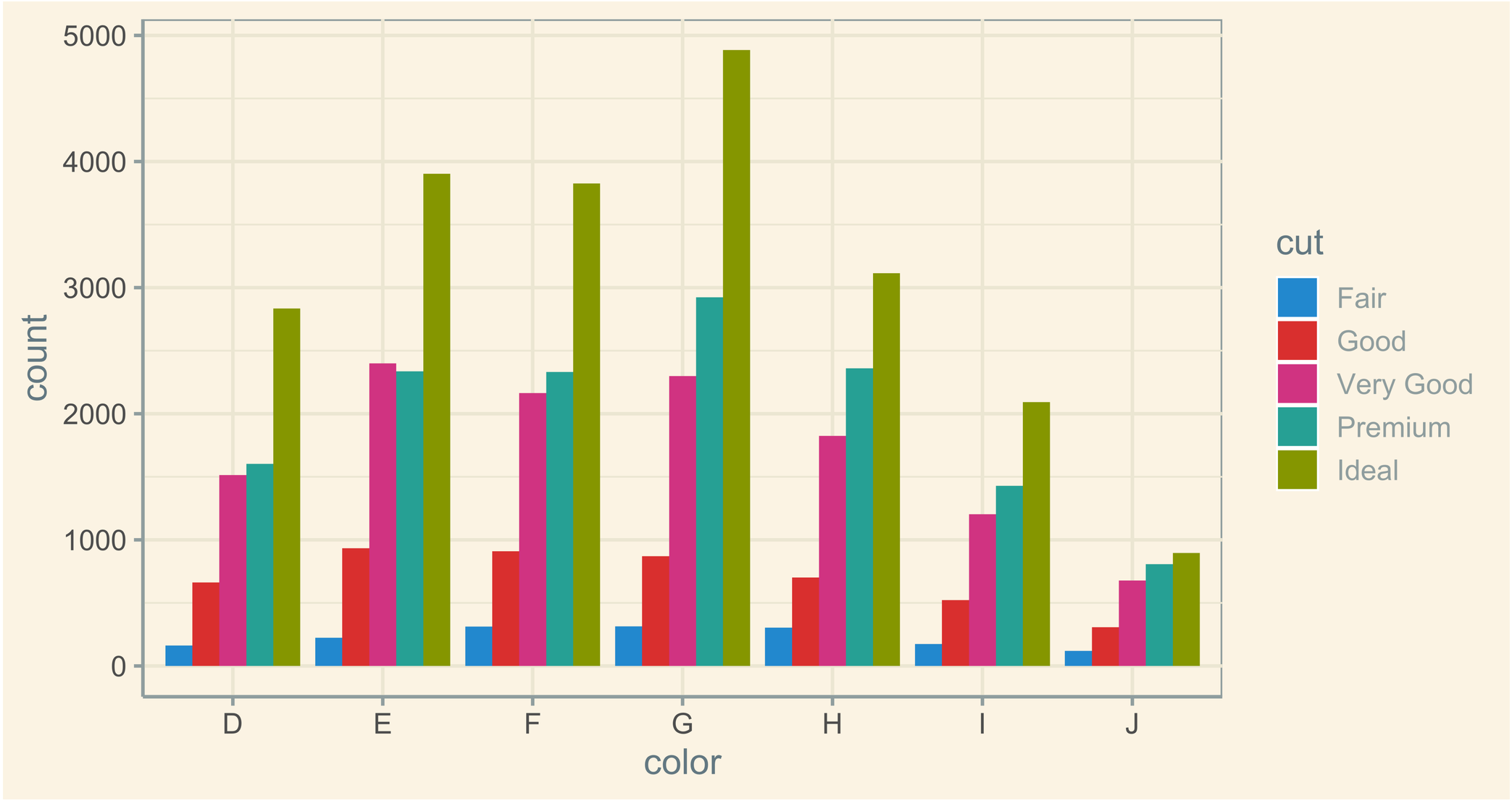

```
p + theme_excel() + scale_fill_excel()
```



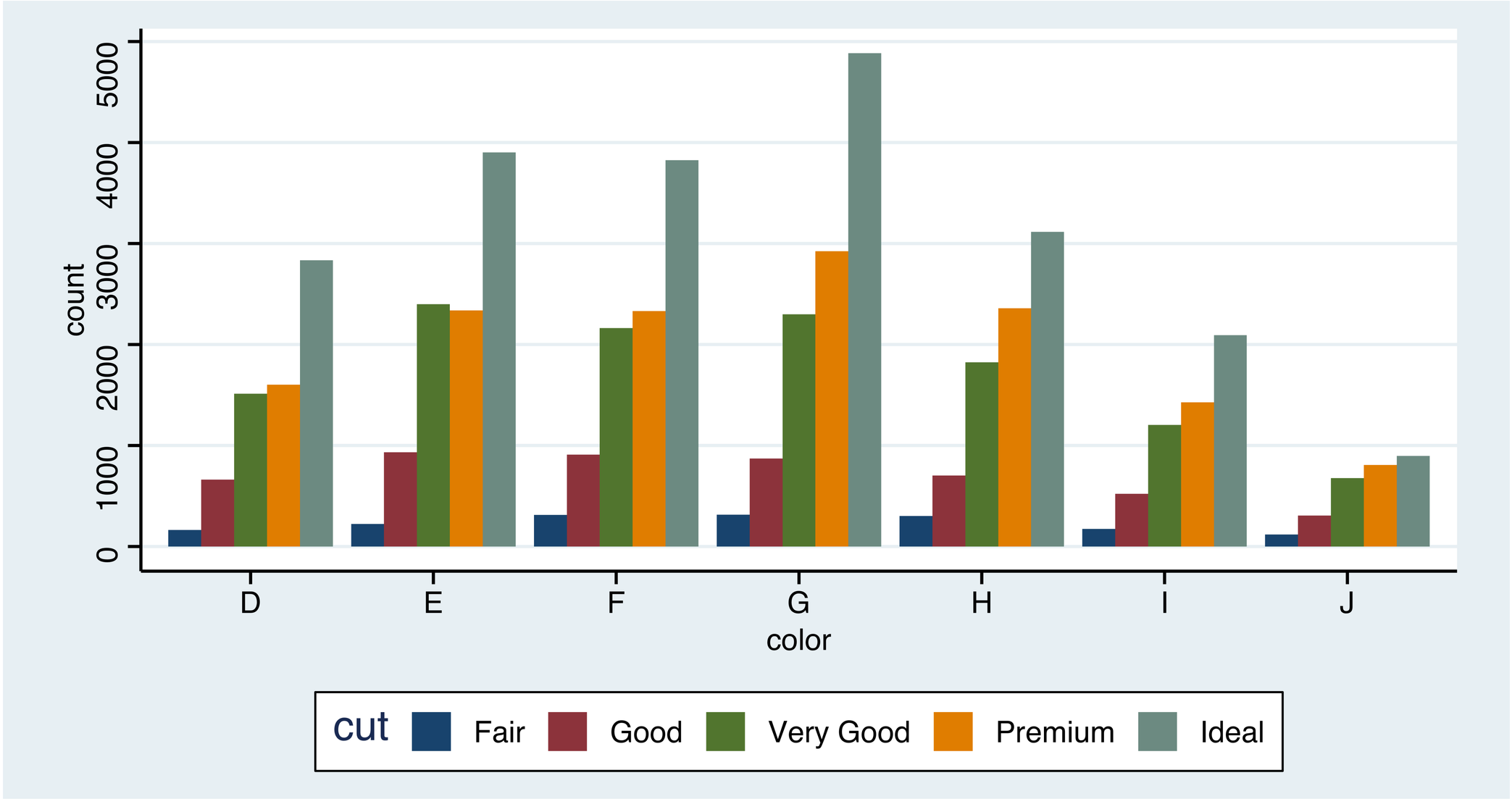

```
p + theme_economist() + scale_fill_economist()
```



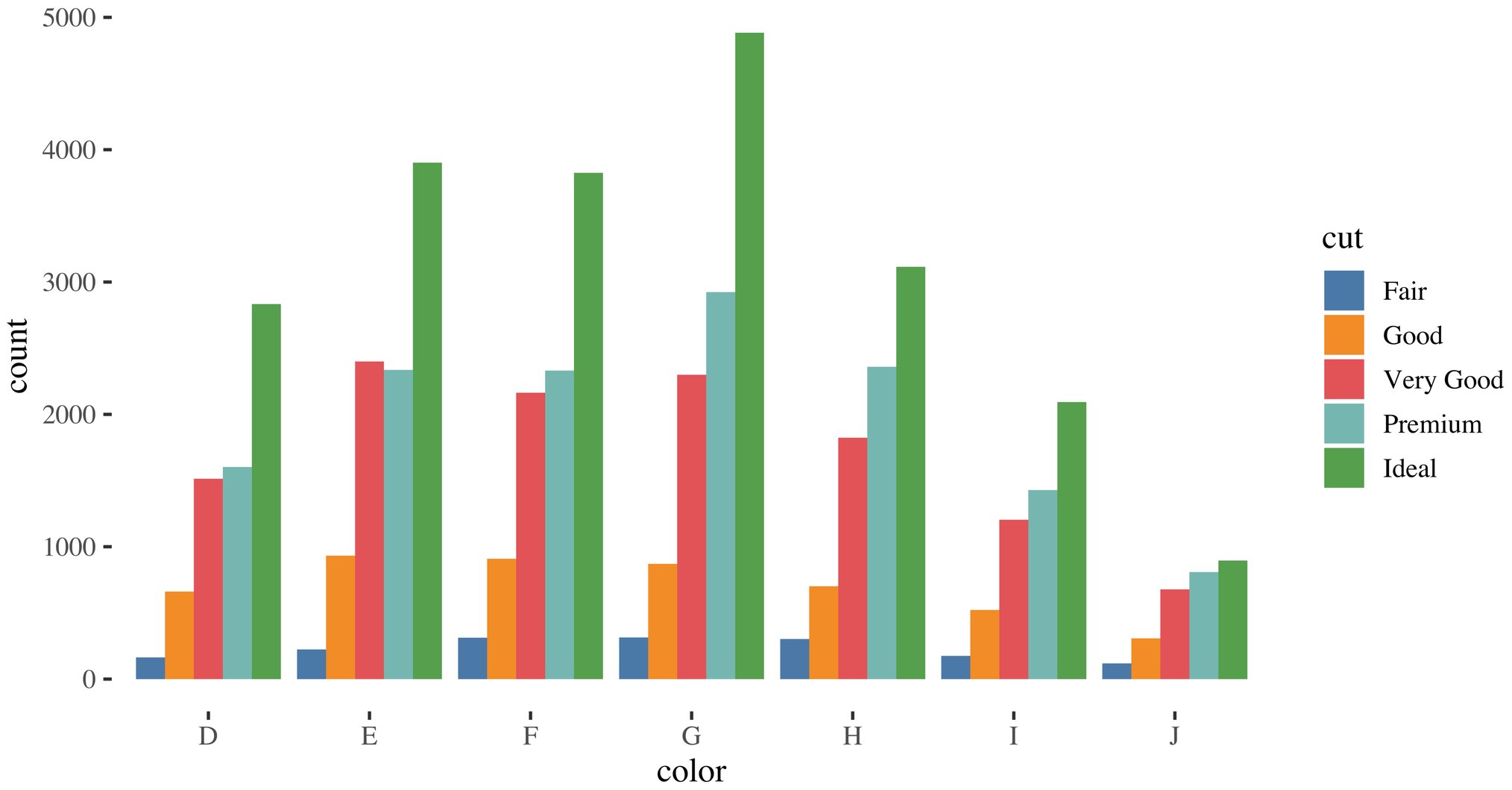

```
p + theme_few() + scale_fill_few()
```



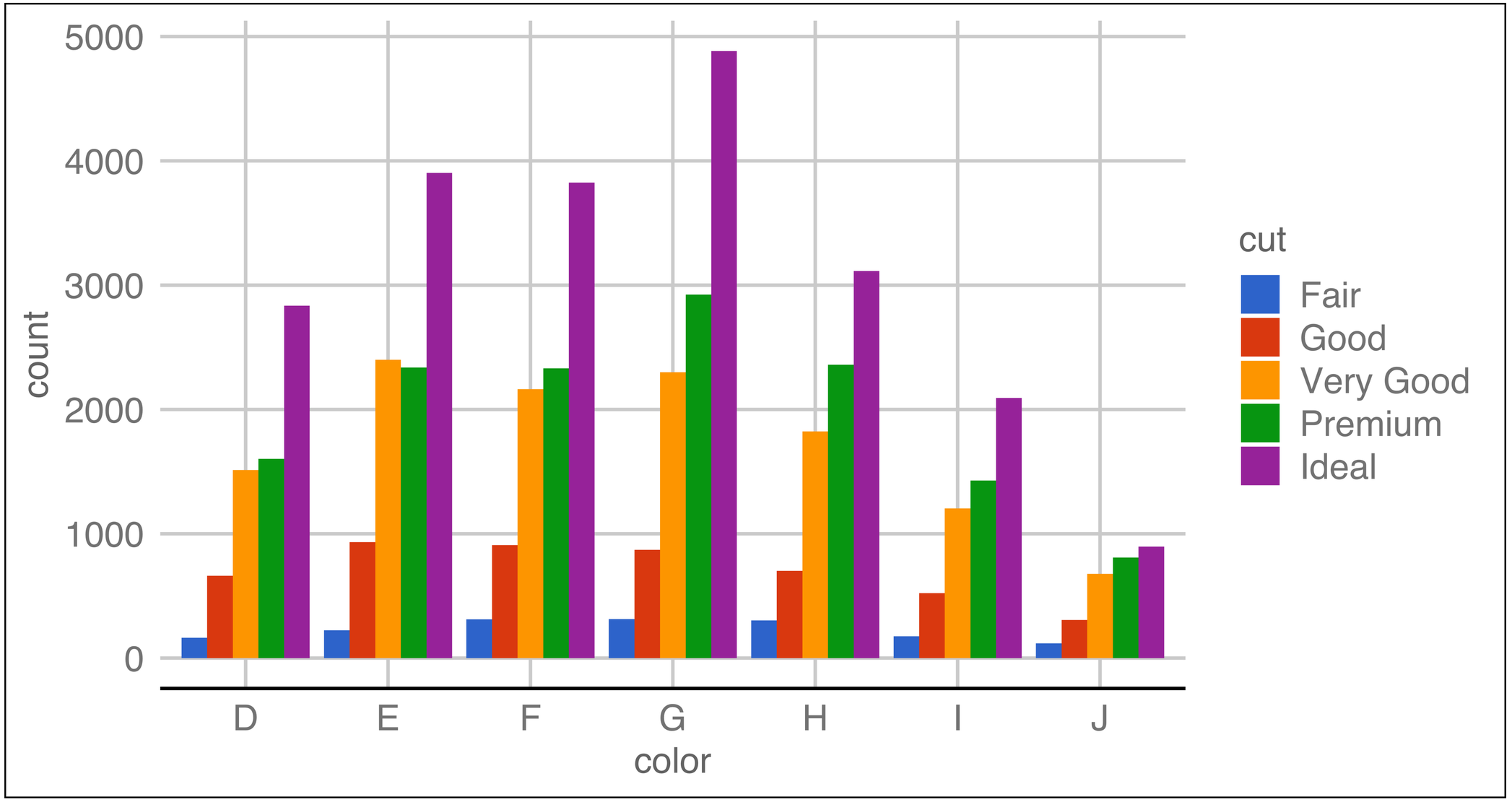

```
p + theme_solarized() + scale_fill_solarized()
```



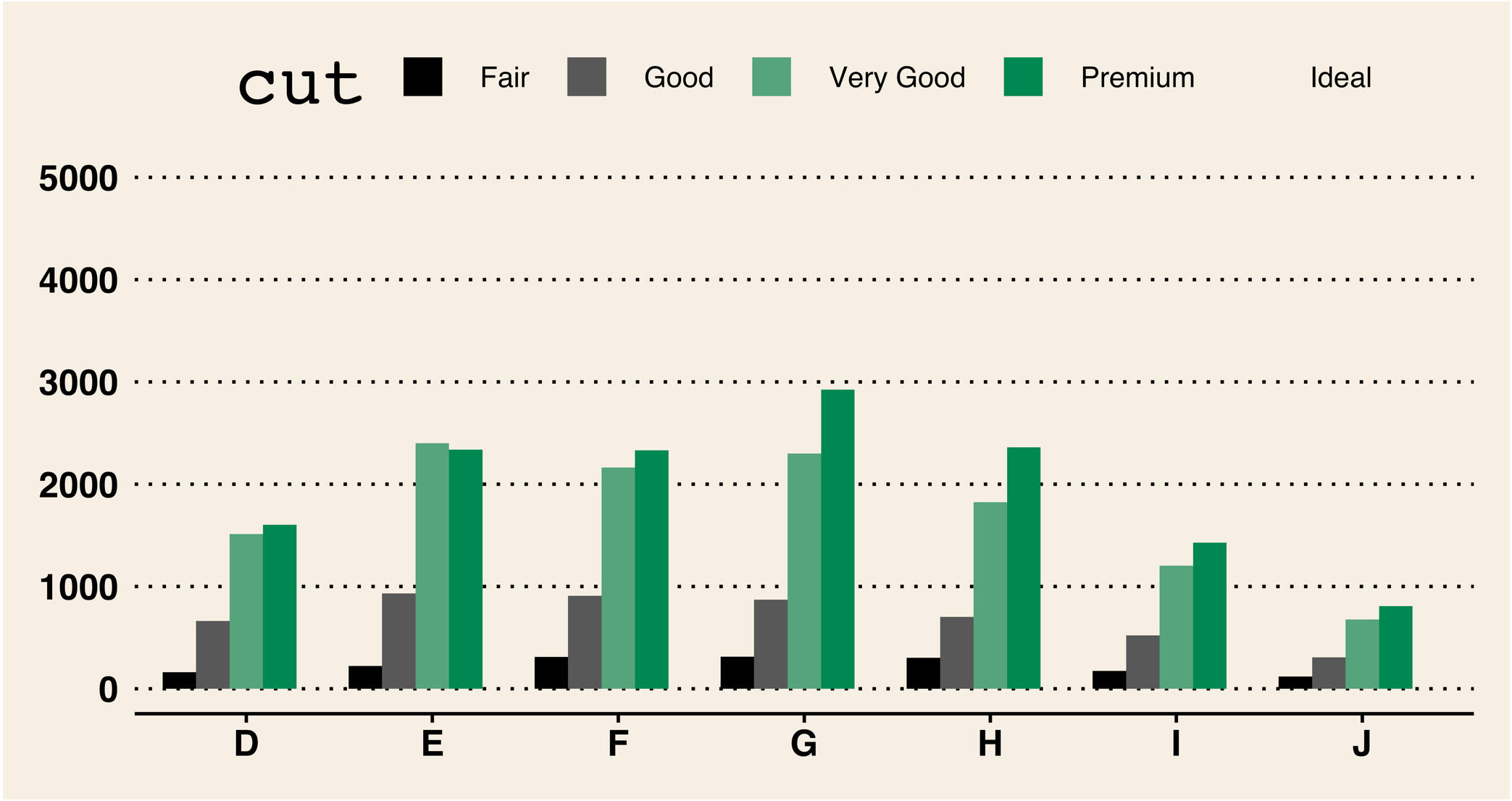

```
p + theme_stata() + scale_fill_stata()
```




```
p + theme_tufte() + scale_fill_tableau()
```




```
p + theme_gdocs() + scale_fill_gdocs()
```




```
p + theme_wsj() + scale_fill_wsj(palette = "black_green")
```

Eixos

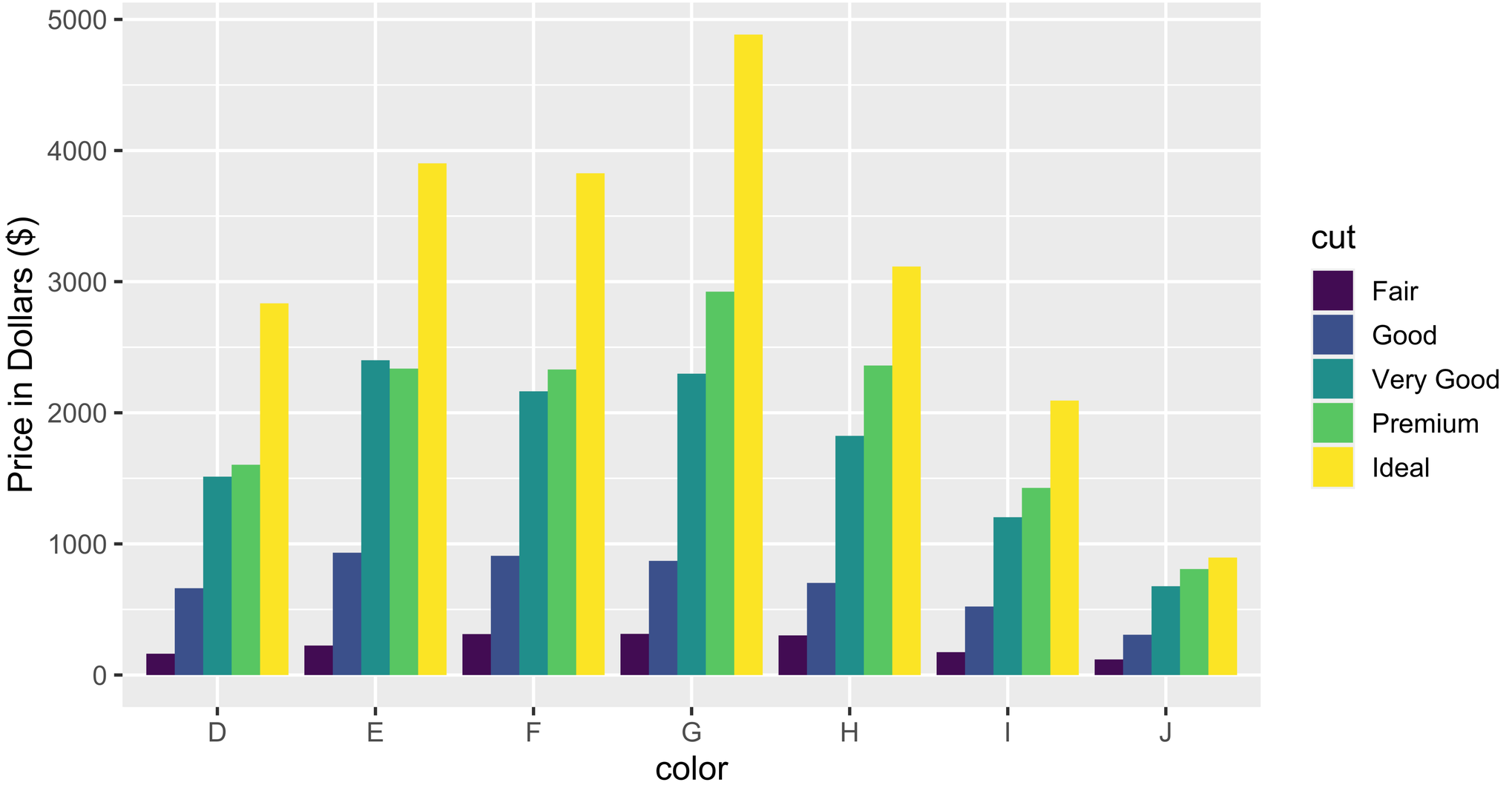
Adicione rótulos aos seus eixos

Eixos

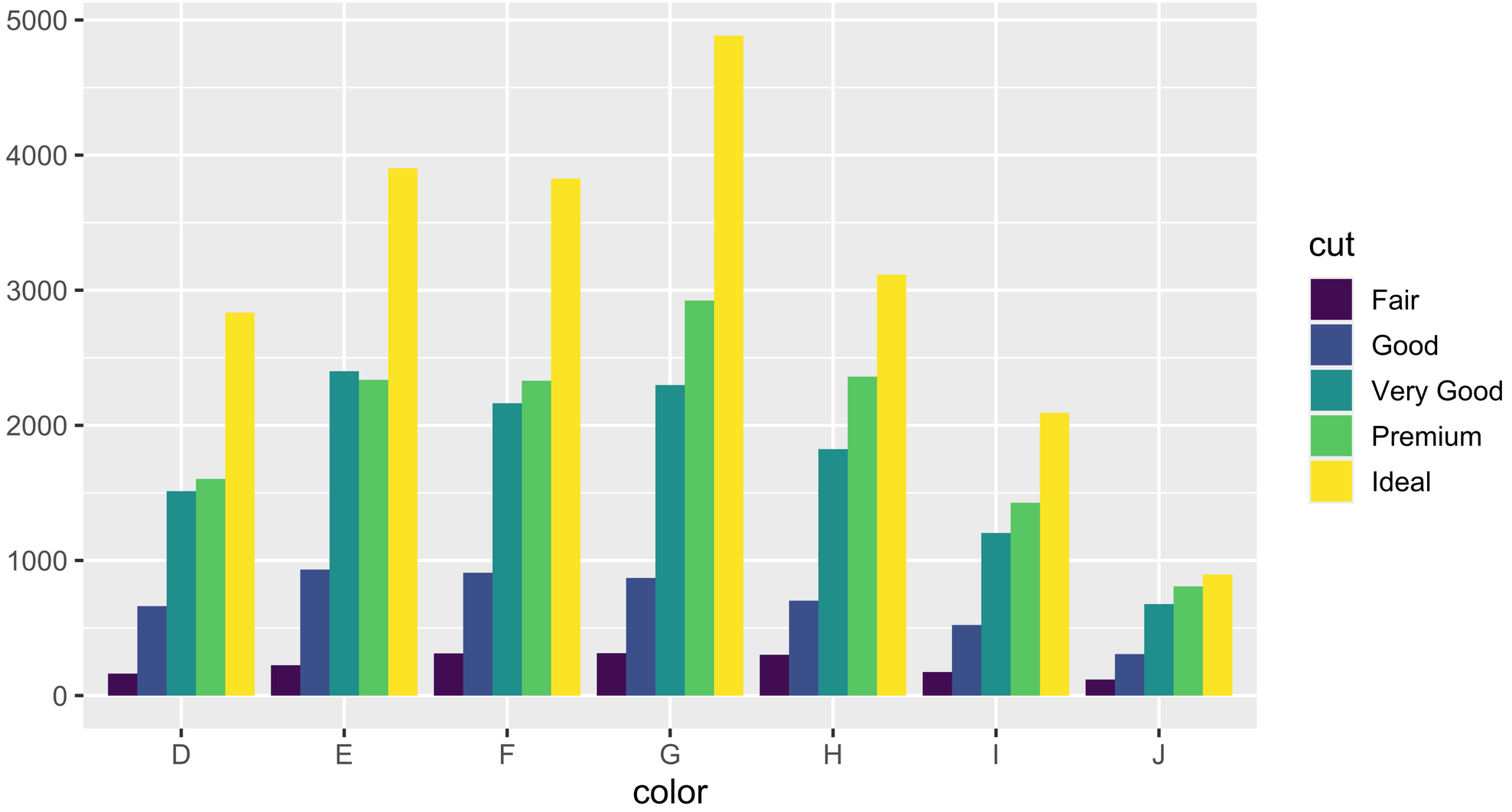
Adicione rótulos aos seus eixos

- Modifique o rótulo dos eixos por meio de `xlab` e `ylab`

```
p + xlab("Rótulo do eixo x")

p + ylab("Rótulo do eixo y")
```




```
p + ylab("Price in Dollars ($)")
```


5000
4000
3000
2000
1000
0
D
E
F
G
H
I
J
color
cut
Fair
Good
Very Good
Premium
Ideal

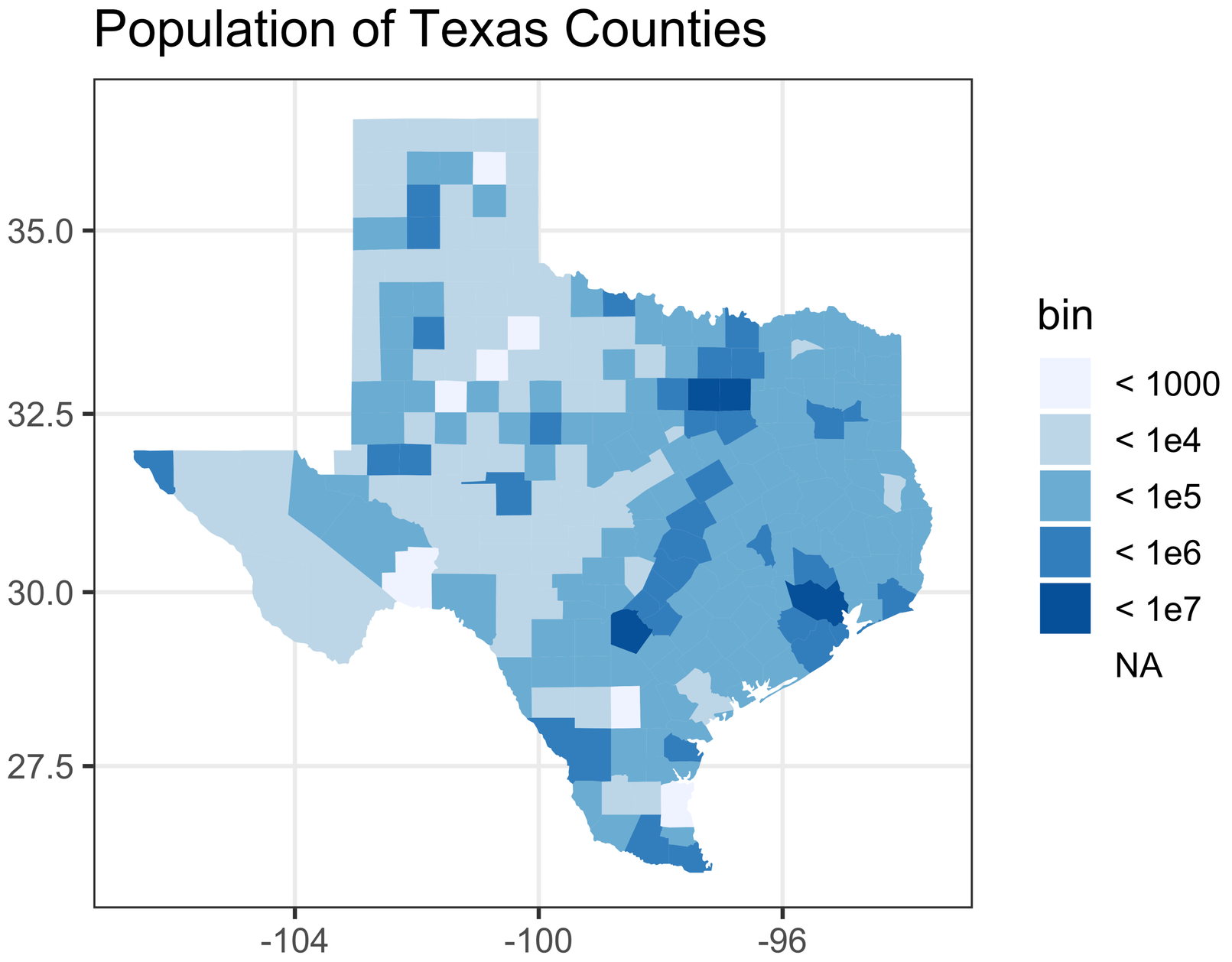
```
p + ylab("")
```

SUA VEZ

- Pratique o que você aprendeu até agora ao banco de dados `tx`
    1. remova os rótulos dos eixos `long` e `lat`
    2. adicione um fundo branco
    3. adicione um título
    4. adicione uma escala de cores `brewer`
    5. altere o sistema de coordenadas para o mais apropriado

SUA VEZ

```
tx + scale_fill_brewer(
  palette = "Blues" ) +
  xlab("") +
  ylab("") +
  theme_bw() +
  coord_map() +
  ggtitle("Population of Texas Counties")
```


Population of Texas Counties
35.0
32.5
30.0
27.5
-104
-100
-96
bin
< 1000
< 1e4
< 1e5
< 1e6
< 1e7
NA

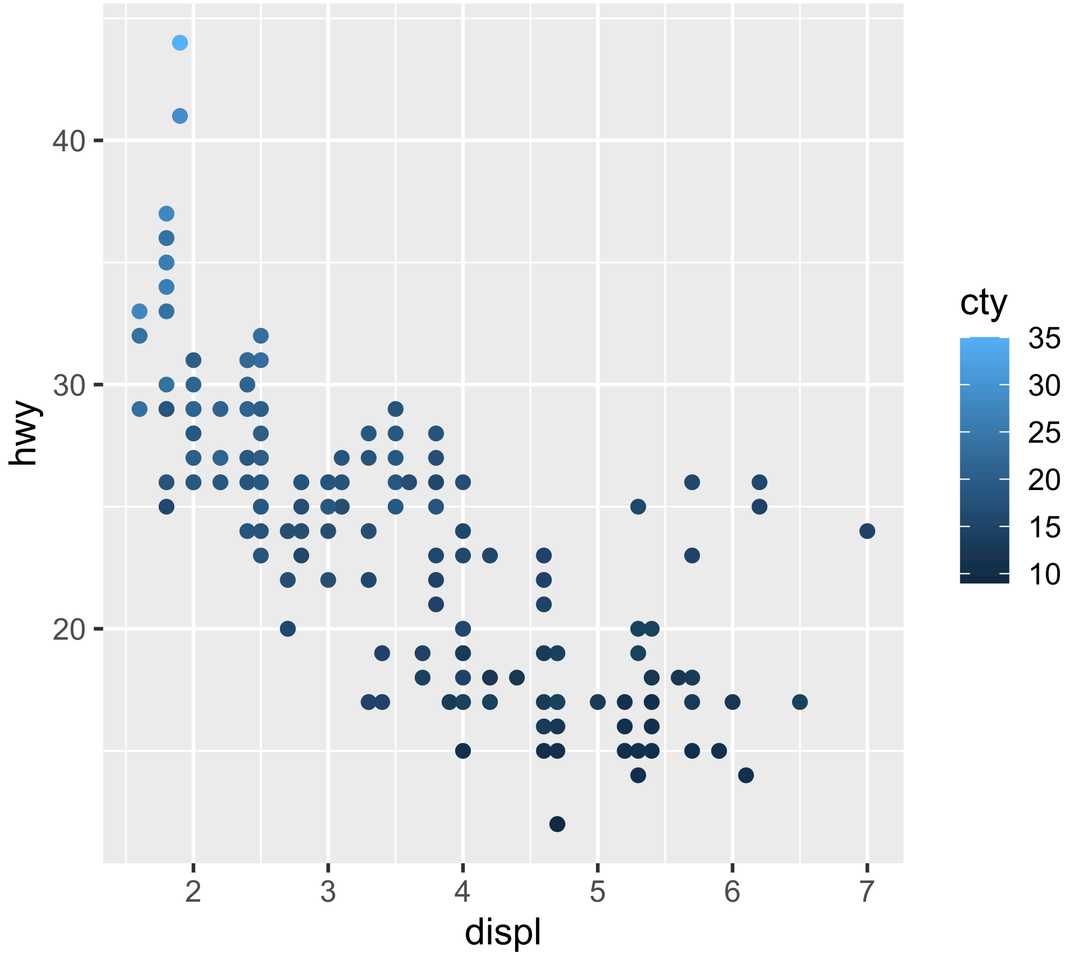
Legendas

Legendas

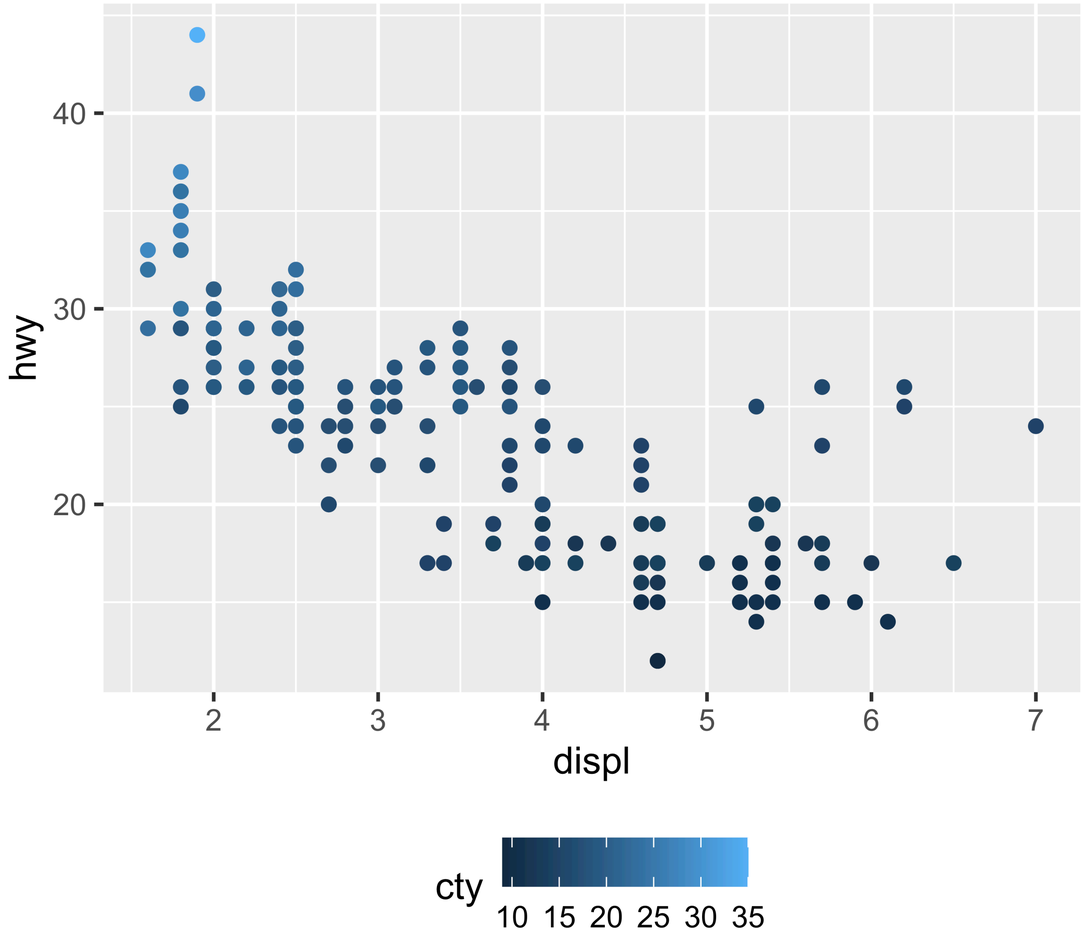
- Para facilitar a visualização dos códigos, vamos definir um gráfico base abaixo

```
q <- ggplot( mpg, aes(displ, hwy) ) +
       geom_point( aes(color = cty) )
```

- A primeira forma de controlar elementos (mudança de posição) das legendas ocorre no próprio tema que você definir




```
q
```




```
q +
  theme(legend.position = "bottom")
```

Tipo de legenda

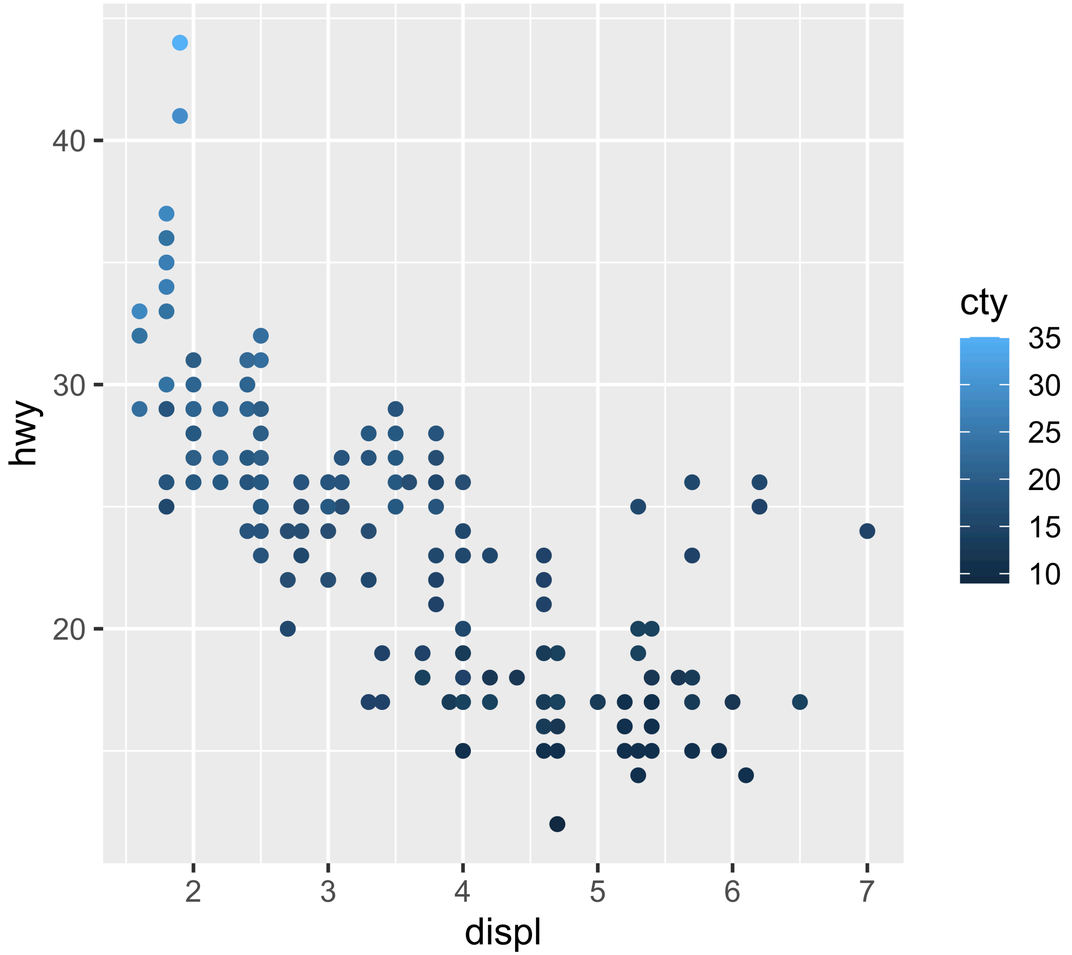
- O tipo de legenda é controlado pelo comando `guides`

Sempre começa com scale

Estética a ser mapeada

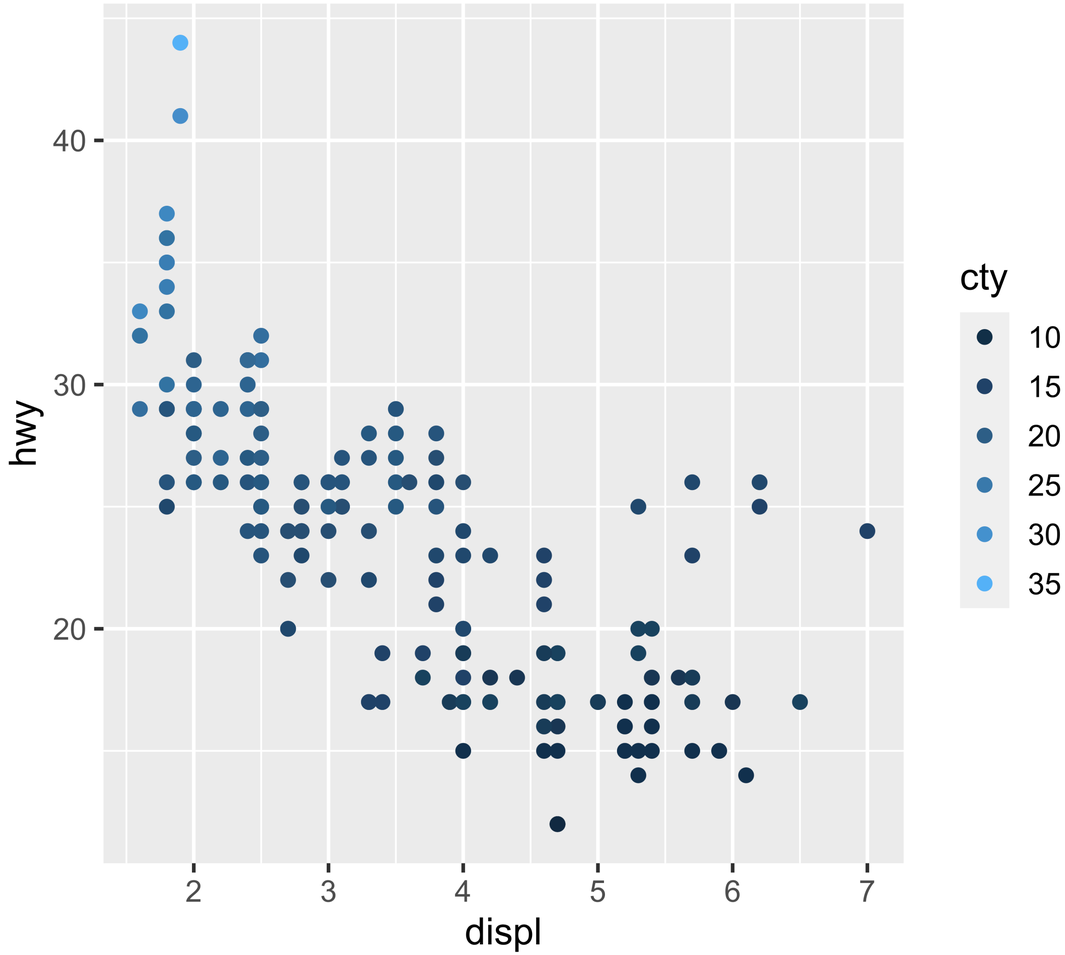
Tipo de legenda a ser utilizada

```
guides( color = "tipo de legenda" )
```

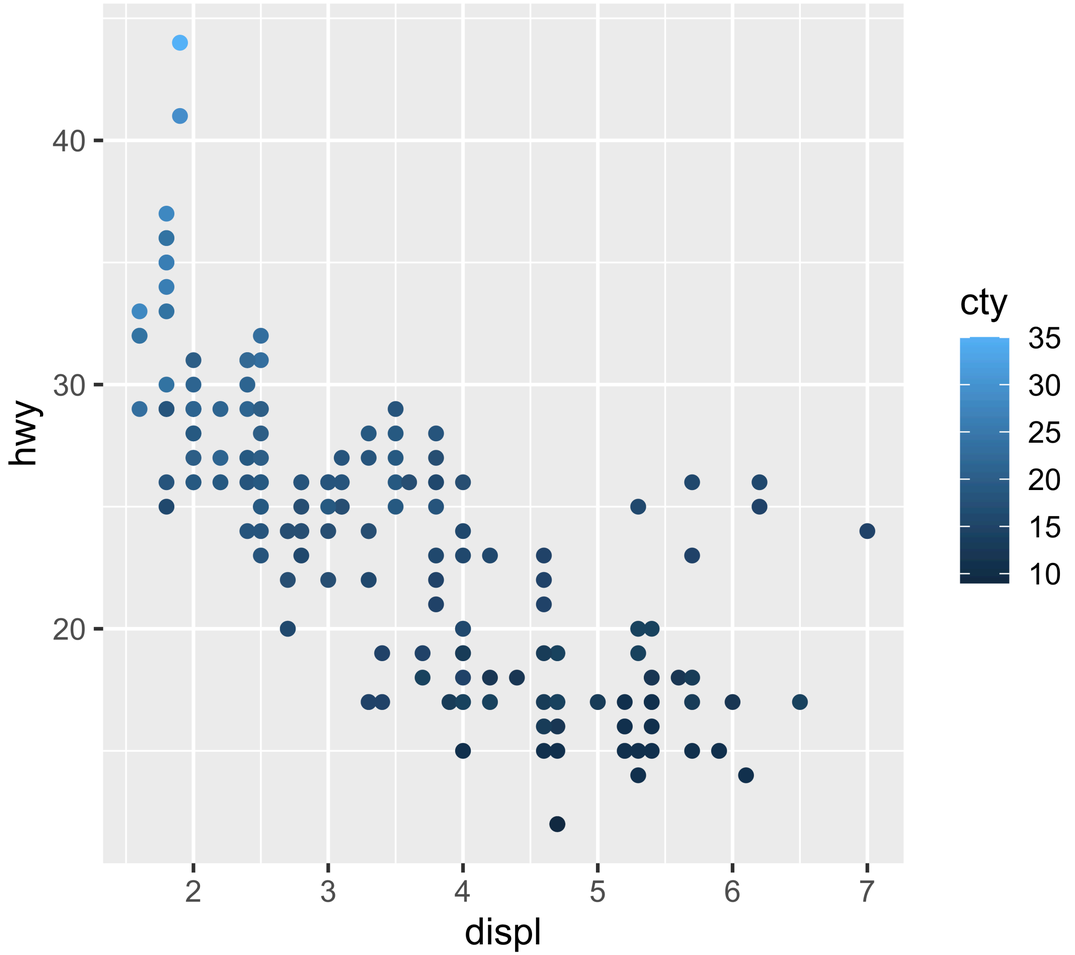



```
q + guides(color = "colorbar")
```

Padrão

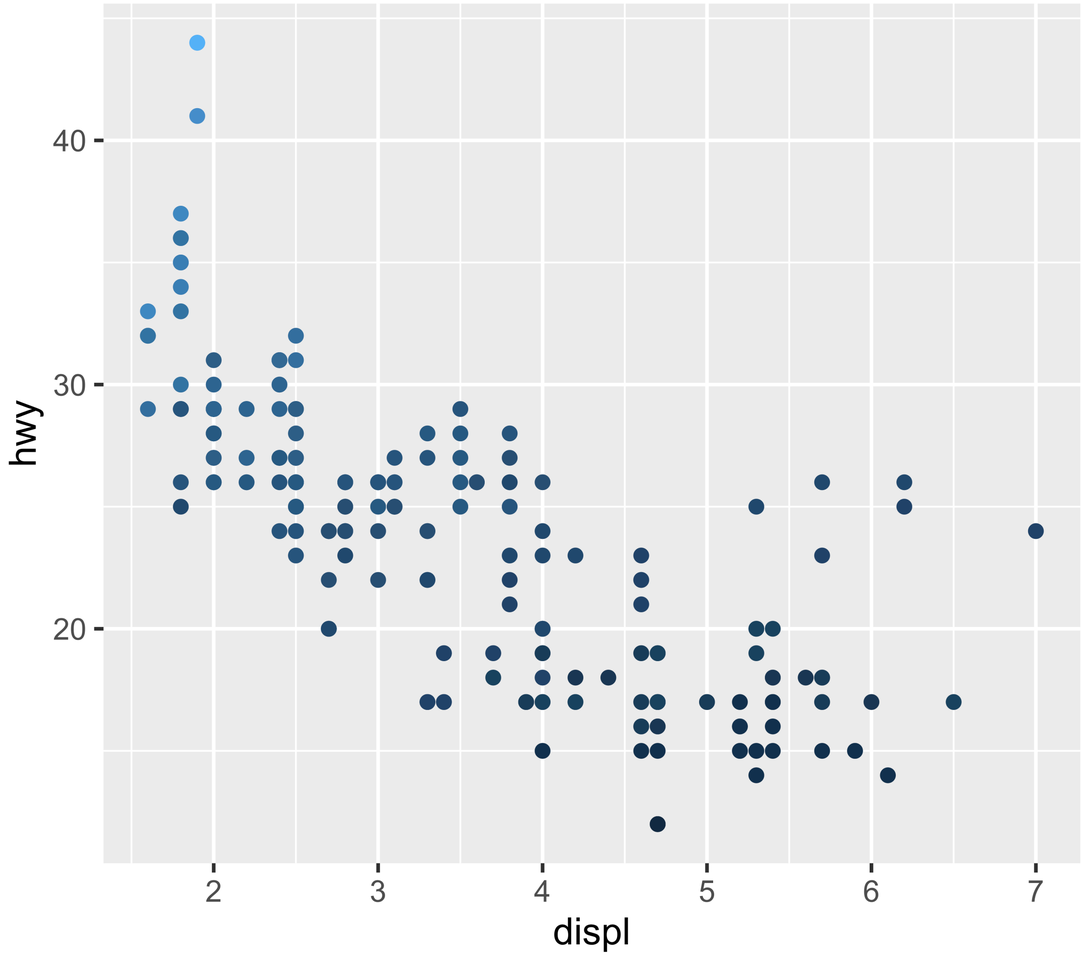



```
q + guides(color = "legend")
```




```
q + guides(color = "colorbar")
```

Padrão



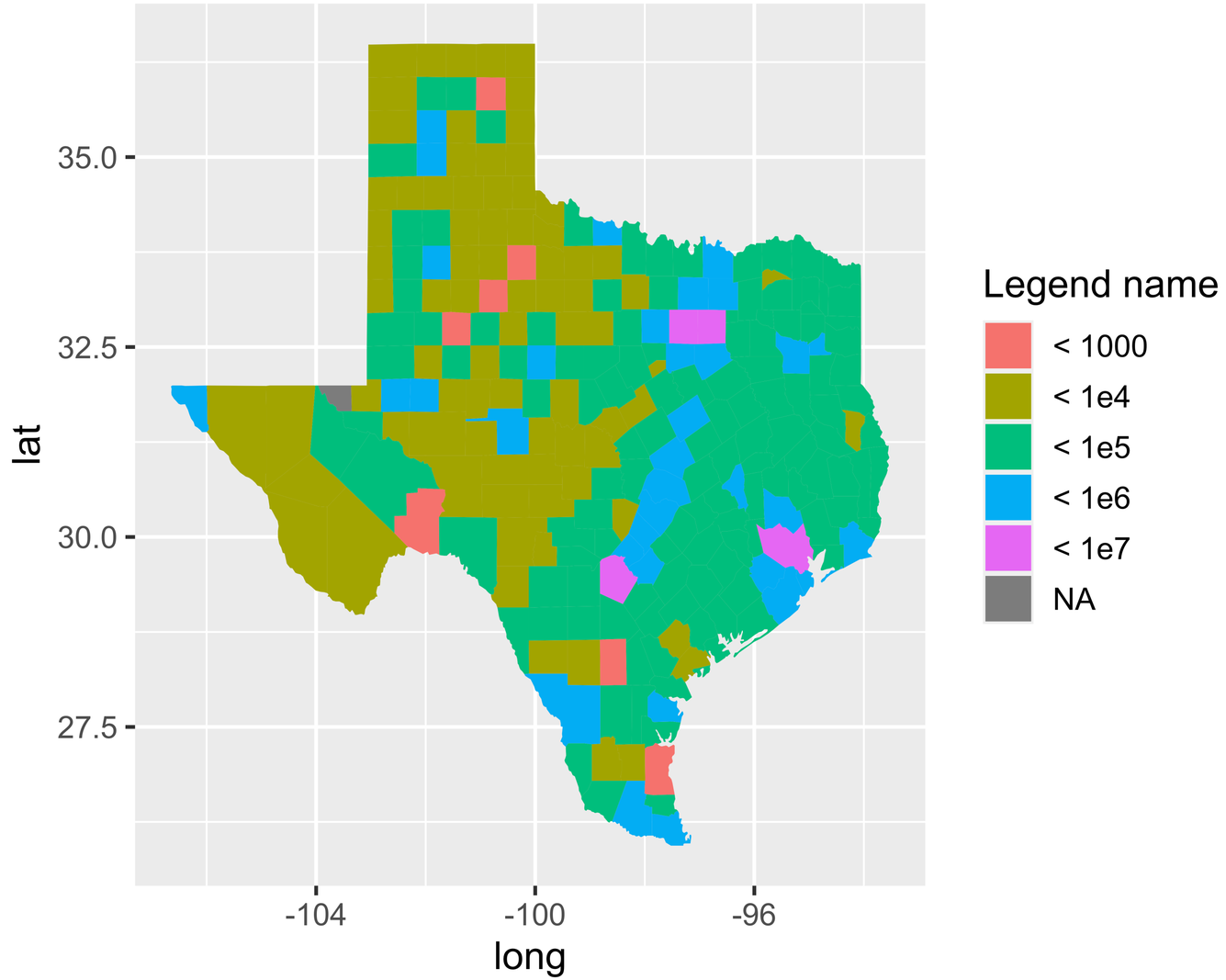

```
q + guides(color = "none")
```

Rótulos das legendas

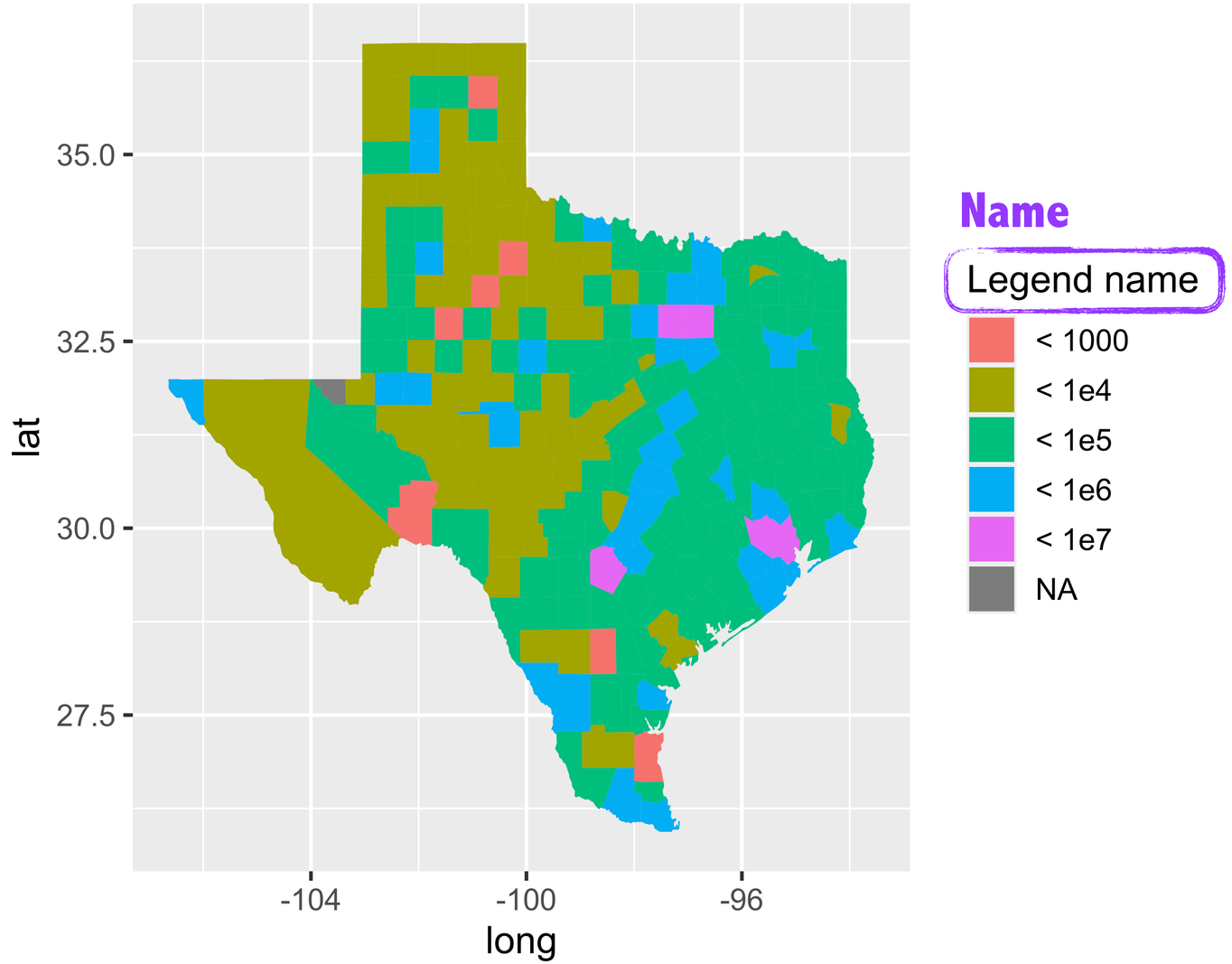
- Você pode alterar o rótulo das legendas a partir das escalas

- Se você quer manter a escala padrão, basta você escolher a escala relativa ao tipo de dados que você está mapeando. Caso contrário, indique a nova escala

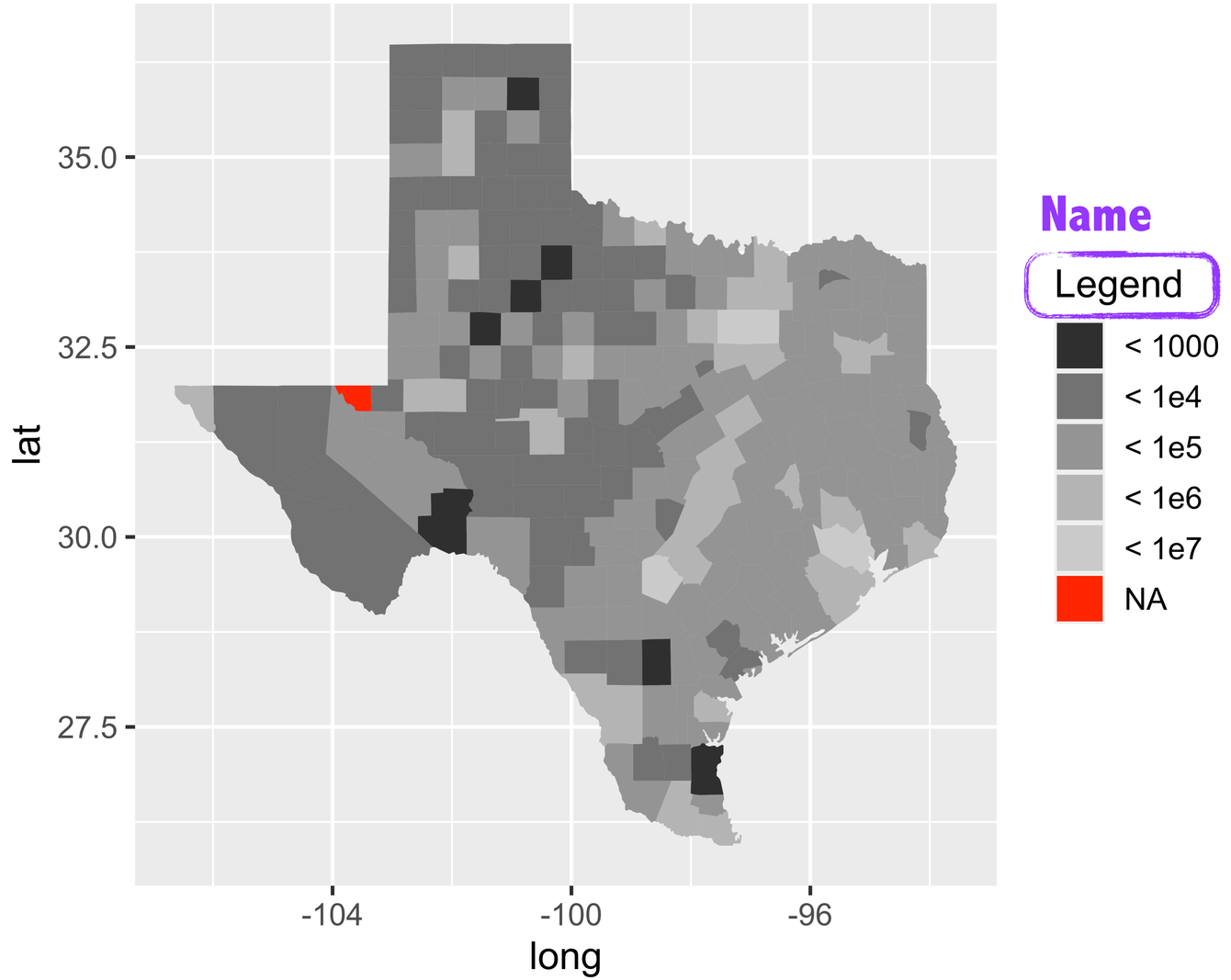
Escala a ser ajustada	Tipo de dados
`scale_*_discrete`	Discreto
`scale_*_continuos`	Contínuo



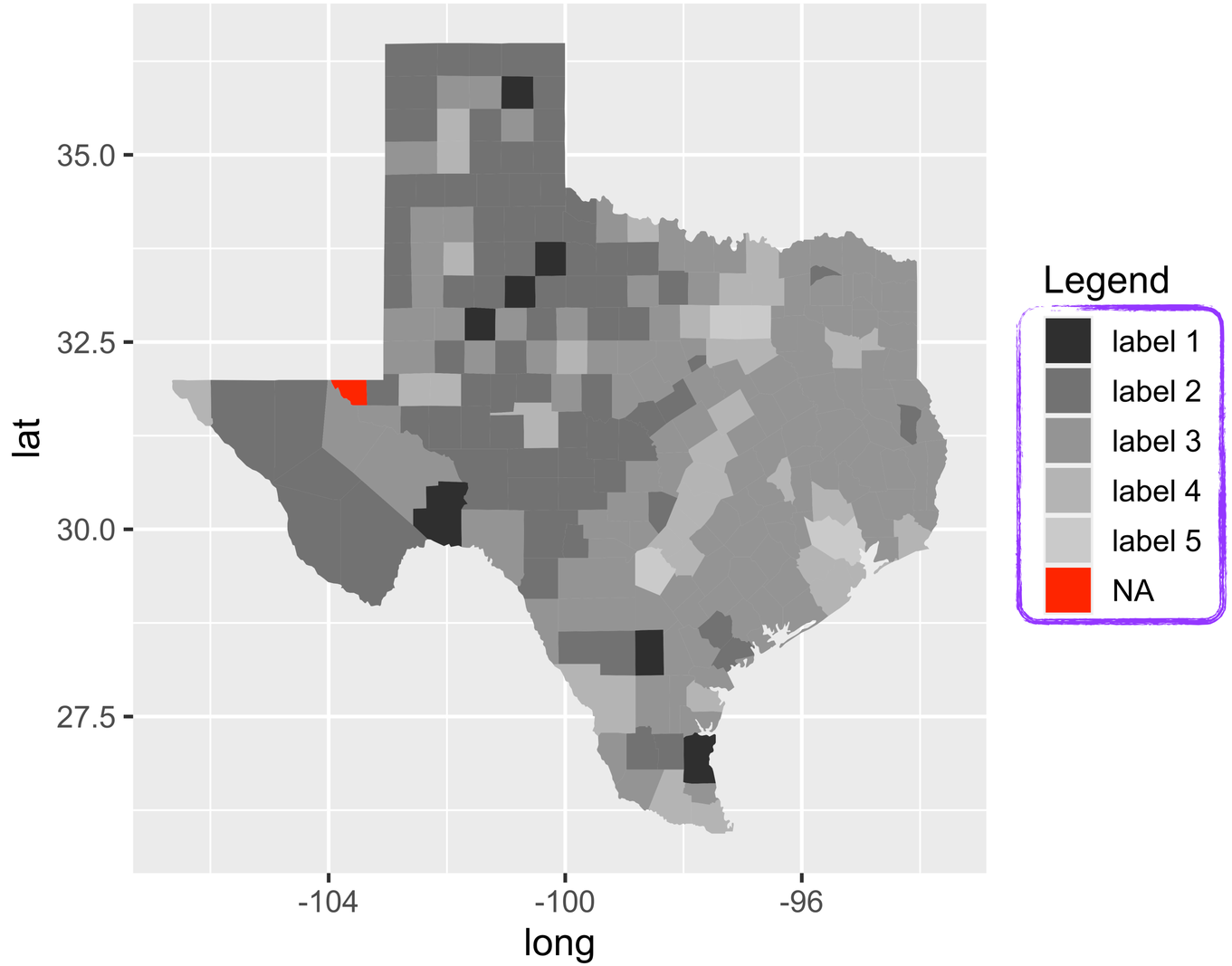

```
tx + scale_fill_discrete( name = "Legend name" )
```




```
tx + scale_fill_discrete( name = "Legend name" )
```



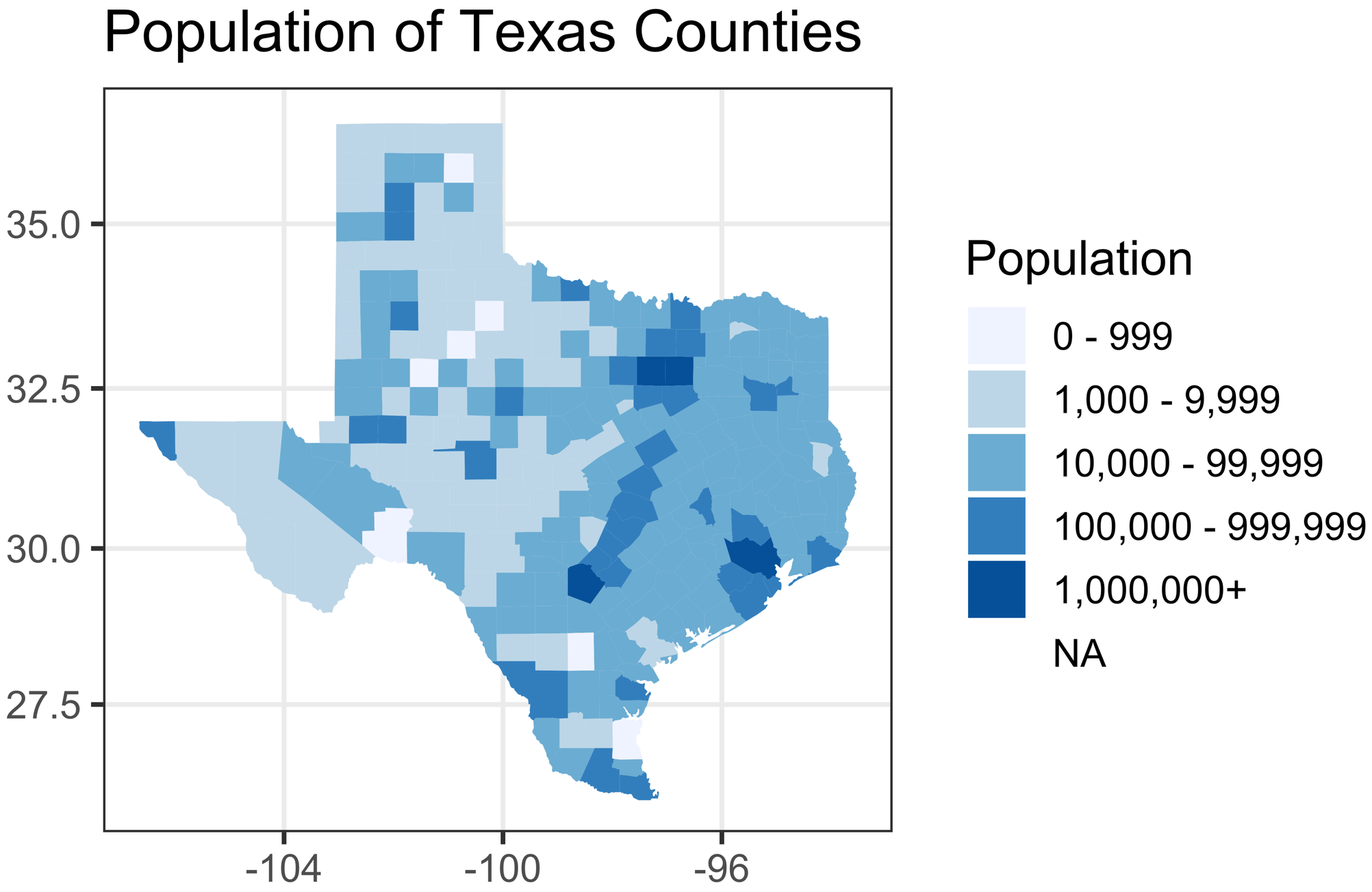

```
tx + scale_fill_grey( name = "Legend" )
```




```
tx + scale_fill_grey(
  name = "Legend",
  labels = c("label 1", "label 2", "label 3","label 4", "label 5") )
```

SUA VEZ

Complete o mapa do Texas que você começou no seu último turno adicionando um título informativo e um conjunto de rótulos à legenda.


Population of Texas Counties
35.0
32.5
30.0
27.5
-104
-100
-96
Population
0 - 999
1,000 - 9,999
10,000 - 99,999
100,000 - 999,999
1,000,000+
NA

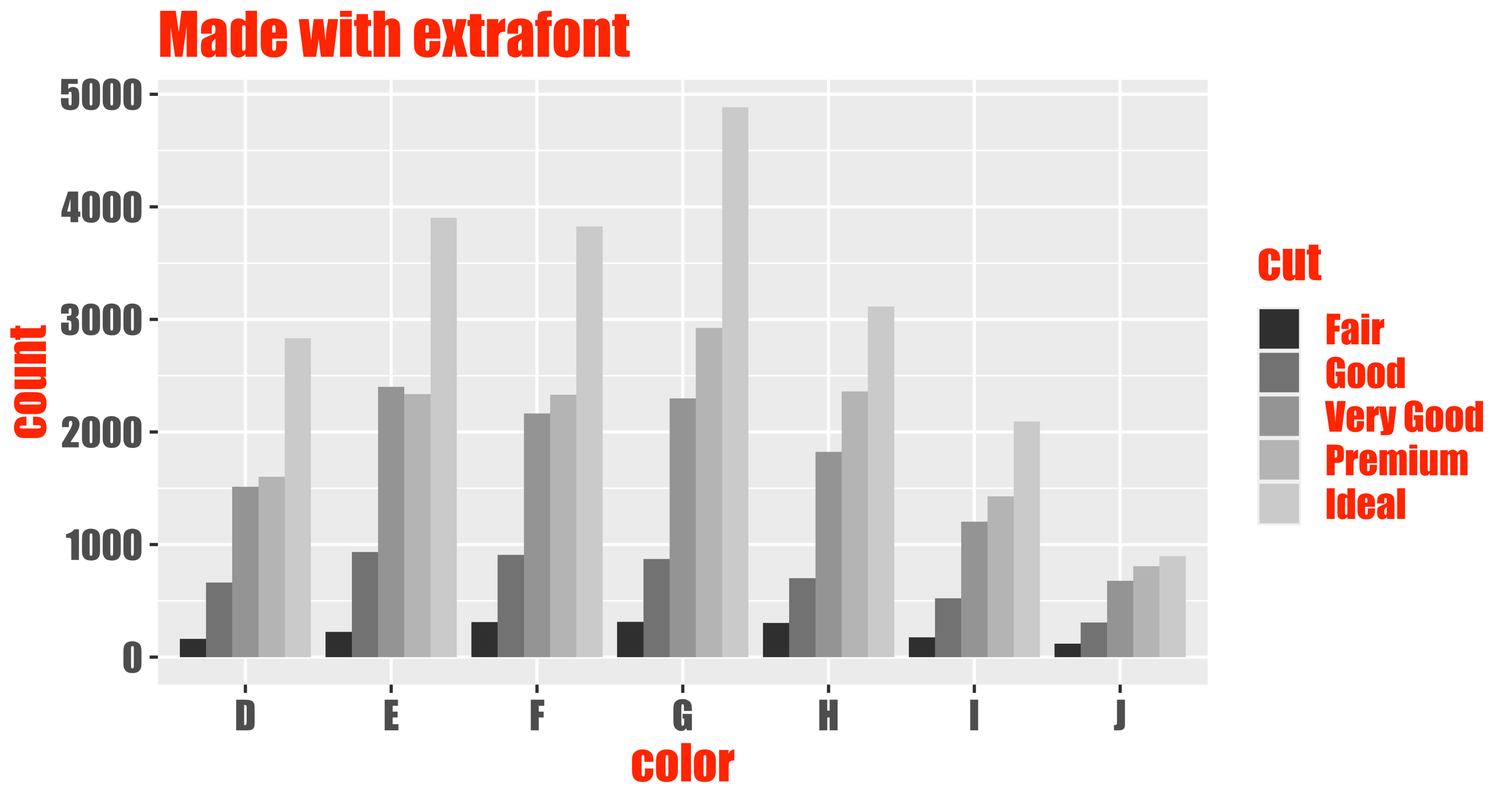
SUA VEZ

```
tx + scale_fill_brewer(
  palette = "Blues",
  name = "Population",
  labels = c("0 - 999", "1,000 - 9,999",
             "10,000 - 99,999", "100,000 - 999,999",
             "1,000,000+")) +
  xlab("") +
  ylab("") +
  theme_bw() +
  coord_map() +
  ggtitle("Population of Texas Counties")
```

Outros pacotes e além

Personalização de fontes

extrafont



extrafont

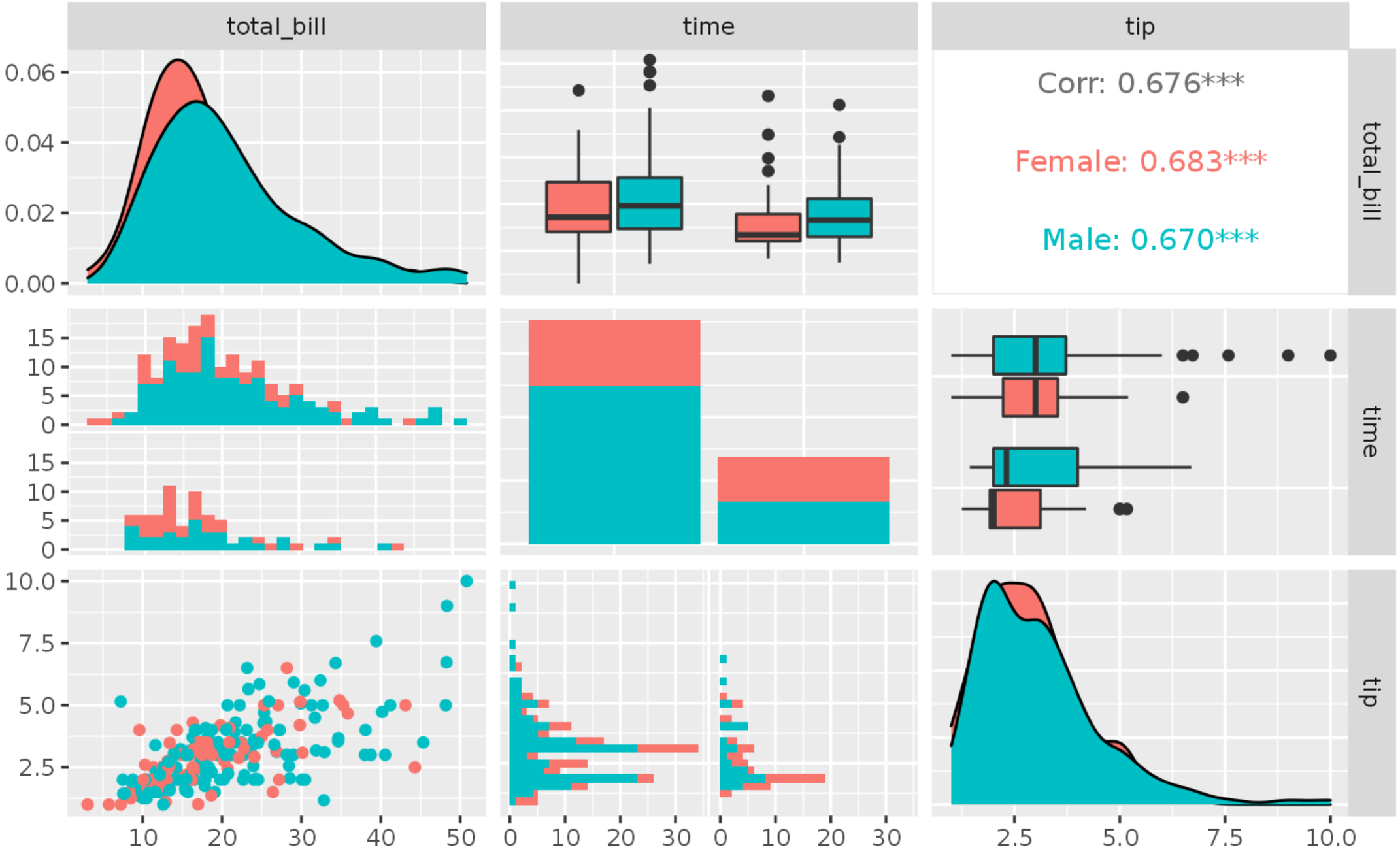
```
p + scale_fill_grey() +
  ggtitle( "Made with extrafont" ) +
  theme( text = element_text( size=16,
                              family = "Impact",
                              colour = "red" ) )
```

Gráficos especializados

GGally

```
install.packages("GGally")
```

https://github.com/ggobi/ggally

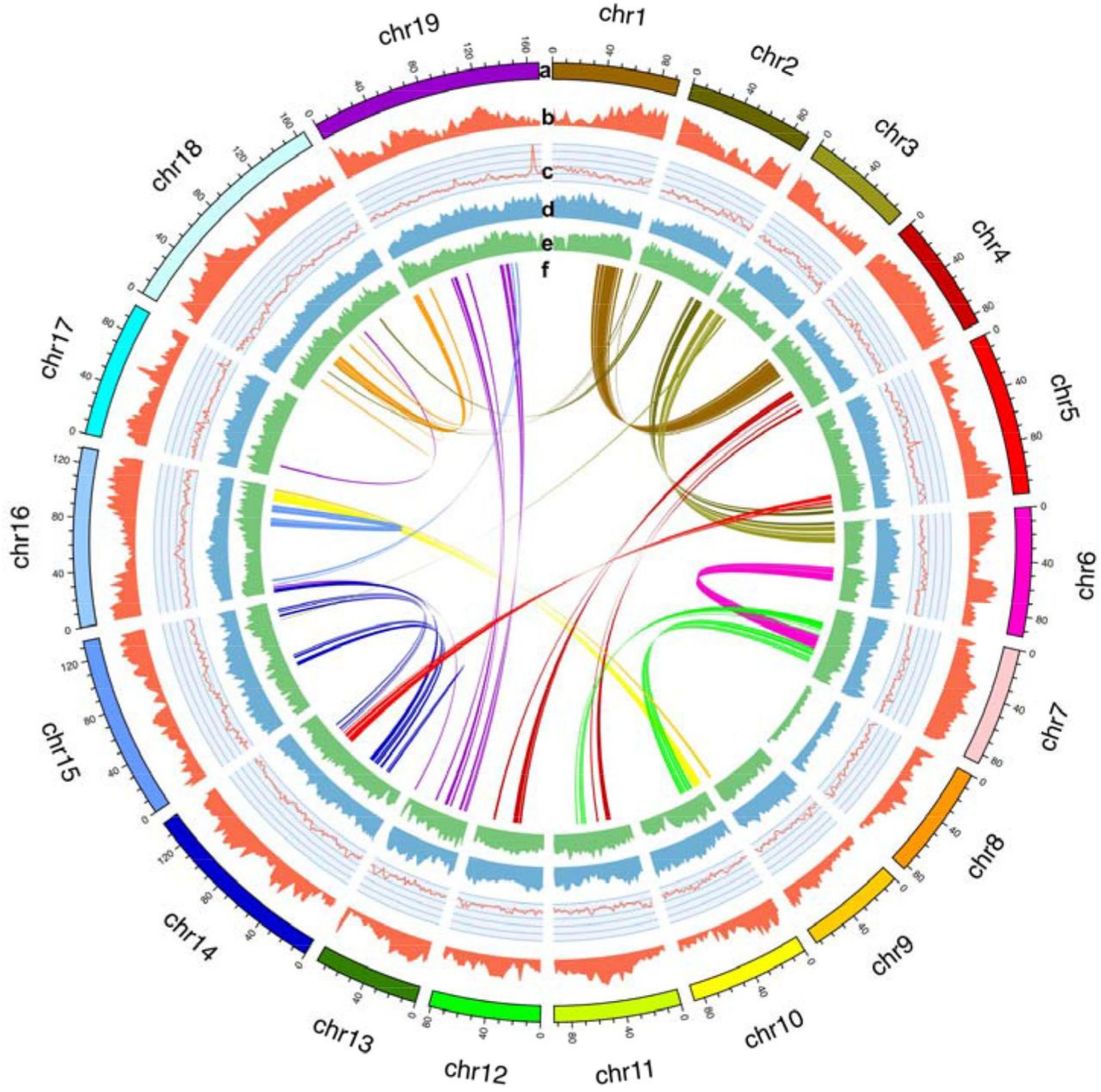

total_bill
time
tip
Corr: 0.676***
Female: 0.683***
Male: 0.670***
total_bill
time
tip
0.06
0.04
0.02
0.00
15
10
5
0
10.0
7.5
5.0
2.5
10
20
30
40
50
0
2.5
5.0
7.5
10.0

ggplot para dados do genoma

ggbio

```
source("http://bioconductor.org/biocLite.R")
biocLite("ggbio")
```

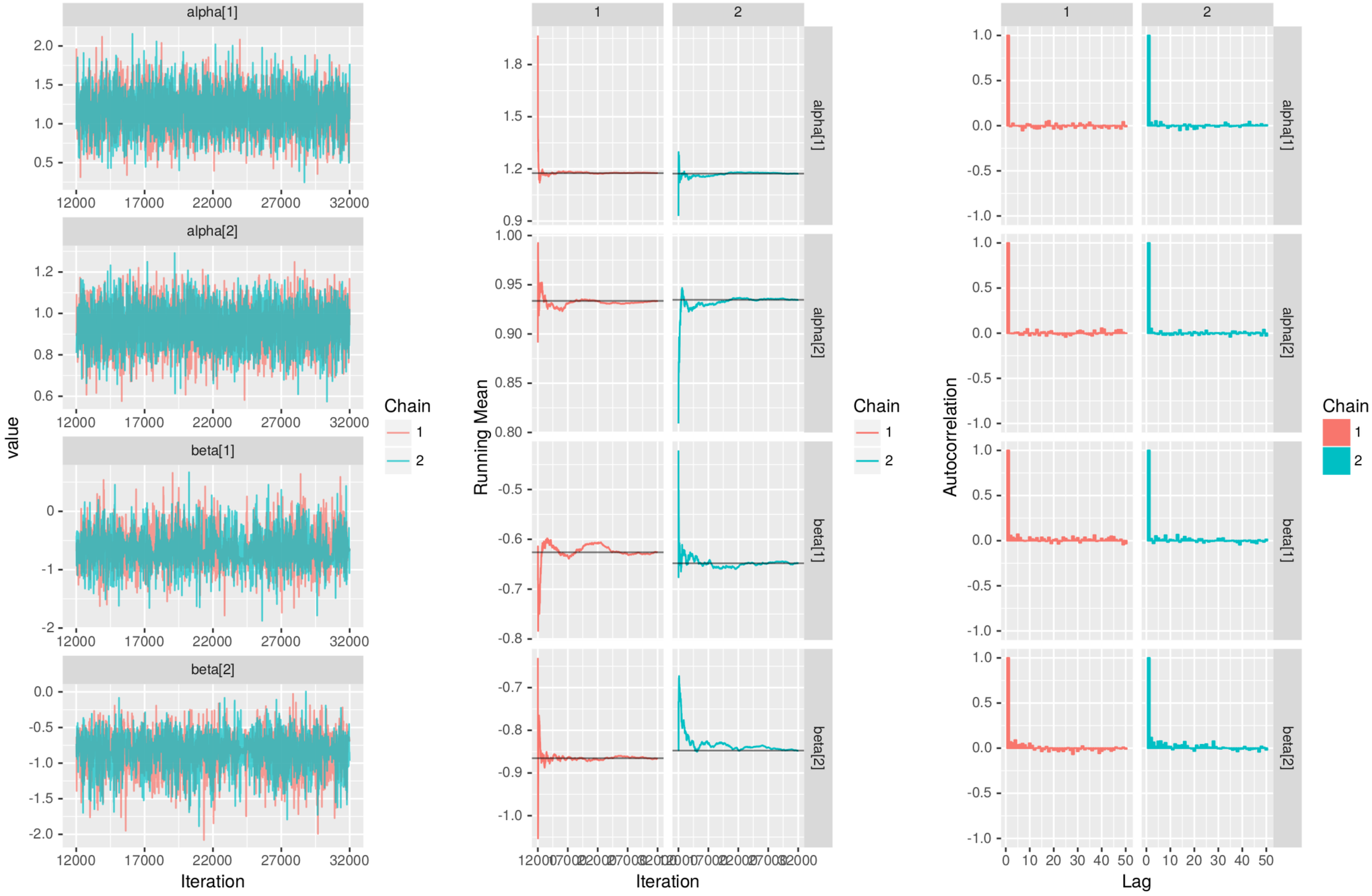
http://tengfei.github.com/ggbio/


chr1
chr2
chr3
chr4
chr5
chr6
chr7
chr8
chr9
chr10
chr11
chr12
chr13
chr14
chr15
chr16
chr17
chr18
chr19
a
b
c
d
e
f

ggmcmc

```
install.packages("ggmcmc")
```

http://xavier-fim.net/packages/ggmcmc/

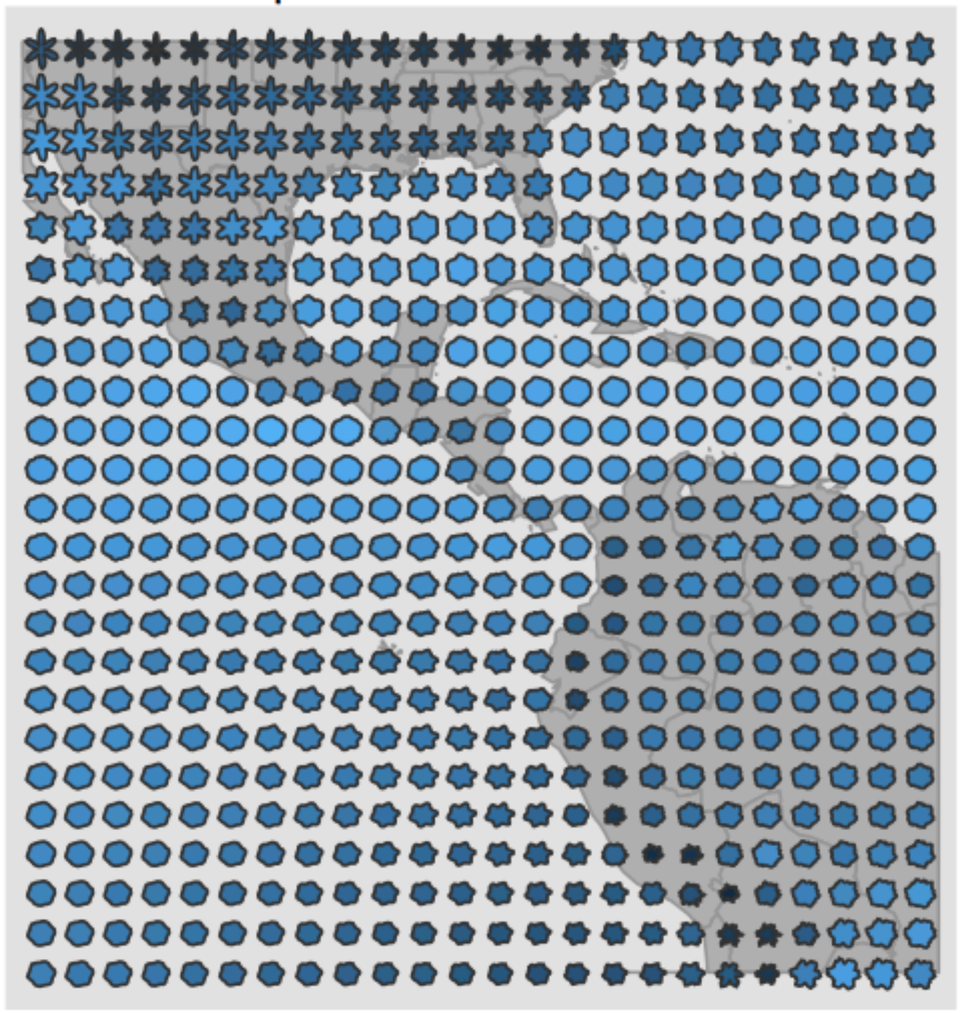

alpha[1]
alpha[2]
beta[1]
beta[2]
value
Iteration
Chain
1
2
Running Mean
1
2
Autocorrelation
Lag

ggsubplot

```
install.packages("ggsubplot")
```

http://blog.revolutionanalytics.com/2012/09/visualize-complex-data-with-subplots.html

Surface temperature fluctuations 1995 – 2001



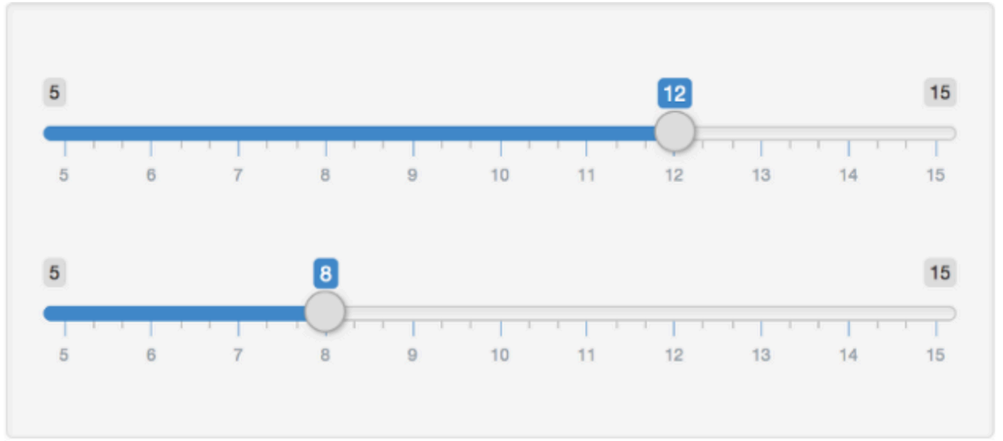
Average Temperature (F)

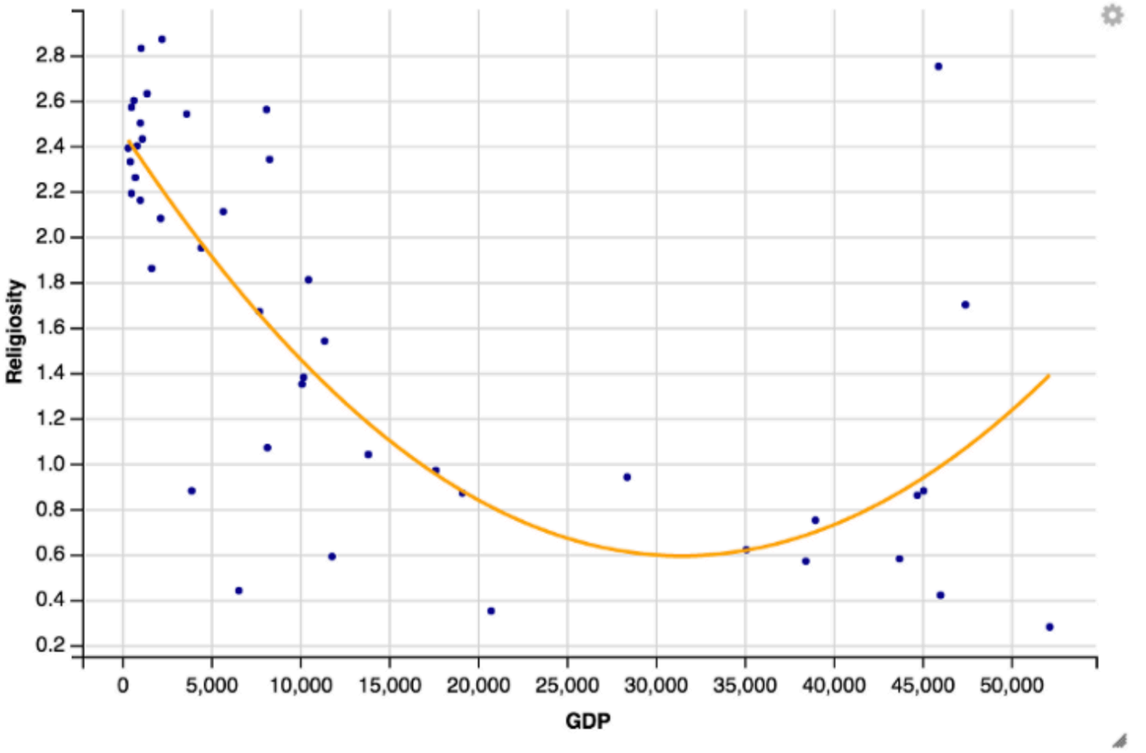
80
75
70
65
60
55

ggvis

```
install.packages("ggvis")
```

http://ggvis.rstudio.com/


5
12
15
5 6 7 8 9 10 11 12 13 14 15
5
8
15
5 6 7 8 9 10 11 12 13 14 15


Religiosity
2.8
2.6
2.4
2.2
2.0
1.8
1.6
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0
5,000
10,000
15,000
20,000
25,000
30,000
35,000
40,000
45,000
50,000
GDP